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INFORMATION SYSTEMS**

MASTER'S THESIS

**PROBABILISTIC MACHINE LEARNING MODELS AND DECISION
OPTIMIZATION FOR HEPATOCELLULAR CARCINOMA DIAGNOSIS**

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**ΤΜΗΜΑ
ΟΙΚΟΝΟΜΙΚΩΝ
ΕΠΙΣΤΗΜΩΝ**

**ΠΜΣ ΔΙΟΙΚΗΣΗ, ΑΝΑΛΥΤΙΚΗ ΚΑΙ ΠΛΗΡΟΦΟΡΙΑΚΑ ΣΥΣΤΗΜΑΤΑ
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**ΑΘΗΝΑ
ΦΕΒΡΟΥΑΡΙΟΣ 2026**

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ABBREVIATIONS

AFP	Alpha-Fetoprotein Level
AP	Average Precision
AUC	Area Under the Curve
AUCPRC	Area Under the Precision-Recall Curve
CBA	Cost-Benefit Analysis
CI	Confidence Interval
CV	Cross-Validation
DCA	Decision Curve Analysis
EDA	Exploratory Data Analysis
FN	False Negative
FP	False Positive
FPR	False Positive Rate
HCC	Hepatocellular Carcinoma
IAF	Intervention Aversion Factor
LFS	Liver Function Score
LR	Logistic Regression
ML	Machine Learning
NCI	National Cancer Institute
NPV	Negative Predictive Value
OM	Overuse Multiplier
PPV	Positive Predictive Value
PR-AUC	Precision-Recall Area Under the Curve
QALYs	Quality-Adjusted Life Years
RAF	Risk Aversion Factor
RF	Random Forest
ROC	Receiver Operating Characteristic
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
TN	True Negative
TP	True Positive
TPR	True Positive Rate
XGB	Extreme Gradient Boosting (XGBoost)

ΠΕΡΙΛΗΨΗ

Η παρούσα διπλωματική εργασία αναπτύσσει ένα ολοκληρωμένο πλαίσιο, το οποίο γεφυρώνει την πιθανοκρατική πρόβλεψη κινδύνου με τη βελτιστοποίηση κλινικών αποφάσεων για τη διάγνωση Ηπατοκυτταρικού Καρκινώματος (HCC).

Στο Μέρος I αξιοποιείται ένα δημόσιο και συνθετικό σύνολο δεδομένων με 5.000 παρατηρήσεις και 14 μεταβλητές, που περιγράφουν δημογραφικές και κλινικές μεταβλητές και βιοδείκτες με μέτρια ανισορροπία κλάσεων (21,8% θετικές περιπτώσεις). Η μεθοδολογία υλοποιείται ως αγωγός ασφαλής από διαρροές με στρωματοποιημένο διαχωρισμό εκπαίδευση / επικύρωση / δοκιμή και σύγκριση Λογιστικής Παλινδρόμησης (LR), Τυχαίου Δάσους (RF) και Ακραίας Ενίσχυσης Κλίσης (XGB). Η αξιολόγηση δεν περιορίζεται στη διάκριση μέσω της Περιοχής Κάτω από την Καμπύλη Λειτουργικού Χαρακτηριστικού Δείκτη (ROC-AUC), αλλά επεκτείνεται στην Περιοχή Κάτω από την Καμπύλη Ακρίβειας-Ανάκλισης (PR-AUC) και τη Μέση Ακρίβεια (AP) και σε αξιοπιστία πιθανοτήτων μέσω βαθμολογία Brier και καμπυλών βαθμονόμησης, με επιλογή μεθόδου βαθμονόμησης (σιγμοειδής ή ισοτονική) βάσει επικύρωσης Brier. Για την αξιολόγηση της απροσδόκητα υψηλής διάκρισης σε συνθετικά δεδομένα, εκτελούνται επιπλέον διαγνωστικά ευρωστίες, συμπεριλαμβανομένης της μετάθεσης ετικετών και δοκιμές αντοχής κατά την αφαίρεση χαρακτηριστικών, καθώς και δύο πιο ρεαλιστικές προσομοιώσεις συνθηκών ανάπτυξης με εισαγωγή θορύβου στα αριθμητικά χαρακτηριστικά και μεταβολή επιπολασμού στο test set. Ενώ η συνθετική φύση των δεδομένων αποδίδει σχεδόν τέλειες μετρήσεις, η παρούσα μελέτη αντιμετωπίζει το σύνολο δεδομένων κυρίως ως ελεγχόμενο περιβάλλον για να καταδείξει τη μεθοδολογική σκοπιμότητα του προτεινόμενου πλαισίου και όχι ως σημείο αναφοράς για την κλινική ετοιμότητα.

Στο Μέρος II, οι βαθμονομημένες πιθανότητες $P(Y = 1|x)$ ενσωματώνονται σε ρητό πλαίσιο θεωρίας αποφάσεων τύπου Pauker-Kassirer, όπου η μετάβαση από πρόβλεψη σε δράση προκύπτει από τη μέγιστη αναμενόμενη ωφέλεια και οδηγεί σε ερμηνεύσιμο βέλτιστο κατώφλι p^* . Ορίζονται δύο περιβάλλοντα ωφελειών, το κλινικό με Ποιοτικά Προσαρμοσμένα Έτη Ζωής (QALYs) και το οικονομικό με Ανάλυση Κόστους-Οφέλους (CBA), ενώ εξετάζονται έξι σενάρια (ισορροπημένο/συντηρητικό/επιθετικό), τα οποία αναπτύσσουν διαφορετικές στάσεις κινδύνου. Τα αποτελέσματα δείχνουν, ότι το p^* μεταβάλλεται σημαντικά ανάλογα με το αντικειμενικό κριτήριο, κυμαινόμενο από 1,4% έως 73,5%. Επιπλέον, μέσω ανάλυσης ευαισθησίας (one-way) των παραμέτρων προτίμησης και κινήτρων, όπως Συντελεστής Αποστροφής Κινδύνου (RAF), Συντελεστής Αποστροφής Παρέμβασης (IAF) και Πολλαπλασιαστής Υπερβολικής Χρήσης (OM), τεκμηριώνεται ότι το p^* μεταβάλλεται συνεχώς και ενίοτε μη γραμμικά. Τελικά, η διατριβή γεφυρώνει το χάσμα πιθανότητας-απόφασης, αποδεικνύοντας ότι η βέλτιστη απόφαση δεν είναι ιδιότητα της AUC, αλλά συνέπεια των συγκεκριμένων συμβιβασμών κόστους/οφέλους που ορίζονται από το ενδιαφερόμενο μέρος.

Λέξεις-Κλειδιά: Ηπατοκυτταρικό Καρκίνωμα (HCC), Πιθανοτική Μηχανική Μάθηση, Βαθμονόμηση Πιθανοτήτων, Θεωρία Αποφάσεων, Ανάλυση Κόστους-Οφέλους, Κλινική Χρησιμότητα

ABSTRACT

This master's thesis develops an end-to-end framework that bridges probabilistic risk prediction with decision optimization for Hepatocellular Carcinoma (HCC) diagnosis.

In Part I, a public and synthetic dataset with 5,000 observations and 14 variables is used, capturing demographic/clinical factors and biomarkers, with moderate class imbalance (21.8% positives). A leakage-safe machine learning pipeline is implemented with stratified train/validation/test splitting, comparing Logistic Regression (LR), Random Forest (RF) and Extreme Gradient Boosting (XGB). Model assessment goes beyond discrimination via the Area Under the Receiver Operating Characteristic Curve (ROC-AUC) by additionally reporting the Area Under the Precision-Recall Curve (PR-AUC) and Average Precision (AP), as well as probability quality through Brier score and calibration curves, while selecting calibration (sigmoid or isotonic) based on validation Brier performance. To contextualize unexpectedly high discrimination on synthetic data, robustness diagnostics are additionally performed, including label permutation and feature-ablation stress, as well as two more realistic simulations of growth conditions with noise injection to the numerical features and prevalence shift in the test set. While the synthetic nature of the data yields near-perfect metrics, this study treats the dataset primarily as a controlled environment to demonstrate the methodological feasibility of the proposed framework rather than as a benchmark for clinical readiness.

In Part II, calibrated probabilities $P(Y = 1|x)$ are embedded into an explicit Pauker–Kassirer decision-theoretic model, where actions are chosen by maximizing expected utility, yielding an interpretable optimal decision threshold p^* . Two utility environments are defined, the clinical using Quality-Adjusted Life Years (QALYs) and the economic using Cost-Benefit Analysis (CBA), while six scenarios (balanced/conservative/aggressive) are constructed to reflect alternative risk attitudes. Results demonstrate that p^* varies substantially with the objective function, ranging from 1.4% to 73.5%. Furthermore, through one-way sensitivity analysis of preference/motivation parameters such as the Risk Aversion Factor (RAF), Intervention Aversion Factor (IAF) and Overuse Multiplier (OM), it is documented that p^* varies continuously and sometimes non-linearly. Ultimately, the thesis bridges the probability-to-decision gap, demonstrating that the optimal decision is not a property of the AUC, but a consequence of the specific cost/benefit trade-offs defined by the stakeholder.

Keywords: Hepatocellular Carcinoma, Probabilistic Machine Learning, Probability Calibration, Decision Theory, Cost-Benefit Analysis (CBA), Clinical Utility

Chapter 1: Introduction

1.1 Hepatocellular Carcinoma (HCC)

Hepatocellular carcinoma (HCC) is the most common type of primary liver cancer and a major contributor to cancer-related mortality worldwide. Clinically, the stage at which the disease is diagnosed is a decisive factor for prognosis. When malignancy is detected at an early stage, therapeutic options such as surgical tumor resection or liver transplantation are often available. This gap between timely and delayed diagnosis constitutes the central focus of the present thesis. The risk of developing HCC is multifactorial. In clinical practice, physicians make decisions based on a combination of demographic information, lifestyle-related exposures, medical history and biomarker signals. The dataset used in this thesis reflects this structure. Specifically, it is a synthetic dataset from Kaggle that is medically realistic and includes 5,000 patient records and 14 variables. It contains demographic variables (age, sex, BMI), lifestyle factors (alcohol consumption, smoking status, physical activity), clinical history indicators (hepatitis B/C, history of cirrhosis, diabetes, family history of cancer) and quantitative clinical measurements (Liver Function Score (LFS) and Alpha-Fetoprotein Level (AFP)). Methodologically, these predictor variables define a risk profile rather than a single diagnostic test and therefore, the task is formulated as a predictive modeling problem that produces risk estimates.

At the same time, the synthetic nature of the dataset implies that relationships may be cleaner than in real surveillance cohorts, potentially yielding unusually high separability and optimistic discrimination metrics. For this reason, the thesis treats the dataset primarily as a controlled environment for methodological demonstration and complements baseline performance with diagnostic robustness evaluations designed to emulate key real-world threats (e.g., measurement uncertainty and prevalence differences).

However, risk estimation alone does not solve the clinical problem. A common limitation in many studies is that they stop after selecting the best classifier and then apply a default threshold, such as 0.50, to produce a binary decision (cancer vs no cancer). In HCC, where the consequences of failing to detect the disease (False Negatives (FN)) may be particularly severe, such a threshold may be clinically inappropriate. For this reason, the present thesis treats prediction and decision-making as two distinct steps that must be optimized jointly.

1.2 Classification and Probabilistic Prediction

This thesis adopts a supervised learning framework in which each observation corresponds to an individual patient profile and the target variable indicates whether liver cancer is present ($\text{liver_cancer} = 1$) or not ($\text{liver_cancer} = 0$).

Although a binary classifier can output a simple “Yes” or “No,” the most valuable information for clinical reasoning is the estimated probability

$$P(\text{liver_cancer} = 1|x),$$

which represents risk on a continuous scale rather than as a strict categorical label.

Probability estimates are clinically useful for several reasons,

- **Clinical interpretability and triage:** a risk score of 0.05 and a risk score of 0.45 should not be treated the same way, even if both are below 0.50.
- **Threshold flexibility:** different clinical settings (conservative vs aggressive decision strategies) may require different decision thresholds.
- **Decision-theoretic integration:** probabilistic estimates are the natural input for expected-utility calculations and formal threshold models, which are developed in Part II.

Consequently, Part I does not aim only at accurate prediction, but also at evaluating models from two complementary perspectives,

- **Discrimination:** the ability to correctly rank and classify patients.
- **Calibration:** the extent to which predicted probabilities correspond to observed outcome frequencies.

For this reason, the evaluation framework extends beyond accuracy and Receiver Operating Characteristic (ROC) curves and includes probability-focused metrics such as calibration curves and the Brier score.

1.3 Overview of the Modeling Approach in Part I

Part I (“Data, Methodology and Probability Evaluation”) is organized around a structured Machine Learning (ML) pipeline that includes dataset exploration, preprocessing, model development and systematic evaluation.

Before modeling, the dataset is examined in terms of its structure, completeness, class balance, distributional characteristics and the associations between predictor variables and the target outcome. This stage informs both the selection of appropriate evaluation metrics and the transformations that may be required.

In this thesis, three widely used and clinically relevant classification models are trained and compared,

- **Logistic Regression (LR):** a baseline probabilistic linear model with high interpretability and a direct probabilistic output.
- **Random Forest (RF):** an ensemble of decision trees capable of capturing non-linearities and predictor interactions.
- **Extreme Gradient Boosting (XGB):** a gradient-boosted tree method that often achieves strong empirical performance by sequentially correcting errors of previous trees.

The selection of the best model is framed as a multi-criteria decision process. Models are compared in terms of discrimination (Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and Precision-Recall Area Under the Curve (PR-AUC)), calibration (Brier score and calibration curves) and threshold-dependent metrics derived from the confusion matrix. Thus, the best model in Part I is defined as the one providing the most reliable probabilistic estimates, which will then be used for decision optimization.

A notable result is that XGB achieves exceptionally high performance on this dataset. To ensure that this near-perfect discrimination is not an artifact of data leakage or an unusually favorable split, diagnostic robustness evaluations are performed. This finding is important but should be interpreted cautiously, because in synthetic datasets, near-perfect discrimination may indicate that the predictor space encodes almost deterministic patterns for the target variable, which may not generalize to real clinical data. While the synthetic nature of the dataset yields near-perfect discrimination metrics (Area Under the Curve (AUC) ≈ 1.0), this thesis treats the data primarily as a controlled environment to demonstrate the methodological framework of bridging probabilistic ML with decision theory, rather than as a benchmark for clinical readiness.

1.4 Overview of Decision Optimization in Part II

Part II (“Decision Optimization in Diagnosis”) starts from a practical clinical question, “Given a predicted probability of HCC, when should clinicians initiate escalation and when should they not?”

A predicted probability can be used meaningfully only after a decision threshold p^* is selected, such that,

- Escalate if $P \geq p^*$
- No Escalation if $P < p^*$

The choice of p^* should not be arbitrary. Instead, Part II applies Decision Theory by introducing a payoff/utility matrix aligned with the confusion matrix outcomes, True Positive (TP), False Positive (FP), True Negative (TN) and False Negative (FN). This mapping makes the clinical meaning of outcomes explicit. The thesis adopts the Pauker and Kassirer threshold framework¹, which defines the optimal threshold as the probability at which the expected utility of escalating equals the expected utility of not escalating.

Let U_{11} denote the utility of escalation when cancer is present, U_{01} the utility of escalation when cancer is absent, U_{10} the utility of non-escalation when cancer is present and U_{00} the utility of non-escalation when cancer is absent.

Then,

$$\begin{aligned}\mathbb{E}[U_{\text{escalate}}] &= P \cdot U_{11} + (1 - P) \cdot U_{01} \\ \mathbb{E}[U_{\text{no_escalation}}] &= P \cdot U_{10} + (1 - P) \cdot U_{00}\end{aligned}$$

The optimal threshold p^* is obtained by solving,

$$\mathbb{E}[U_{\text{escalate}}] = \mathbb{E}[U_{\text{no_escalation}}],$$

which yields an expression of p^* as a function of relative utility differences.

A central contribution of Part II is that it does not rely on a single set of utilities. Instead, it recognizes that the optimal threshold depends on the objective function and risk attitude. For this reason, six decision-making scenarios are defined along two axes,

- **Utility type:** clinical (Quality-Adjusted Life Years (QALYs)) and economic (Cost–Benefit Analysis (CBA))
- **Risk attitude:** conservative, balanced and aggressive

An optimal threshold is computed for each scenario. This is precisely the step that transforms the thesis from simple model training into a structured decision support framework. ML models provide calibrated probability estimates, while decision theory provides actionable thresholds aligned with clinical or economic priorities.

In addition to these six reference scenarios, Part II includes one-way sensitivity analyses that systematically vary preference and incentive parameters (e.g. Risk Aversion Factor (RAF) /

¹ Pauker, S. G., & Kassirer, J. P. (1980). The threshold approach to clinical decision making. *New England Journal of Medicine*, 302(20), 1109-1117.

Intervention Aversion Factor (IAF) in the clinical QALYs framework and Overuse Multiplier (OM) in the economic framework).

Below, Figure 1 illustrates the integrated framework of this thesis, demonstrating how the calibrated risk probabilities from Part I serve as the necessary input for the decision-theoretic model in Part II, thereby bridging the probability-to-decision gap.

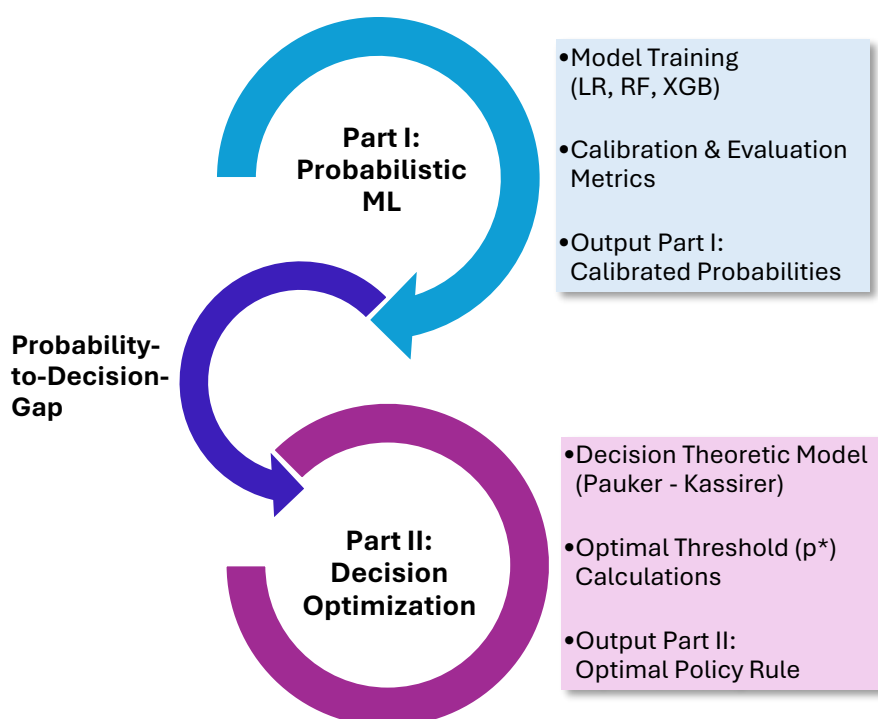


Figure 1: Thesis Workflow Diagram

1.5 Related Literature, Research Gap and Contributions of the Thesis

In recent years, research activity has increased substantially around clinical prediction models that use (ML) methods for estimating disease risk or supporting diagnostic decision-making. At the same time, it has become clear that a model does not appear to perform well on a single discrimination index, but requires transparent and methodical development, documentation and evaluation, so that the findings are reproducible and clinically usable. In this direction, recent TRIPOD+AI guidelines² emphasize systematic reporting of prediction model development and evaluation (with regression or ML), while PROBAST+AI³ extends the qualitative assessment of risk of bias and applicability to models that use artificial intelligence techniques. In addition, contemporary guides for the development and evaluation of clinical prediction models stress the

² Collins GS, et al. (2024). TRIPOD+AI statement: Updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*, 385

³ Moons KGM, et al. (2025). PROBAST+AI: An updated quality, risk of bias, and applicability assessment tool for clinical prediction models that use regression or artificial intelligence methods. *BMJ*, 389

need not only for internal and external validation, but also for assessment of clinical usefulness, so that performance is interpreted in terms of consequences for the patient and the healthcare system (Efthimiou et al., 2024, Riley et al., 2024).

In the field of HCC, the ML literature has produced a wide range of risk estimation and diagnostic support models, leveraging electronic health records (EHRs), clinical characteristics and biomarkers, as well as more complex sources (e.g., radiomics). For example, HCC risk prediction models have been proposed for patients with chronic hepatitis B or cirrhosis, along with models aiming at personalized prediction using relatively minimal feature sets to enhance practical applicability (Kim et al., 2022, Liang et al., 2021, Lee et al., 2023, Xu et al., 2024). Moreover, systematic reviews in HCC (especially in radiomics applications) suggest that evaluation often focuses on classification metrics, with emphasis on discrimination, while documentation of probabilistic reliability and clinical utility is less consistently reported across studies (Mohd Haniff et al., 2024). In other words, the overall conclusion of the related literature is that ML models show promising discrimination performance, but the level of methodological documentation and the translation of model performance into clinically applicable decision policies remain heterogeneous (Collins et al., 2024, Mohd Haniff et al., 2024, Riley et al., 2024). An additional issue highlighted in HCC (as in many clinical problems) is that datasets are often imbalanced, which affects how a model should be evaluated. Using ROC-AUC as the sole criterion can be misleading in strongly imbalanced settings, whereas metrics such as PR-AUC (AUPRC) often provide a more direct view of performance in detecting rare positive cases. Relevant studies have examined properties of the ROC curve and ROC-AUC – PR-AUC comparisons under imbalance and different sampling schemes, emphasizing that metric selection must be linked to the real-world problem and to the type of error that is considered more critical (Kim & Hwang, 2022, Richardson et al., 2024).

However, even when discrimination is assessed appropriately, a crucial step for clinical use is moving from good rankings to reliable risk probabilities. This concerns calibration, i.e. the extent to which predicted probabilities correspond to the actual event frequencies. The literature provides both theoretical guidance on calibration measures and calibration models, as well as comparative analyses of calibration techniques (e.g., sigmoid/Platt scaling, isotonic regression), stressing that probabilistic quality is a distinct property from discrimination (Huang et al., 2020, Ojeda et al., 2023, Silva Filho et al., 2023). In the same spirit, metrics such as the Brier score function as overall measures of probabilistic prediction quality, linking prediction to probability reliability and indirectly to clinical utility (Zhu et al., 2025).

The need to connect probabilistic predictions with clinical decision-making takes us to the next level, namely, that how good the model is, is not limited to AUC performance, but depends on what decision is made based on probability and what the consequences of errors are (FP/FN). Tools such as Decision Curve Analysis (DCA) and the concept of net benefit approach exactly

this problem, that is, they assess whether a model adds clinical utility over a realistic range of thresholds, compared to simple “treat-all” or “treat-none” policies (Piovani et al., 2023, Sadatsafavi et al., 2021, Vickers & Holland, 2021). Thus, the literature converges on the view that risk prediction and decision-making are two distinct steps that must be connected explicitly, first, a reliable probability estimate and then an explicit choice of threshold.

In summary, the related work shows that ML has provided strong predictive tools for HCC and has matured methodologically with respect to discrimination, metrics under class imbalance and calibration techniques (Huang et al., 2020, Mohd Haniff et al., 2024, Richardson et al., 2024). Nevertheless, a critical gap remains in the systematic translation of probabilistic prediction into clinically and economically actionable decisions.

The gap that emerges is the tendency of a substantial part of the literature to conclude analysis at the level of selecting the best classifier and then adopt default thresholds (e.g., 0.5) or threshold selection criteria that are not explicitly linked to clinical usefulness, cost or preferences, particularly in problems such as HCC, where the cost of a false negative may be disproportionately severe. This creates a probability-to-decision gap, where even a well-trained model may lead to suboptimal practice if the decision threshold is not defined based on the consequences of FP/FN and the target objective (clinical or economic) (Riley et al., 2024, Sadatsafavi et al., 2021, Vickers & Holland, 2021).

Based on the above research gap, the present thesis does not treat prediction as an end in itself, but as the first stage of a chain that is only complete when an interpretable and well-justified decision policy is derived.

Overall, this thesis contributes the following,

- Consistent development and evaluation of prediction models for HCC, with emphasis on rigorous methodological documentation aligned with modern reporting and quality assessment directions (reporting and risk-of-bias considerations).
- Probabilistic evaluation beyond discrimination, so that predictions act as reliable risk probabilities, using appropriate metrics under class imbalance and assessment and improvement of calibration (calibration models, Brier-based measures).
- Diagnostic robustness evaluation to contextualize unexpectedly high discrimination on synthetic data, including leakage sanity checks (label permutation and repeated stratified splits).
- Stress testing via biomarker ablation (AFP, LFS and joint removal) to assess whether performance is dominated by a small subset of features.
- Additional robustness experiments to emulate deployment-relevant threats, like measurement uncertainty via noise injection and screening-like low-prevalence evaluation via prevalence shift (5% test prevalence) with repeated resampling.

- Explicit bridging from probability to decision by deriving actionable thresholds through decision theory, treating the threshold as a policy-driven choice rather than a technical detail.
- Integration of probabilistic prediction and decision policy into a coherent framework, from calibrated risk estimation to interpretable and justified clinical action.

In this way, the thesis positions itself at the intersection of contemporary practices in clinical prediction model development and decision-analytic evaluation, it maintains the necessary rigor in probabilistic reliability, while simultaneously translating ML outputs into clinically and economically interpretable decisions.

1.6 Structure of the Thesis

This thesis is organized into an introductory chapter (the present chapter), followed by two main parts.

Part I: Data, Methodology and Probability Evaluation establishes the empirical and methodological foundations for probabilistic prediction of liver cancer risk. Chapter 2 (Dataset Exploration) presents the dataset and Exploratory Data Analysis (EDA), including descriptive statistics. Chapter 3 describes the preprocessing strategy and predictive modeling pipeline, including development and tuning of LR, RF and XGB models, as well as the Cross-Validation (CV) process. Chapter 4 presents and compares predictive performance in terms of discrimination and probability quality, leading to the selection of the most suitable probabilistic model for decision-making.

Part II: Decision Optimization in Diagnosis translates probabilistic predictions into actionable diagnostic decisions. Chapter 5 introduces the decision-theoretic framework, including payoff/utility modeling and the Pauker–Kassirer threshold approach. Chapters 6 and 7 define the clinical and economic utility environments, respectively, under different risk attitudes (conservative, balanced, aggressive). Chapter 8 consolidates the payoff matrices and optimal thresholds across all scenarios and operationalizes them on the held-out test set by reporting escalation rates and confusion-matrix outcomes.

The thesis concludes with Chapter 9, where findings from both parts are summarized, key limitations are highlighted and directions for future research and extensions are proposed.

Part I: Data, Methodology and Probability Evaluation

Chapter 2: Data Exploration

2.1 Dataset Description

The dataset used in the present thesis, as mentioned in the introductory chapter, comes from the public Kaggle repository titled “Predict Liver Cancer from Clinical Features” (Miadul, 2023). Although it is a synthetic dataset, it has been designed to be medically realistic. It includes 5,000 observations and 14 variables, describing anonymized patient profiles created for the study of HCC. Each record corresponds to a unique patient and includes demographic characteristics, lifestyle factors, clinical history indicators and quantitative laboratory measurements. The target variable `liver_cancer` is binary and encodes the presence (`liver_cancer = 1`) or absence (`liver_cancer = 0`) of cancer.

The input variables include,

- Demographic characteristics: `age`, `gender`, `bmi`
- Lifestyle factors: `alcohol_consumption`, `smoking_status`, `physical_activity_level`
- Medical history indicators: `hepatitis_b`, `hepatitis_c`, `cirrhosis_history`, `family_history_cancer`, `diabetes`
- Clinical biomarkers: `liver_function_score`, `alpha_fetoprotein_level`

Overall, the above variables represent both environmental and biological factors that are associated with the risk of developing liver cancer.

age	gender	bmi	alcohol_consumption	smoking_status	hepatitis_b	hepatitis_c	liver_function_score
68	Female	18.10	Regular	Former	0	0	51.90
81	Female	19.90	Occasional	Never	0	0	41.60
58	Female	25.50	Never	Never	0	0	76.00
44	Male	16.00	Never	Former	0	0	50.30
72	Male	21.00	Occasional	Former	0	0	39.50
alpha_fetoprotein_level	cirrhosis_history	family_history_cancer	physical_activity_level	diabetes	liver_cancer		
16.4400	0	0	Low	0	0		
8.0900	0	0	Moderate	1	0		
0.6400	0	0	Moderate	0	0		
19.0900	0	0	Low	1	0		
4.9500	1	0	Low	1	1		

Table 1: First five rows of the dataset (see Appendix A, Code A.3)

Table 1 presents the combination of numerical and categorical predictor variables, as well as the target variable (liver_cancer). The data types, as shown in Table 2, include integers for binary indicators, floats for continuous measurements and objects for categorical variables.

Shape (rows, cols): (5000, 14)

Dtypes:	
age	int64
gender	object
bmi	float64
alcohol_consumption	object
smoking_status	object
hepatitis_b	int64
hepatitis_c	int64
liver_function_score	float64
alpha_fetoprotein_level	float64
cirrhosis_history	int64
family_history_cancer	int64
physical_activity_level	object
diabetes	int64
liver_cancer	int64
dtype:	object

Table 2: Dataset dimensions and data types (see Appendix A, Code A.3)

This heterogeneous structure requires type-aware preprocessing, standardization of numerical variables and encoding of categorical variables during model training.

An initial assessment showed that no variable contains missing values. All variables have `missing_count = 0` and `missing_rate = 0.0000`, as shown in the following table. This allows models to be trained directly without the need for missing value imputation. Nevertheless, the preprocessing pipeline has been designed to support imputation, in order to ensure robustness when working with future real-world data.

	missing_count	missing_rate
age	0	0.0000
gender	0	0.0000
bmi	0	0.0000
alcohol_consumption	0	0.0000
smoking_status	0	0.0000
hepatitis_b	0	0.0000
hepatitis_c	0	0.0000
liver_function_score	0	0.0000
alpha_fetoprotein_level	0	0.0000
cirrhosis_history	0	0.0000
family_history_cancer	0	0.0000
physical_activity_level	0	0.0000
diabetes	0	0.0000
liver_cancer	0	0.0000

Table 3: Missing values per variable (see Appendix A, Code A.4)

2.2 Variable Catalog and Descriptive Statistics

Following the structural check, the variables are grouped into numerical and categorical variables in order to support distinct analytical approaches. The numeric predictor variables include age, bmi, liver_function_score and alpha_fetoprotein_level. The categorical predictor variables and binary indicator variables include all remaining attributes, such as gender, alcohol_consumption, smoking_status and the clinical history variables. This grouping ensures that continuous numerical variables can be standardized and analyzed using continuous metrics, while categorical features can be encoded and compared using frequency-based analyses. Next, as in the subsequent sections, detailed descriptive statistics and exploratory visualizations are presented for each category.

The target variable liver_cancer indicates whether a patient is diagnosed with liver cancer (liver_cancer = 1) or not (liver_cancer = 0).

Table 4 below presents the distribution of the two classes.

liver_cancer	count	ratio
0	3911	0.7822
1	1089	0.2178

Table 4: Target balance (counts and ratios) (see Appendix A, Code A.4)

The dataset exhibits moderate class imbalance, with approximately 21.8% positive cases and 78.2% negative cases. This asymmetry makes it appropriate to use evaluation metrics that remain reliable under unequal class proportions, such as PR-AUC. In addition, to reduce bias in favor of the majority class during model fitting, the LR model is trained using class-balanced weighting. Table 5 presents descriptive statistics for all numeric variables. Overall, the observed distributions fall within plausible clinical ranges, supporting the medically realistic nature of the dataset.

- **Age:** The mean age is approximately 57 years ($SD \approx 15.8$), with an adult range of 30–84 years. This range is generally consistent with typical risk age groups for liver cancer.
- **BMI:** The mean BMI is approximately 25.9, corresponding to the “overweight” category according to World Health Organization (WHO) standards, with moderate variability ($SD \approx 5.1$).
- **LFS:** The mean is approximately 65 ($SD \approx 14.9$), indicating variation in liver function across individuals.
- **AFP:** The mean is approximately 15.1, with pronounced skewness (skewness = 1.95), a finding consistent with clinical expectations for AFP as a biomarker in positive cancer cases.

- **Binary medical indicators** (e.g., hepatitis B/C, cirrhosis, diabetes): These binary indicator variables take values 0/1 and show skewness > 1 , reflecting their relatively low prevalence in the sample.

	count	mean	std	min	25%
age	5000.00	57.22	15.80	30.00	44.00
bmi	5000.00	25.90	5.11	8.10	22.50
hepatitis_b	5000.00	0.15	0.36	0.00	0.00
hepatitis_c	5000.00	0.10	0.29	0.00	0.00
liver_function_score	5000.00	65.20	14.88	5.50	54.80
alpha_fetoprotein_level	5000.00	15.11	15.20	0.00	4.36
cirrhosis_history	5000.00	0.30	0.46	0.00	0.00
family_history_cancer	5000.00	0.24	0.43	0.00	0.00
diabetes	5000.00	0.20	0.40	0.00	0.00
	50%	75%	max	skew	kurtosis
age	57.00	71.00	84.00	-0.02	-1.18
bmi	25.90	29.30	43.70	-0.01	-0.00
hepatitis_b	0.00	0.00	1.00	1.95	1.79
hepatitis_c	0.00	0.00	1.00	2.75	5.55
liver_function_score	65.20	75.20	115.70	0.01	-0.03
alpha_fetoprotein_level	10.34	21.01	136.80	1.95	5.18
cirrhosis_history	0.00	1.00	1.00	0.86	-1.25
family_history_cancer	0.00	0.00	1.00	1.21	-0.54
diabetes	0.00	0.00	1.00	1.51	0.27

Table 5: Descriptive statistics for numeric variables (see Appendix A, Code A.5)

The observed skewness of AFP and the binary indicators suggests right-tailed distributions. This distributional form is expected to be handled effectively by the non-parametric and ensemble-based models employed in later stages of the analysis.

variable	n_unique	n_missing	top_category	top_freq
alcohol_consumption	3	0	Never	2014
smoking_status	3	0	Never	2407
physical_activity_level	3	0	Moderate	2027
gender	2	0	Female	2506

Table 6: Descriptive statistics for categorical variables (see Appendix A, Code A.6)

Table 6 presents the number of distinct categories and the most frequent category for each categorical variable. Overall, the variables show low to moderate categorical diversity, with clearly dominant levels,

- **Gender:** Two categories (Male/Female), with a slight predominance of females ($\approx 50\%$).
- **Alcohol consumption:** Three categories, with “Never” being the most frequent ($\approx 40\%$).
- **Smoking status:** Three categories, with “Never” also being the dominant class.

- **Physical activity level:** Three categories (“Low,” “Moderate,” “High”), with “Moderate” marginally prevailing.

All categorical variables contain no missing values and present sufficiently balanced distributions to support model training and robust one-hot encoding.

These exploratory summaries confirm that the dataset is well-structured, complete and clinically realistic. The next section examines potential data quality issues, such as unrealistic values or extreme outliers, before exploring distributions and associations.

2.3 Data Quality Assessment: Missingness and Validity Checks

Following the descriptive overview, a systematic check is conducted to identify unreasonable or degenerate values in the numerical variables. Given that no missing values were observed in Section 2.1, this stage focuses on the proportion of zero values within both the numerical and the binary indicator variables. Although zero values are expected and meaningful for binary variables (e.g., hepatitis B, diabetes), in continuous variables they may indicate encoding conventions or measurement anomalies.

variable	n_zero	zero_rate
hepatitis_c	4521	0.90
hepatitis_b	4244	0.85
diabetes	4004	0.80
family_history_cancer	3793	0.76
cirrhosis_history	3492	0.70
alpha_fetoprotein_level	1	0.00
age	0	0.00
liver_function_score	0	0.00
bmi	0	0.00

Table 7: Zero-value scan across numeric variables (see Appendix A, Code A.7)

The results in Table 7 show that,

- Binary clinical indicators, such as hepatitis C (90.4%), hepatitis B (84.9%) and diabetes (80.1%), exhibit high proportions of zero values.
- Family history of cancer (75.9%) and history of cirrhosis (69.8%) also show the expected low prevalence of positive cases.
- The continuous variables (age, bmi, liver_function_score, alpha_fetoprotein_level) do not display structural zeros, indicating consistent measurement scales.

These findings confirm that there is no information loss due to missingness. Therefore, all variables can be safely retained for the remainder of the analysis, without requiring corrective interventions.

2.4 Numeric Exploratory Analysis

This section presents the univariate EDA of the numeric variables, using two complementary plots for each variable. For every numerical variable, the following visualizations are provided,

- i. Marginal distribution of the variable across the full sample, using histograms.
- ii. Class-conditional distribution, stratified by the binary target variable, using violin plots.

This combination serves two main purposes. First, it provides a descriptive summary of the range of values, central tendency, dispersion and potential skewness or outliers for each numerical variable. Second, it allows an initial qualitative assessment of whether a variable exhibits distributional separation between the two target classes, which serves as an early indication of potential predictive utility.

Histograms summarize how values are distributed in the overall sample and are particularly informative for continuous measurements such as age, BMI and LFS. Violin plots, on the other hand, depict how the distribution changes as a function of cancer status. Their width reflects the density, while the internal markers provide a compact summary of central tendency and variability. It is important to emphasize that, given the moderate class imbalance, class-wise visualizations should be interpreted as descriptive comparisons and not as evidence of causal relationships.

Age

The age variable spans from early adulthood to older age (approximately 30 to 84 years) and shows a relatively wide distribution, with central tendency around the late 50s. The histogram indicates substantial coverage of middle-aged and older individuals, suggesting that the sample includes meaningful representation of age groups typically associated with increased liver cancer risk.

When stratified by the target variable, the violin plot reveals a clear shift toward higher ages for the cancer class (`liver_cancer = 1`). This result aligns with clinical observations in oncology, according to which risk often increases with age. Nevertheless, the two distributions still overlap, indicating that age alone is not sufficient for accurate classification.

liver_cancer	count	mean	std	median	min	max
0	3911	55.4127	15.6032	54	30	84
1	1089	63.7236	14.7610	66	30	84

Table 8: Target-wise numeric summary table - Age (see Appendix A, Code A.11)

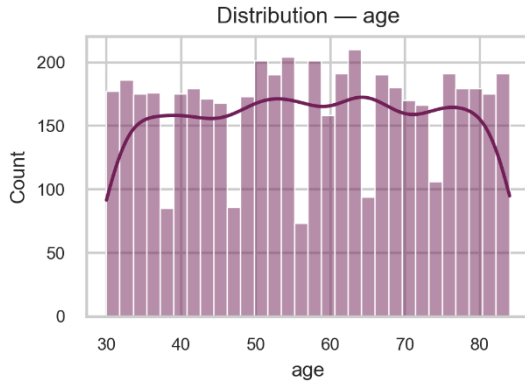


Figure 2: Distribution of Age
(see Appendix A, Code A.8)

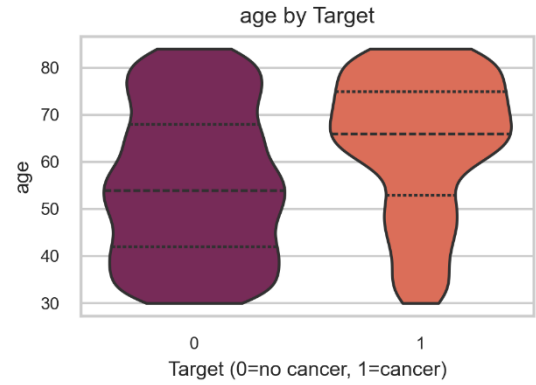


Figure 3: Age by Target
(see Appendix A, Code A.9)

Body Mass Index (BMI)

liver_cancer	count	mean	std	median	min	max
0	3911	25.448	4.8797	25.5	8.3	43
1	1089	27.5446	5.5442	27.9	8.1	43.7

Table 9: Target-wise numeric summary table - BMI (see Appendix A, Code A.11)

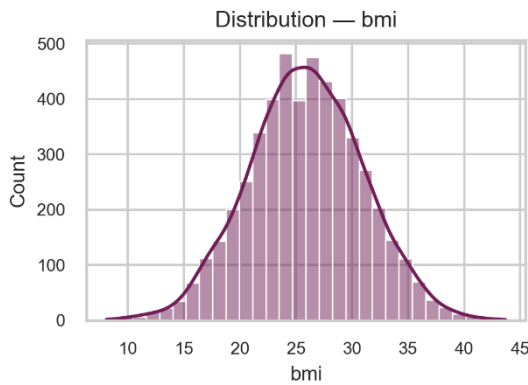


Figure 4: Distribution of BMI
(see Appendix A, Code A.8)

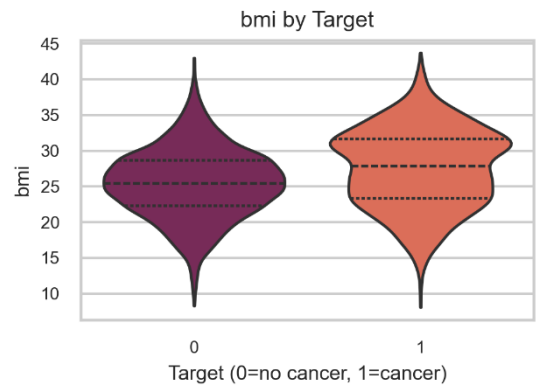


Figure 5: BMI by Target
(see Appendix A, Code A.9)

BMI follows an approximately normal distribution, concentrated around values in the mid-20s, indicating that most observations fall within the normal-to-overweight range. The histogram shows only mild skewness and is broadly consistent with a unimodal distribution.

In the target-stratified violin plot, the cancer-positive class shows a moderate shift toward higher values compared to the non-cancer class. Specifically, for `liver_cancer = 1`, both the median and the upper quantiles are higher, with the density extending further into the overweight range.

However, the two distributions overlap substantially, suggesting that BMI provides limited discriminatory power when examined in isolation.

Liver Function Score

liver_cancer	count	mean	std	median	min	max
0	3911	66.3509	14.3425	66.2	13.9	115.7
1	1089	61.0846	15.9917	60.7	5.5	103.9

Table 10: Target-wise numeric summary table - Liver Function Score (see Appendix A, Code A.11)

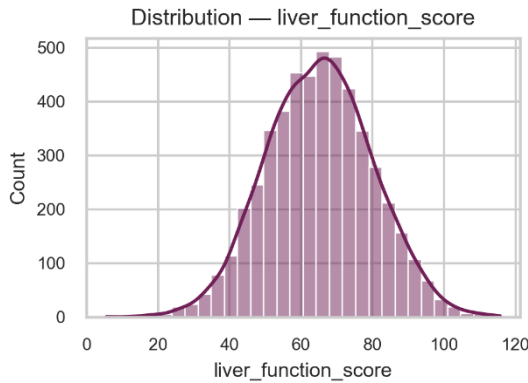


Figure 6: Distribution of Liver Function Score
(see Appendix A, Code A.8)

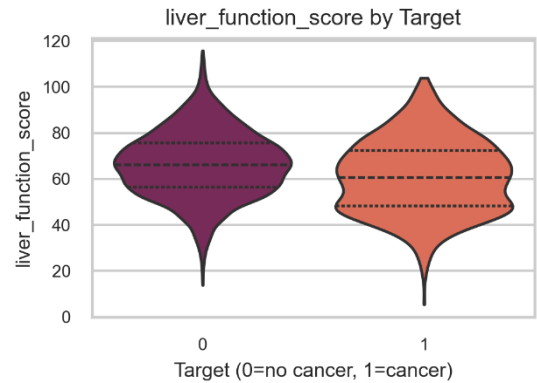


Figure 7: Liver Function Score by Target
(see Appendix A, Code A.9)

The liver function score follows an approximately normal distribution, centered around the mid-60s, with a relatively wide range. A small number of low-score values and a mild tail toward higher values are also observed. This distribution indicates substantial variability in liver function within the population, which is clinically meaningful.

Stratification by the target variable reveals a systematic shift toward lower values for the cancer class (`liver_cancer = 1`), while the non-cancer class is concentrated at higher values. This represents an important descriptive pattern, as it conceptually aligns with the relationship between impaired liver function and increased likelihood of severe liver disease. Although the two distributions still overlap, the shift is visually consistent and suggests that the liver function score may provide useful information for subsequent predictive modeling.

Cirrhosis History

Cirrhosis history is a binary variable and therefore its distribution is concentrated at values 0 and 1, with substantially more observations at 0 (absence of cirrhosis) than at 1 (presence of cirrhosis). This imbalance is typical for disease-history variables in general-population datasets.

However, the target-stratified visualization indicates a strong association with the target variable. The cancer class shows a higher density at value 1 compared to the non-cancer class. In other

words, the probability of liver cancer is substantially higher among individuals with a history of cirrhosis. This finding is not only clinically plausible, but also statistically informative, as binary variables that shift markedly across classes often constitute important predictive factors.

liver_cancer	count	mean	std	median	min	max
0	3911	0.2401	0.4272	0	0	1
1	1089	0.5225	0.4997	1	0	1

Table 11: Target-wise numeric summary table - Cirrhosis History (see Appendix A, Code A.11)

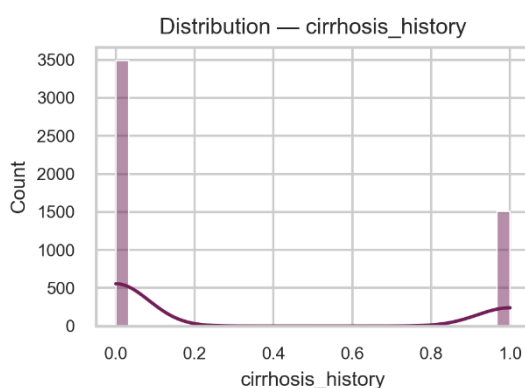


Figure 8: Distribution of Cirrhosis History (see Appendix A, Code A.8)

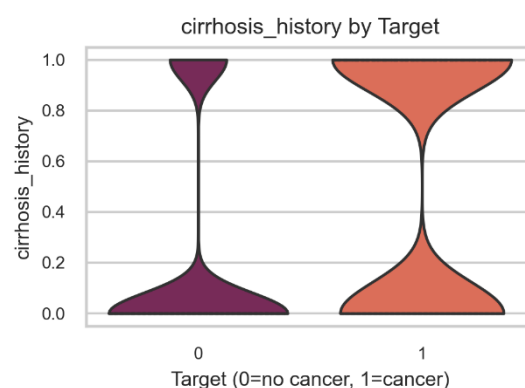


Figure 9: Cirrhosis History by Target (see Appendix A, Code A.9)

Diabetes

liver_cancer	count	mean	std	median	min	max
0	3911	0.1501	0.36	0	0	1
1	1089	0.3756	0.48	0	0	1

Table 12: Target-wise numeric summary table - Diabetes (see Appendix A, Code A.11)

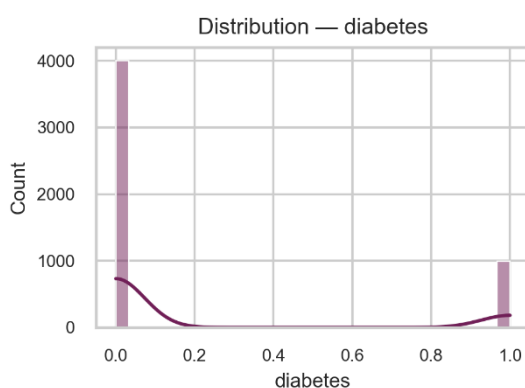


Figure 10: Distribution of Diabetes (see Appendix A, Code A.8)

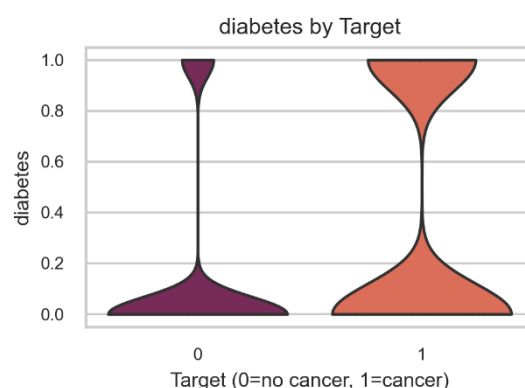


Figure 11: Figure 12: Diabetes by Target (see Appendix A, Code A.9)

The diabetes indicator is also binary and shows dominant concentration at value 0, indicating that most individuals in the dataset do not report diabetes.

Despite this moderate imbalance, the violin plot reveals a clear association between diabetes and the cancer class. Specifically, within the cancer class there is a higher concentration of value 1 (diabetes = 1) compared to the non-cancer class, indicating higher prevalence of diabetes among cancer cases in this dataset. This suggests that diabetes is positively associated with the target variable and may contribute predictive information. It also indicates that diabetes may improve predictive performance when combined with liver-related and lifestyle variables.

Family History of Cancer

Family history of cancer is encoded as a binary indicator and therefore takes only values 0 and 1. Its distribution suggests that most individuals do not report a family history.

When stratified by the target variable, the cancer class shows higher density at value 1 compared to the non-cancer class, indicating that family history is more common among positive cases. This result is consistent with the assumption that inherited susceptibility may be associated with increased cancer risk. Although this feature does not specify the type of cancer or the degree of relatedness, it can provide useful background risk information in multivariable models.

liver_cancer	count	mean	std	median	min	max
0	3911	0.189	0.3915	0	0	1
1	1089	0.4298	0.4953	0	0	1

Table 13: Target-wise numeric summary table - Family History of Cancer (see Appendix A, Code A.11)

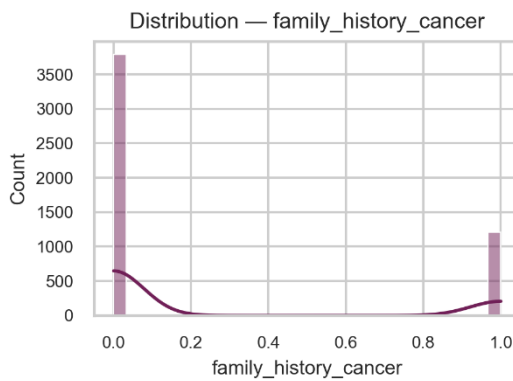


Figure 12: Distribution of Family History of Cancer (see Appendix A, Code A.8)

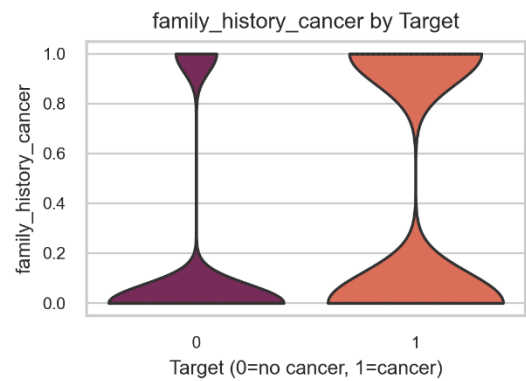


Figure 13: Family History of Cancer by Target (see Appendix A, Code A.9)

Hepatitis B

Hepatitis B is a binary indicator and shows strong class imbalance in its values, with most observations at 0. This suggests low prevalence of hepatitis B in the dataset population.

The violin plot indicates a higher prevalence of value 1 within the cancer class (`liver_cancer = 1`) compared to the non-cancer class, meaning hepatitis B occurs more frequently among positive cases. This finding is clinically plausible given the well-documented association between chronic hepatitis B and long-term liver disease.

From a predictive modeling perspective, such binary risk indicators may have substantial predictive strength because they can sharply increase estimated risk when present. At the same time, the rarity of positive values highlights the need for careful validation and regularization, in order to reduce the risk of overfitting in more flexible models.

liver_cancer	count	mean	std	median	min	max
0	3911	0.114	0.32	0	0	1
1	1089	0.2847	0.45	0	0	1

Table 14: Target-wise numeric summary table - Hepatitis B (see Appendix A, Code A.11)

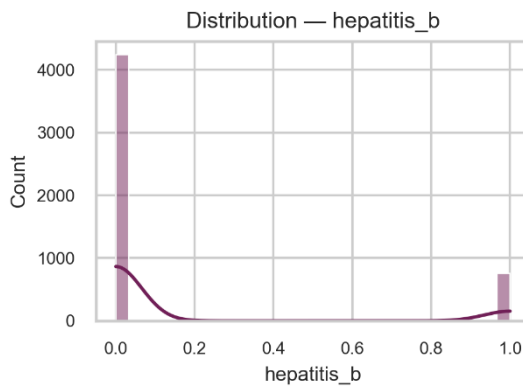


Figure 14: Distribution of Hepatitis B
(see Appendix A, Code A.8)

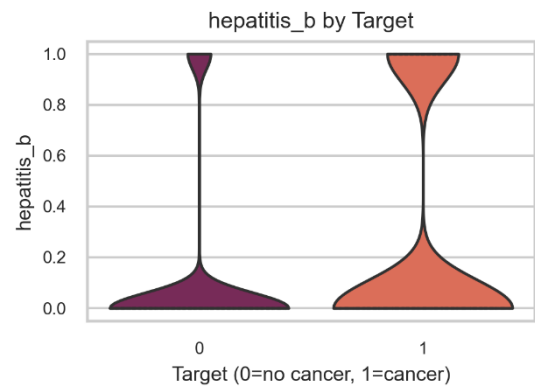


Figure 15: Hepatitis B by Target
(see Appendix A, Code A.9)

Hepatitis C

Hepatitis C is also encoded as a binary indicator and shows even lower prevalence compared to hepatitis B, since the vast majority of observations are at 0.

Nevertheless, the violin plot shows that the cancer class has higher density at value 1 compared to the non-cancer class, suggesting a positive association with liver cancer. As with hepatitis B, this result is clinically plausible because hepatitis C is recognized as an important risk factor for progressive liver disease and HCC.

Within probabilistic classifiers, hepatitis C can provide highly informative signal when present. In addition, it may interact with other predictor variables, amplifying predicted probabilities.

liver_cancer	count	mean	std	median	min	max
0	3911	0.067	0.25	0	0	1
1	1089	0.1993	0.3996	0	0	1

Table 15: Target-wise numeric summary table - Hepatitis C (see Appendix A, Code A.11)

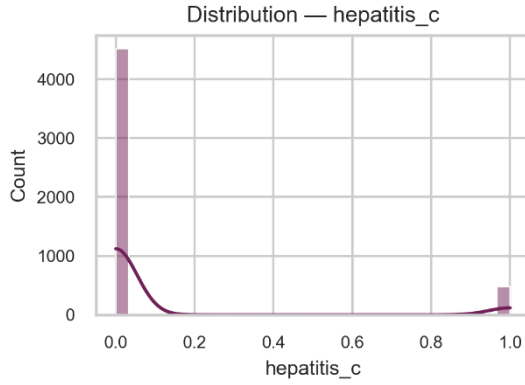


Figure 16: Distribution of Hepatitis C
(see Appendix A, Code A.8)

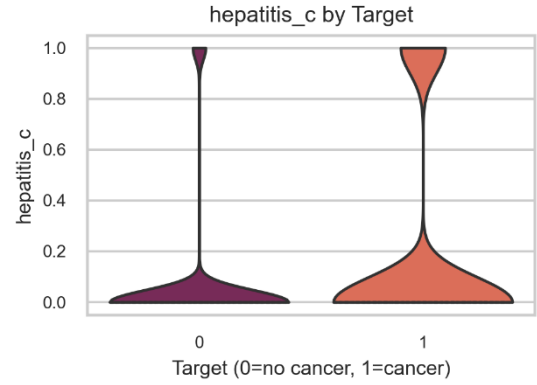


Figure 17: Figure 18: Hepatitis C by Target
(see Appendix A, Code A.9)

Alpha-Fetoprotein Level (AFP)

liver_cancer	count	mean	std	median	min	max
0	3911	13.5384	14.2	9.26	0	136.8
1	1089	20.7516	17.2	18.01	0.02	100.36

Table 16: Target-wise numeric summary table - AFP (see Appendix A, Code A.11)

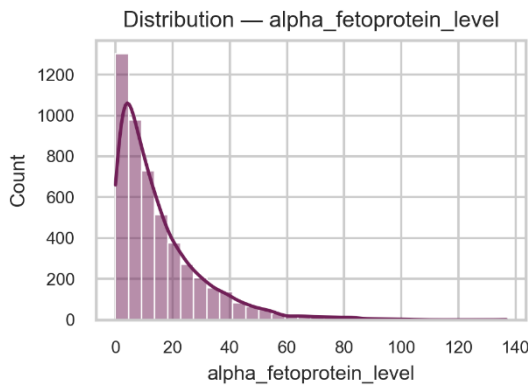


Figure 18: Distribution of AFP
(see Appendix A, Code A.8)

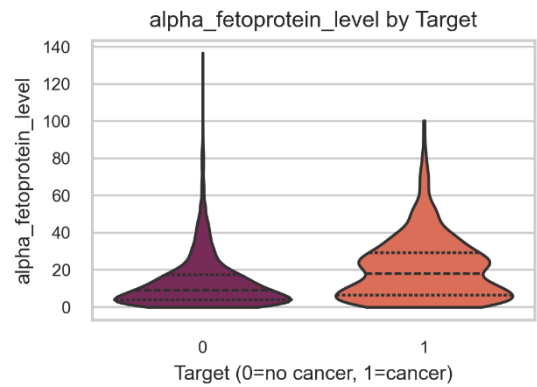


Figure 19: AFP by Target
(see Appendix A, Code A.9)

AFP levels exhibit strong right skewness, with a large proportion of observations concentrated at low values and a long right tail extending to substantially higher values. This skewness is typical for tumor-related biomarkers in clinical data, where most individuals display low baseline levels.

The violin plot demonstrates strong and clinically meaningful separation. The cancer class is shifted toward higher values (increased central tendency) and shows greater dispersion, with higher density at medium to high AFP values.

Such distributions highlight the importance of robust preprocessing and careful calibration assessment, since highly predictive biomarkers may lead to overly extreme (overconfident) probabilities if they are not appropriately transformed and validated.

Summary Conclusion

Across the numerical features, the univariate exploratory EDA indicates that liver cancer status is associated with systematic distributional differences in both continuous and binary variables. Among continuous variables, age shows a clear shift toward higher values in the cancer class, while the liver function score exhibits a directional shift consistent with differentiation in liver status across target outcomes. BMI shows weaker but visible differentiation, suggesting a smaller discriminatory contribution.

Among binary indicators, cirrhosis history, diabetes, family history of cancer and hepatitis B/C show higher prevalence in the cancer class, reinforcing their relevance as risk factors in this dataset.

Finally, AFP stands out due to its pronounced right skewness and especially strong separation with respect to the target variable, consistent with its role as a clinically important biomarker.

2.5 Categorical Exploratory Analysis

To explore the relationship between the categorical variables and the target variable (liver_cancer), two complementary visualizations are used for each categorical variable.

First, Counts by Target are presented, which display the absolute counts of observations per category and per target class. This visualization is essential because it shows the sample size within each category, ensuring that observed differences are not driven by very small subsets.

Second, within-category proportions by target are shown, obtained by normalizing within each category (row-wise normalization). This visualization reflects the prevalence of cancer (liver_cancer = 1) within each category and facilitates comparisons across categories, independently of their overall frequency in the dataset.

Gender

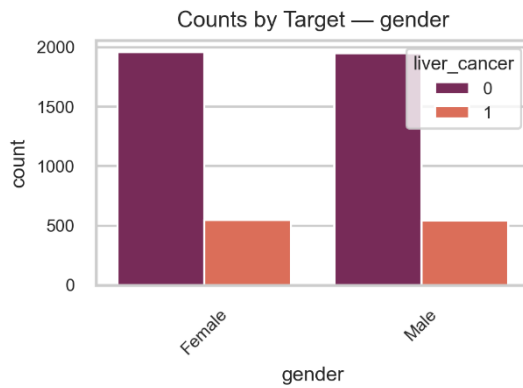


Figure 20: Counts of gender by target (see Appendix A, Code A.10)

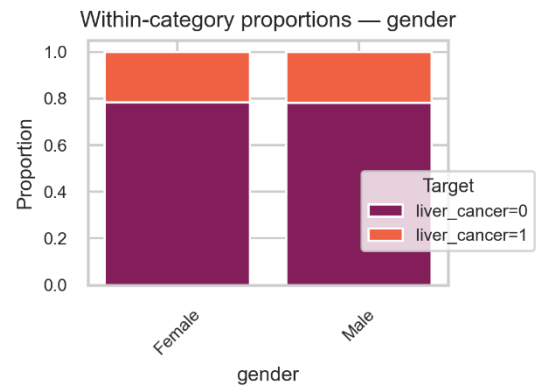


Figure 21: Within-category proportions by target: gender (see Appendix A, Code A.15)

The distribution of gender categories in the dataset is nearly balanced and the Counts by Target plot shows that both male and female categories include comparable sample sizes.

The within-category proportions indicate that liver cancer prevalence is similar across the two genders. This suggests that, within the context of this dataset, gender alone does not appear to be a strong discriminator for the cancer class compared to other clinical variables, at least at the univariate level.

Alcohol Consumption

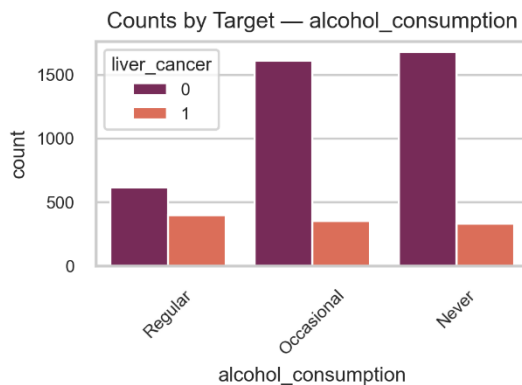


Figure 22: Counts of alcohol consumption by target (see Appendix A, Code A.10)

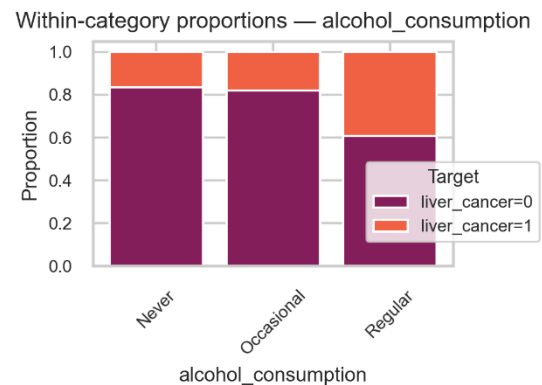


Figure 23: Within-category proportions by target: alcohol consumption (see Appendix A, Code A.15)

Alcohol consumption shows a non-uniform distribution across categories, with “Never” and “Occasional” occurring more frequently than “Regular”. The Counts by Target plot indicates that all categories contain a sufficient number of observations, enabling more reliable descriptive comparisons.

The within-category proportions reveal a notable differentiation. The “Regular” category exhibits clearly higher cancer prevalence compared to “Never” and “Occasional”. This finding suggests that regular alcohol consumption is associated with increased observed prevalence of liver cancer in the present dataset.

From an exploratory perspective, this variable may provide useful discriminatory information for classification, although these differences should not be interpreted as causal. Rather, they suggest that alcohol consumption may function as a categorical predictor variable when combined with other risk factors in multivariable models.

Smoking Status

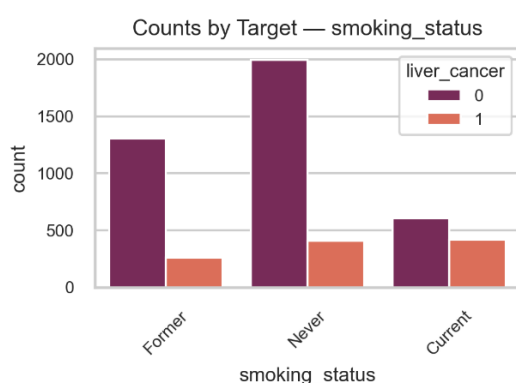


Figure 24: Counts of smoking status by target
(see Appendix A, Code A.10)

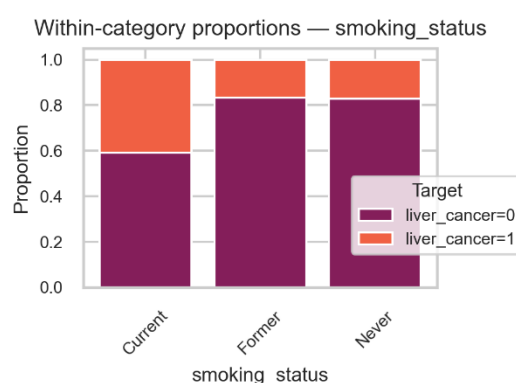


Figure 25: Within-category proportions by target:
smoking status
(see Appendix A, Code A.15)

Smoking status is categorized as “Never”, “Former” and “Current” with “Never” being the most frequent category. The target-stratified counts show adequate sample support in each category, although “Current” includes fewer observations compared to “Never” and “Former.”

The within-category proportions highlight strong differentiation across smoking categories. The “Current” category shows substantially higher observed frequency of liver cancer cases compared to the other categories. This descriptive pattern suggests that current smoking status is strongly associated with increased observed prevalence of liver cancer in the dataset and may provide meaningful predictive value.

Given that smoking_status is also one of the categorical features with higher association values in subsequent statistical tests (e.g., chi-square / Cramer’s V), the visual separation observed here is consistent with the quantified relationship between the variable and the target variable.

Physical Activity Level

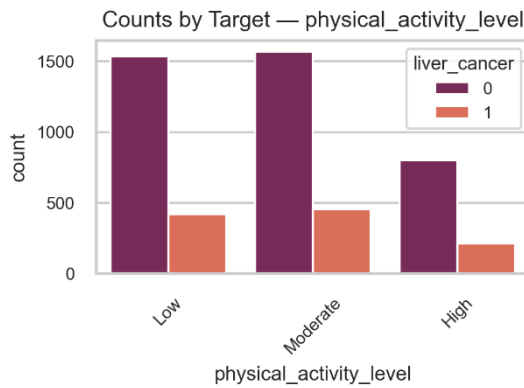


Figure 26: Counts of physical activity level by target (see Appendix A, Code A.10)

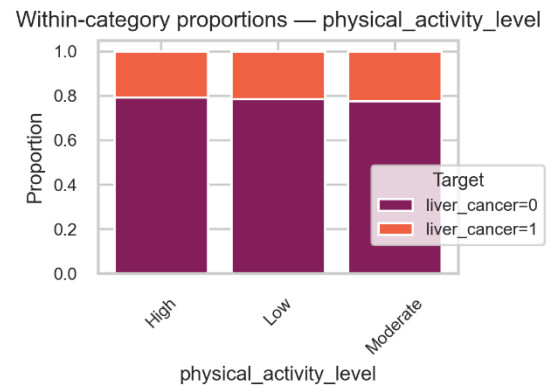


Figure 27: Within-category proportions by target: physical activity level (see Appendix A, Code A.15)

Physical activity level includes three categories, “Low”, “Moderate” and “High” with “High” occurring less frequently than the other two. The Counts by Target plot shows that all categories contain a sufficient number of observations to support comparison.

In the within-category proportions plot, cancer prevalence appears relatively similar across activity categories. Compared with alcohol consumption and smoking status, physical activity level shows weaker visual differentiation across categories, suggesting that physical activity may have a limited univariate effect in this dataset. This does not exclude a potential contribution in a multivariable setting, but descriptively its standalone impact appears smaller.

Summary Conclusion

Overall, the categorical EDA suggests that smoking status and alcohol consumption exhibit the strongest differences in liver cancer prevalence across categories, making them potentially valuable categorical predictor variables for classification models. In contrast, gender and physical activity level show weaker differentiation and may contribute less when examined in isolation, although they may still add value at the multivariable level.

These conclusions are based on descriptive comparisons supported both by absolute category-by-target counts (sample support) and by within-category proportions of cancer prevalence.

2.6 Correlation Analysis and Variable-Target Relationships

Following the univariate exploration of both numerical and categorical variables, the next step is to examine the relationships among predictor variables as well as the association of each predictor variable with the binary target variable (liver_cancer).

Correlation among Numeric Variables

To study pairwise correlations among the numerical variables, Spearman's rank correlation coefficient, ρ_s , is computed. Spearman correlation is preferred over Pearson in an exploratory setting when variables may not follow a normal distribution or when relationships may be monotonic rather than strictly linear, as is often the case with skewed biomarkers (e.g., `alpha_fetoprotein_level`).

In general form, Spearman's ρ_s is defined as the Pearson correlation between the ranks of the two variables,

$$\rho_s = \text{corr}(\text{rank}(X), \text{rank}(Y))$$

In the special case with no ties, it can also be expressed equivalently as,

$$\rho_s = 1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}$$

where d_i is the difference between the ranks of the i -th observation in the two variables and n is the sample size.

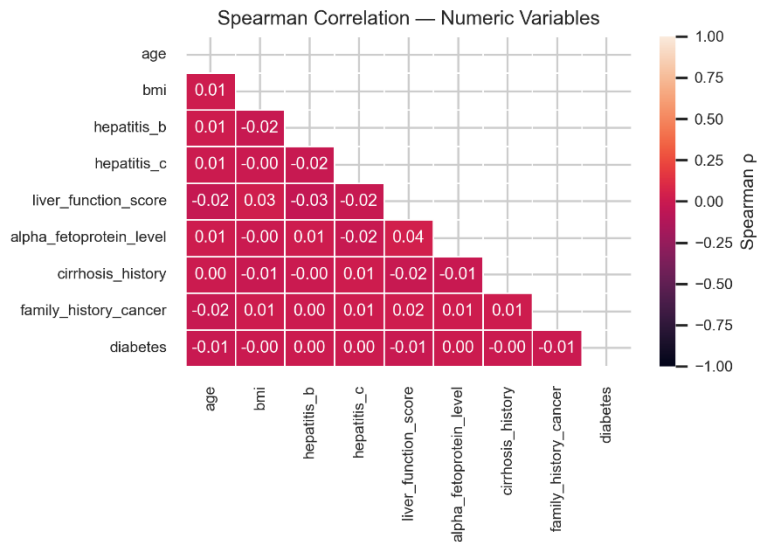


Figure 28: Spearman correlation heatmap - numeric variables
(see Appendix A, Code A.12)

The heatmap shows that pairwise Spearman correlations among the numeric variables are extremely small (approximately within $[-0.03, 0.04]$). This indicates that the numerical variables in the dataset are largely independent.

In practical terms, this implies,

- **Low risk of multicollinearity:** linear models (e.g. LR) are less likely to produce unstable coefficient estimates due to strong correlations among predictors.
- **Limited need for correlation-based variable removal:** there is no indication that a numeric feature duplicates the information of another (i.e. redundancy).
- **Complementary information:** low correlations suggest that features may provide complementary information to the models rather than encoding the same patterns.

It is noted that the absence of strong correlations among predictor variables does not imply a weak relationship with the target variable. It simply indicates that predictors are not highly correlated with one another, an attribute often considered desirable in predictive modeling.

Correlation Between Numeric Features and Target Variable

To quantify the relationship between each numerical variable and the binary target variable $Y \in \{0,1\}$, the point-biserial correlation coefficient, r_{pb} is used. The point-biserial correlation is equivalent to the Pearson correlation when one variable is continuous and the other is binary.

Its standard form is,

$$r_{pb} = \frac{\mu_1 - \mu_0}{s_x} \sqrt{pq}$$

Where,

- μ_1 and μ_0 are the mean values of the numerical variable X in the classes $Y=1$ (cancer) and $Y=0$ (no cancer), respectively,
- s_x is the standard deviation of X in the full sample,
- $p = P(Y = 1)$ and $q = P(Y = 0) = 1 - p$

The sign of r_{pb} indicates the direction of the relationship,

- $r_{pb} > 0$: higher values of X are observed more often (on average) in class $Y=1$.
- $r_{pb} < 0$: higher values of X are observed more often (on average) in class $Y=0$.

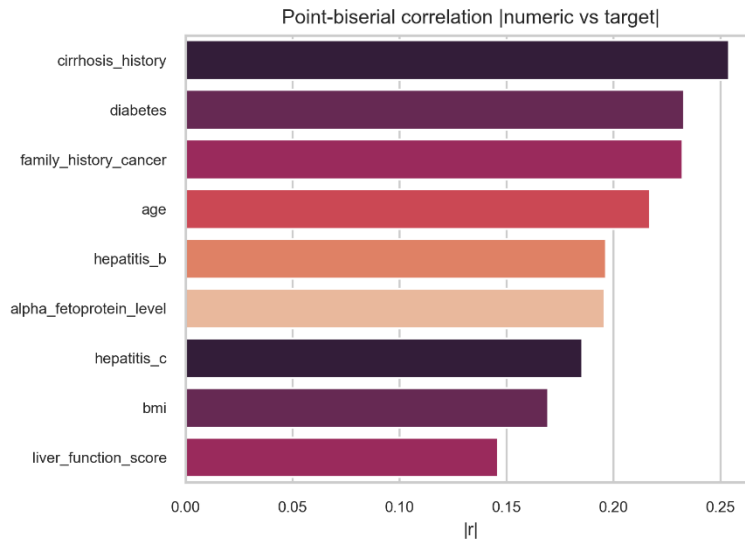


Figure 29: Point-biserial correlations (numeric vs target)
(see Appendix A, Code A.13)

variable	r	abs_r
cirrhosis_history	0.2540	0.2540
diabetes	0.2330	0.2330
family_history_cancer	0.2323	0.2323
age	0.2171	0.2171
hepatitis_b	0.1966	0.1966
alpha_fetoprotein_level	0.1959	0.1959
hepatitis_c	0.1855	0.1855
bmi	0.1695	0.1695
liver_function_score	-0.1461	0.1461

Table 17: Point-biserial coefficients (see Appendix A, Code A.13)

The point-biserial analysis ranks predictor variables based on the absolute magnitude of their correlation with the target variable liver_cancer. The ranking shows that the variables with the strongest correlations are,

- cirrhosis_history (| r |= 0.254)
- diabetes (| r |= 0.233)
- family_history_cancer (| r |= 0.232)
- age (| r |= 0.217)
- hepatitis_b (| r |= 0.196)
- alpha_fetoprotein_level (| r |= 0.196)
- hepatitis_c (| r |= 0.185)
- bmi (| r |= 0.169)
- liver_function_score (| r |= 0.146)

The direction of the correlations is consistent with the descriptive statistics reported earlier. The cancer class exhibits, on average, higher age, BMI and AFP levels, while the LFS is, on average, lower in the cancer class (consistent with a negative r_{pb}). Similarly, binary clinical indicators such as cirrhosis, hepatitis, diabetes and family history show positive correlations, meaning they occur more frequently among positive cancer cases.

Although r_{pb} is useful for ranking predictor variables, it is a bivariate measure and does not capture interactions. A variable with a moderate $|r_{pb}|$ may become highly important when combined with others. Therefore, these results should be interpreted as a preliminary overview rather than as a strict variable selection rule.

Association Between Categorical Features and Target Variable

For categorical variables (e.g., `smoking_status`, `alcohol_consumption`, `gender`, `physical_activity_level`), the standard approach for testing statistical independence between a variable and the target variable is the Chi-square test of independence, using a contingency table. The Chi-square statistic is defined as,

$$\chi^2 = \sum_{i=1}^R \sum_{j=1}^C \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} denotes observed counts and E_{ij} the expected counts under the independence assumption, for R rows and C columns.

Because χ^2 is influenced by sample size, it is also useful to report an effect size measure. A common choice is Cramér's V , defined as,

$$V = \sqrt{\frac{\chi^2}{n(k-1)}}, \quad \text{where } k = \min(R, C)$$

and n is the total number of observations. Cramér's V ranges in $[0, 1]$, with larger values indicating stronger association.

variable	chi2	dof	cramers_v
smoking_status	273.3028	2	0.2330
alcohol_consumption	233.1374	2	0.2150
gender	0.0031	1	0.0000
physical_activity_level	1.1907	2	0.0000

Table 18: Chi-square and Cramér's V for categorical variables vs target (see Appendix A, Code A.14)

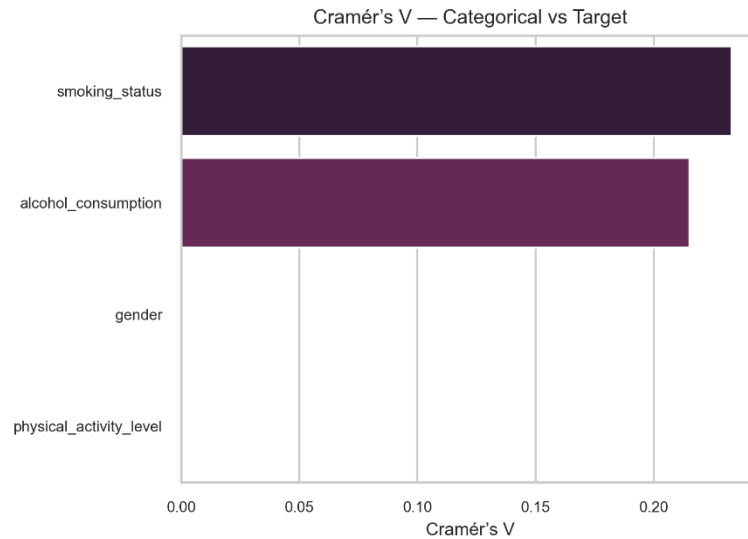


Figure 30: Cramér's V association strength
(see Appendix A, Code A.14)

Two categorical variables show the strongest association with the target variable, as reflected by Cramér's V,

- smoking_status ($V = 0.233$)
- alcohol_consumption ($V = 0.215$)

In contrast, gender and physical_activity_level yield values of V equal to zero or very close to zero, indicating negligible statistical association with the target variable in this dataset. This result is consistent with the descriptive findings of Section 2.5.

Overall, these findings provide a concise summary that, among categorical variables, lifestyle variables such as smoking and alcohol consumption appear to carry more information than demographic variables (such as gender) or activity level.

Chapter 3: Model Comparison Pipeline with Probability Calibration

3.1 Purpose and Overview of the Pipeline

This chapter presents an end-to-end ML pipeline, developed to compare and evaluate three candidate classifiers, LR, RF and XGB. Beyond discrimination performance, the pipeline emphasizes the reliability of the predicted probabilities, through probability calibration, so that these predictions can be interpreted as clinically meaningful.

The main objectives are,

- Fair model comparison under a unified and consistent preprocessing and evaluation framework.
- Strong discrimination performance evaluated using ROC-AUC and Precision-Recall Area Under the Curve (PR-AUC).
- Reliable probabilistic predictions evaluated using calibration curves and the Brier score.
- Uncertainty quantification through bootstrap Confidence Intervals (CI) for ROC-AUC and PR-AUC.
- Model selection on development data (training and validation), with the test set reserved for final reporting.

Given the exceptionally high discrimination observed for some models on this synthetic dataset, the pipeline additionally incorporates a set of lightweight robustness and sanity checks, including label-permutation tests, repeated stratified splits and feature ablation stress tests, in order to contextualize performance and reduce the risk of spurious conclusions. These include label-permutation tests, repeated stratified splits, feature ablation stress tests and two additional robustness experiments, noise injection into numerical measurements to simulate measurement error and prevalence-shift evaluation on a low-risk test cohort (5% prevalence) to assess behavior under stronger class imbalance.

Implementation details are provided in Appendix B, where the full pipeline code and corresponding computational steps are included (see Appendix B, Code B.1 – Code B.10). The additional code modifications used for the feature ablation stress tests are documented separately in Appendix C (Code C.1–C.3) and the robustness code for noise injection and prevalence shift evaluation is documented in Appendix D (Code D.1–D.4).

3.2 Data Loading, Feature Typing and Train/Validation/Test Split

As presented in Chapter 2, the dataset is loaded from a CSV file and the target variable is defined as $\text{liver_cancer} \in \{0, 1\}$. Predictor variables are separated into numerical and categorical groups based on their data types (dtypes), enabling the appropriate preprocessing steps, scaling/standardization for numerical variables and one-hot encoding for categorical variables. This separation aims to avoid manual errors and maintain the robustness of the pipeline (see Appendix B, Code B.2).

To estimate generalization performance, while preventing methodological bias in model selection the dataset is divided into training (60%), validation (20%) and test (20%) sets using a stratified split with respect to the target variable (see Appendix B, Code B.3). The split is performed in two stages, first, a held-out test set (20%) is created and kept completely untouched until the final evaluation and then, the remaining development set (80%) is further split into training (75% of DEV = 60% of total) and validation (25% of DEV = 20% of total). Stratification is particularly important under class imbalance, as it preserves similar class proportions across subsets, reducing the risk of a non-representative target distribution.

The training set is used for cross-validated hyperparameter tuning and primary model comparison (ROC-AUC), while the validation set is used to select the probability calibration method, based on the Brier score and to provide a secondary criterion for model selection when needed. The test set is used exclusively for the final unbiased performance report.

3.3 Leakage-Free Pipeline Construction

To ensure a valid and fair model comparison, preprocessing is integrated into a unified scikit-learn pipeline, so that all transformations are fitted exclusively on the training folds during Cross-Validation (CV), thereby eliminating data leakage from the evaluation data into the training data (see Appendix B, Code B.3).

The preprocessing of numerical variables (numerical preprocessing) includes,

- Median imputation for missing values. Although this dataset contains no missing values, this choice strengthens the robustness and reproducibility of the pipeline for future datasets.
- Standardization using StandardScaler, so that each numerical variable is transformed to have zero mean and unit standard deviation,

$$z = \frac{x - \mu}{\sigma}$$

As mentioned above, standardization is particularly important for models that are sensitive to input scaling, such as LR (see Appendix B, Code B.3).

The preprocessing of categorical variables (categorical preprocessing) includes,

- Most frequent imputation for missing values.
- One-hot encoding, which converts a categorical variable with K levels into K binary indicator variables.

These components are combined using a ColumnTransformer, resulting in a unified preprocessor that is applied consistently across all models (see Appendix B, Code B.3).

3.4 Predictive Models

This section describes the three candidate models evaluated in the pipeline.

Logistic Regression

LR is a linear probabilistic classifier that models:

$$P(y = 1 | x) = \sigma(w^T x + b)$$

where $\sigma(t) = \frac{1}{1+e^{-t}}$ is the logistic sigmoid function, w is the weight vector and b is the intercept.

Training typically corresponds to maximizing the log-likelihood or equivalently minimizing the negative log-likelihood. With L2 regularization, the objective function can be written as,

$$\min_{w,b} \left\{ - \sum_{i=1}^n [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] + \lambda \|w\|_2^2 \right\}$$

where $p_i = P(y = 1 | x_i)$ and λ is the regularization coefficient.

Because the dataset exhibits class imbalance, LR is configured with `class_weight="balanced"`, reweighting the loss so that the minority class receives greater emphasis (see Appendix B, Code B.4). Hyperparameter tuning is performed with respect to the inverse regularization strength C ($C = 1/\lambda$) and the solver (see Appendix B, Code B.5).

Random Forest

RF is an ensemble of decision trees trained using bootstrap aggregation (bagging). Each tree is trained on a bootstrap sample of the training set and at each split only a random subset of features is considered (feature subsampling).

In binary classification, the final probability estimate is obtained as the average of the individual tree probabilities,

$$\hat{p}(y = 1 | x) = \frac{1}{T} \sum_{t=1}^T \hat{p}_t(y = 1 | x)$$

where T is the number of trees and $\hat{p}_t$ is the probability estimate produced by tree t .

This structure reduces variance and typically improves robustness compared to a single decision tree. In the present pipeline, hyperparameter tuning focuses on the number of trees ($n_estimators$) and maximum depth (max_depth), which reflect the core bias–variance trade-off (see Appendix B, Code B.4 – B.5).

Extreme Gradient Boosting

XGB is a gradient boosting model that builds an additive ensemble of decision trees,

$$\hat{y}(x) = \sum_{t=1}^T f_t(x), \quad f_t \in \mathcal{F}$$

where f_t is a regression tree from the function space $\mathcal{F}$.

For binary classification, XGB applies a logistic transformation to obtain probabilities,

$$P(y = 1 | x) = \sigma(\hat{y}(x))$$

Training minimizes a regularized objective function,

$$\mathcal{L} = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t)$$

where ℓ is the logistic classification loss and $\Omega(\cdot)$ is a regularization term penalizing model complexity (e.g., number of leaves and leaf weights). Regularization improves generalization and reduces the risk of overfitting.

To address class imbalance, XGB is configured using,

$$scale_pos_weight = \frac{\text{number of negatives}}{\text{number of positives}}$$

computed from the training set to avoid leakage (see Appendix B, Code B.4). Hyperparameter tuning explores the number of estimators/trees, maximum depth and learning rate (see Appendix B, Code B.5).

3.5 Cross-Validation and Hyperparameter Optimization

As mentioned in Section 3.1, the pipeline aims at fair model comparison under a unified framework, so each model is tuned using GridSearchCV with stratified 5-fold CV and ROC-AUC as the primary scoring metric (see Appendix B, Code B.5). Stratification preserves similar class proportions in each fold, which is particularly important under class imbalance. In addition, the CV procedure is made reproducible by using a shuffled StratifiedKFold with a fixed random seed (see Appendix B, Code B.5).

ROC-AUC is a threshold-independent metric and measures ranking/discrimination ability across the full range of possible decision thresholds. In medical screening contexts, where operating thresholds may vary depending on clinical and economic trade-offs, this property makes ROC-AUC a reasonable initial criterion for model comparison at the tuning stage.

The GridSearchCV procedure returns the best cross-validated ROC-AUC and the corresponding hyperparameter configuration for each model. With `refit=True`, GridSearchCV automatically refits the estimator on the full training set using the optimal hyperparameters identified. As mentioned in section 3.2, the test set is not used during tuning, instead, the tuned estimators proceed to the probability calibration stage and are evaluated on the test set only at the final reporting stage (see Appendix B, Code B7-B.8).

3.6 Robustness and Sanity Checks for Unexpectedly High Discrimination

Given the exceptionally high discrimination observed for XGB on this synthetic dataset, a set of sanity checks are included to contextualize performance and to reduce the risk that the result is driven by trivial artifacts. These checks are diagnostic only and do not contribute to model selection.

Label-Permutation Test

First, a label-permutation test was performed, where the training labels are randomly permuted and the tuned XGB pipeline was re-evaluated under the same stratified cross-validation protocol. In the absence of target leakage, performance is expected to collapse to chance level ($AUC \approx 0.5$). This test provides a direct check against predictor-label shortcuts that would artificially inflate discrimination performance (Appendix B, Code B.6).

Repeated Stratified Development Splits

Second, repeated stratified hold-out splits within the development set are used to assess sensitivity to the random partition. Consistently high AUC across repeated splits indicates that performance is not driven by a single favorable split (Appendix B, Code B.6).

Feature Ablation Stress Test

In addition to the above checks, a feature ablation stress test is conducted to evaluate whether near-perfect discrimination is disproportionately driven by a small subset of highly informative biomarkers.

Specifically, the full pipeline is re-trained and re-evaluated under three ablation scenarios,

- i. Removal of `alpha_fetoprotein_level`.
- ii. Removal of `liver_function_score`.
- iii. Removal of both variables.

In all ablation experiments, the remaining components of the pipeline remain unchanged, i.e. the same stratified train/validation/test splits, CV protocol, hyperparameter search space, calibration strategy and evaluation metrics are used. The only modification concerns the predictor variable set, implemented via a column drop step prior to predictor variable typing and preprocessing (Appendix C, Code C.1-C.3).

This stress test does not aim to improve performance, but rather to assess robustness and interpretability by quantifying how much discrimination and probabilistic accuracy degrade when key biomarkers are removed. In the context of a synthetic dataset, such analysis helps distinguish between genuine multivariate signal distributed across predictor variables and near-deterministic behavior driven by a small number of dominant variables.

Robustness to Measurement Error via Noise Injection

To reduce the unrealistically clean separability of the synthetic dataset and emulate measurement uncertainty common in clinical measurements (e.g., laboratory variability, assay noise), Gaussian noise is injected into numerical variables. This is implemented as a custom transformer inserted into the numerical preprocessing branch between imputation and scaling (Appendix D, Code D.1–D.2), with the noise scale learned from the training data only (per fold), ensuring leakage-free evaluation.

Robustness under Low Prevalence

Medical screening often operates under substantially lower disease prevalence than balanced or enriched datasets. To evaluate model behavior under stronger class imbalance, a prevalence-shift

experiment is performed on the test set by constructing low-prevalence test cohorts with a target prevalence of 5%. Under a fixed dataset size, this is implemented by retaining all negative cases and downsampling the positive class to reach the desired prevalence (Appendix D, Code D.3). To quantify instability induced by the reduced number of positive cases, the procedure is repeated across multiple random seeds, producing a distribution of performance estimates and reporting empirical uncertainty intervals across seeds (Appendix D, Code D.4). This robustness experiment evaluates the degradation of precision–recall performance and probabilistic accuracy under conditions that are more representative of low-risk populations.

3.7 Probability Calibration Strategy

As mentioned in section 3.1, discrimination alone is not sufficient when probabilities are intended for decision-making. A model may rank samples well (high ROC-AUC/PR-AUC) but still produce miscalibrated probabilities, such as systematically overconfident predictions. For this reason, the present thesis applies post hoc probability calibration using `CalibratedClassifierCV`, evaluating two calibration approaches (see Appendix B, Code B.7).

Sigmoid Calibration (Platt Scaling)

This method fits a logistic model that maps the raw classifier scores s to calibrated probabilities,

$$\hat{p} = \frac{1}{1 + \exp (As + B)}$$

It is a parametric and generally stable approach, which often performs well when miscalibration can be corrected by a simple monotonic transformation.

Isotonic Regression

Isotonic calibration fits a non-parametric monotonic function $g(\cdot)$, such that,

$$\hat{p} = g(s)$$

This approach is flexible and may perform better when the score–probability relationship is more complex, but it is more prone to overfitting, especially when the available calibration sample is limited.

For each tuned model, both calibration methods are trained using a stratified 5-fold cross-validation object (for reproducibility) and compared based on the Brier score on the validation set, as introduced in Section 3.2. The calibration method with the lowest validation Brier score is selected for each model (see Appendix B, Code B.7). Finally, after selecting the calibration

method, the calibrated model is refit on the full development data (training and validation) to maximize the available training signal, while keeping the test set untouched for the final evaluation (see Appendix B, Code B.7–B.8).

3.8 Evaluation Metrics and Confidence Intervals

Model evaluation is performed on the hold-out test set, using the final calibrated models, focusing on both discrimination performance and probability quality. For concise reporting, the metrics ROC-AUC, Average Precision (AP) as an implementation of PR-AUC and the Brier score are used, while uncertainty for ROC-AUC and AP is quantified using bootstrap confidence intervals (see Appendix B, Code B.8).

ROC-AUC

ROC-AUC summarizes the ROC curve, which plots the True Positive Rate (TPR) against the False Positive Rate (FPR) across all possible decision thresholds.

AUC can be interpreted as,

$$\text{AUC} = P(\hat{s}(x^+) > \hat{s}(x^-))$$

that is, the probability that a randomly selected positive instance x^+ receives a higher score $\hat{s}(\cdot)$ than a randomly selected negative instance x^- .

PR-AUC / AP

Precision–Recall analysis is particularly informative under class imbalance, especially for the positive class. Precision and Recall are defined as,

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

In the pipeline, AP is used as a summary metric of the PR curve. AP approximates the area under the PR curve and emphasizes the model’s ability to correctly identify positive cases (see Appendix B, Code B.8).

Brier Score

The Brier score evaluates the probabilistic accuracy of predictions,

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (\hat{p}_i - y_i)^2$$

where $\hat{p}_i$ is the predicted probability for the positive class and $y_i \in \{0,1\}$ is the true outcome. Lower values indicate that predicted probabilities are, on average, closer to observed outcomes, combining elements of calibration and refinement.

Bootstrap Confidence Intervals

To quantify uncertainty in ROC-AUC and AP on the test set, the pipeline applies bootstrap resampling using the percentile method, with a large number of resamples. Specifically, $B = 2000$ bootstrap resamples are drawn with replacement from the test set and the metric is computed in each sample, with the 2.5th and 97.5th percentiles of the bootstrap distribution yielding the 95% CI (See Appendix B, Code B.8).

In summary,

- Sample n observations from the test set with replacement.
- Compute the metric (AUC or AP) on the bootstrap sample.
- Repeat the procedure for B resamples and take the percentiles $(\alpha/2, 1 - \alpha/2)$ as the 95% CI.

This procedure is implemented as a reusable function `bootstrap_ci` (see Appendix B, Code B.8).

3.9 Model Selection

As mentioned in previous sections, the test set is retained as a final hold-out set and is not used for model selection. Model selection is conducted deterministically on development data, combining discrimination performance estimated via CV on the training set and probability quality assessed on the validation set.

The pipeline produces,

- A performance summary table on the test set for reporting, including ROC-AUC, 95% CI for ROC-AUC, AP, 95% CI for AP and Brier score, enabling simultaneous evaluation of discrimination and probability quality (see Appendix B, Code B.8).
- Three comparative overlay plots across all three models on the test set, including ROC curves, PR curves and calibration curves. Each plot enables visual comparison of both ranking and calibration (see Appendix B, Code B.9).

Final model selection follows a transparent multi-criteria rule (see Appendix B, Code B.10),

- **Primary criterion:** maximize cross-validated ROC-AUC (from GridSearchCV), reflecting discrimination/ranking ability.
- **Tie-breaker:** minimize validation Brier score, after calibration-method selection, reflecting probability quality.

After selecting the calibration method per model on validation, as described in Section 3.7, each calibrated model is refit on the full development data (training and validation) and evaluated once on the test set solely for the final unbiased report (see Appendix B, Code B.7–B.8).

Chapter 4: Experimental Results and Comparative Evaluation

4.1 Cross-Validated Model Tuning Results

As described in Chapter 3, the first level of comparison is conducted during the hyperparameter tuning on the training set using stratified 5-fold CV, with ROC-AUC as the primary metric. The results, as presented in the table below, show that all three classifiers achieve very high cross-validated performance, indicating that the dataset contains strong predictive information.

model	cv_best_roc_auc	best_params
LogisticRegression	0.9657	{'clf__C': 1.0, 'clf__solver': 'lbfgs'}
RandomForest	0.9861	{'clf__max_depth': 8, 'clf__n_estimators': 600}
XGBoost	0.9993	{'clf__learning_rate': 0.07, 'clf__max_depth': ...}

Table 19: Cross-validation tuning table

As observed in Table 19, LR serves as a strong linear baseline, suggesting that a substantial part of the discriminatory structure can be captured through a linear separation in feature space after preprocessing, particularly after one-hot encoding and standardization.

RF further improves ROC-AUC, a finding consistent with its ability to capture nonlinearities and interactions through an ensemble of trees, while simultaneously reducing variance compared to individual trees.

XGB achieves near-perfect performance in cross-validation terms, indicating that boosted trees exploit additional predictive structure beyond what is captured by bagging, resulting in an exceptionally strong classification model.

A consolidated robustness summary combining these sanity checks with the feature-ablation tests and the additional noise injection and prevalence shift experiments is presented in Section 4.8.

4.2 Calibration Method Selection

After selecting the optimal hyperparameters, each classifier undergoes post hoc probability calibration using two alternative methods, sigmoid (Platt scaling) and isotonic regression. The calibration approach is selected based on the Brier score on the held-out validation set introduced in Section 3.2. This selection reflects the pipeline’s focus on producing probabilistically reliable predictions beyond discrimination performance.

model	picked	val_brier
LogisticRegression	sigmoid	0.0587
RandomForest	isotonic	0.0276
XGBoost	sigmoid	0.0010

Table 20: Calibration selection table

As shown in Table 20, LR selects sigmoid calibration, which is a common outcome when only mild correction of the probability scale is required. Although LR is inherently probabilistic, practices such as class weighting and L2 regularization can influence the calibration of the produced probabilities, making a parametric correction beneficial.

RF selects isotonic regression, which is consistent with the observation that tree ensembles often exhibit systematic calibration deviations, such as compressed probabilities due to the nature of leaf estimates. Isotonic regression, as a non-parametric monotonic mapping, can flexibly correct such deviations.

XGB selects sigmoid calibration and in fact achieves an exceptionally low validation Brier score, suggesting that the score–probability mismatch can be adequately corrected with a simple parametric transformation. At the same time, such low Brier values indicate that the model produces probabilities that are already very close to the true outcomes in this particular dataset.

4.3 Test Set Summary

The final evaluation is performed once on the retained hold-out test set using the final calibrated models, after refitting each calibrated model on the full development data (training and validation), as mentioned in Sections 3.7 and 3.8. The summary table below reports discrimination performance via ROC-AUC, PR performance via AP and probability quality via the Brier score. For ROC-AUC and AP, 95% bootstrap CI are additionally reported, reflecting estimation uncertainty.

model	auc	auc_ci	ap	ap_ci	brier
XGB	1.0000	(0.9999, 1.0)	1.0000	(0.9998, 1.00)	0.0013
RF	0.9926	(0.9884, 0.9962)	0.9788	(0.9677, 0.9883)	0.0273
LR	0.9671	(0.9571, 0.9763)	0.8961	(0.8619, 0.9267)	0.0593

Table 21: Test-set summary table

Overall, all models demonstrate strong performance on the test set, however, a clear ranking emerges.

XGB achieves perfect discrimination on this specific split ($AUC = 1.00$ and $AP = 1.00$) and simultaneously an exceptionally low Brier score, indicating near-perfect probabilistic accuracy after calibration. The extremely narrow CI near 1 suggest that performance remains very high even under bootstrap resampling of the test set. This should be interpreted in the context of the sanity checks reported in Section 4.8.

RF also records very high discrimination ($AUC \approx 0.993$, $AP \approx 0.980$) and substantially better probability quality than LR (Brier ≈ 0.026) but remains below XGB.

LR maintains strong performance as a linear baseline ($AUC \approx 0.967$, $AP \approx 0.896$), yet exhibits the highest Brier score, indicating that even after calibration its probabilities are, on average, less accurate compared to the tree ensembles.

It is worth noting that the ranking by Brier score is consistent with the ranking by ROC-AUC/AP, XGB yields the lowest squared probability error, RF follows and LR ranks third. This supports the conclusion that improved discrimination is not achieved at the expense of probabilistic reliability in this dataset but rather coincides with better probability quality.

4.4 ROC Curve Analysis

As discussed in Chapter 3, ROC curves illustrate the trade-off between the TPR and the FPR across the full range of possible decision thresholds. The closer a curve lies to the upper-left corner of the plot, the better the model's discrimination/ranking ability, while the dashed diagonal corresponds to a random classifier, i.e. $AUC = 0.5$.

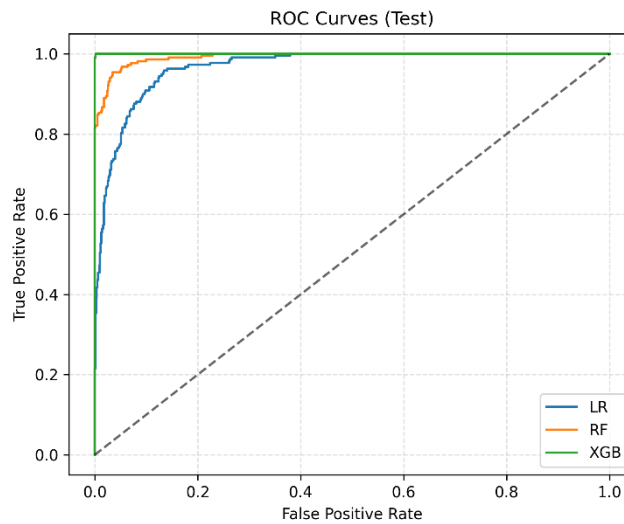


Figure 31: ROC Curves (Test)

As shown in the above figure, the XGBoost curve essentially reaches the upper-left boundary, indicating near-perfect class separation on the test set and consistent with $ROC-AUC \approx 1$. In practical terms, this means that even at very low FPR values it already achieves very high TPR values, i.e. it identifies almost all positive cases with minimal false alarms.

RF outperforms LR across most thresholds, exhibiting systematically higher TPR at the same FPR level. This finding aligns with RF's higher AUC and is consistent with its ability to model nonlinearities and interactions.

LR presents a strong ROC curve, however, the lower AUC suggests weaker ranking performance relative to the tree ensemble models.

In screening settings, ROC analysis is particularly useful because the operating threshold can be selected based on clinical priorities, such as emphasizing sensitivity (reducing false negatives) or emphasizing specificity (specificity = $1 - \text{FPR}$, reducing FP).

4.5 Precision-Recall Curve Analysis

Given that the positive class (cancer) is the minority, PR curves provide particularly useful complementary information, as they focus directly on performance in detecting positive cases and controlling false positives. In contrast to ROC curves, PR curves are directly affected by the prevalence of the positive class and are often more informative in class imbalance problems.

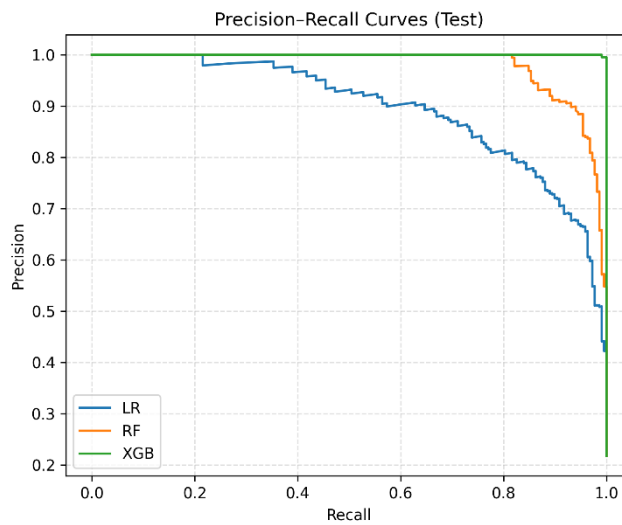


Figure 32: Precision-Recall Curves

As shown in the above figure, XGB exhibits near-ideal behavior, maintaining very high precision across the entire range of recall. This is consistent with $\text{AP} \approx 1$, meaning that the model achieves both almost complete retrieval of positive cases (high recall) and a very low proportion of false positives among predicted positives (high precision).

RF also demonstrates extremely strong performance, with a curve that remains close to the upper boundary of the plot across a wide range of recall. Its high AP indicates effective retrieval of positive cases with a relatively limited number of false alarms.

LR shows a more pronounced drop in precision as recall increases, which is also reflected in its lower AP (≈ 0.896). In operational terms, this means that in order to retrieve a very large proportion of positive cases, the model tends to label more negative instances as positive (increased FP).

In medical settings, PR curves are often more informative than ROC curves because performance is directly linked to the number of false positives among patients labeled as positive. This becomes particularly important under prevalence shift as examined in Section 4.8.

4.6 Calibration Curves and Brier Score Interpretation

Calibration curves (reliability diagrams) compare predicted probabilities with the observed frequencies of the positive class across discretized probability bins. The diagonal line $y = x$ corresponds to perfect calibration, meaning that predicted probabilities $\hat{p}$ match the empirical event probability.

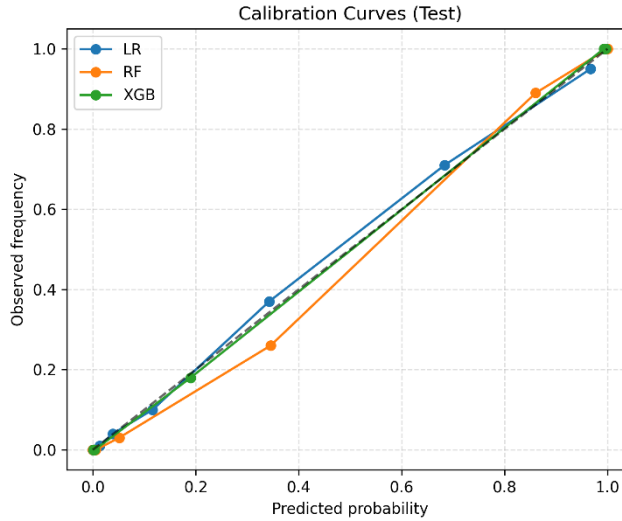


Figure 33: Calibration Curves

As shown in the Figure 33, XGB displays a curve that aligns very closely with the diagonal across most of the probability range. This finding is consistent with its exceptionally low Brier score, indicating that the calibrated probabilities are not only discriminatively strong, but also quantitatively reliable as risk estimates.

RF, after isotonic calibration, shows clear improvement in predicted–observed agreement, yet retains small deviations from the diagonal in parts of the range. These deviations align with RF’s higher Brier score compared to XGB and suggest that, although calibration is good, some residual miscalibration remains relative to the optimal model.

LR exhibits the largest deviations from the diagonal and the highest Brier score among the three models. This suggests that its probabilistic outputs are overall less accurate, either due to the limited expressiveness of the linear form or due to effects such as class weighting and regularization, which can influence the probability scale even after calibration.

Finally, interpretation of the Brier score is directly linked to the above diagram, since it measures the mean squared error $(\hat{p} - y)^2$, it penalizes both highly confident incorrect classifications and systematic deviations of probabilities from empirical frequencies. Therefore, the observed

ranking (XGB best, RF intermediate, LR worst) is consistent with how closely each model’s curve lies to the diagonal.

4.7 Best Model Selection

According to the model selection rule defined in Section 3.9, the best model is selected without using the test set. Specifically, selection is performed on development data using training cross-validated ROC-AUC as the primary criterion, with validation Brier score as a tie-breaker after calibration-method selection (see Appendix B, Code B.10). Under this rule, the selected model is XGB.

model	cv_best_roc_auc	best_params	val_brier
LogisticRegression	0.9657	{'clf__C': 1.0, 'clf__solver': 'lbfgs'}	0.0587
RandomForest	0.9861	{'clf__max_depth': 8, 'clf__n_estimators': 600}	0.0276
XGBoost	0.9993	{'clf__learning_rate': 0.07, 'clf__max_depth':...	0.001

Table 22: Model selection table

This selection is also consistent with the final test-set reporting results (Sections 4.3–4.6), where XGB achieves the strongest discrimination (ROC-AUC and AP) and the lowest Brier score among the three models. However, it is methodologically important to interpret perfect or near-perfect performance with caution, particularly in synthetic datasets.

In this case, the near-perfect discrimination is consistent with high separability induced by the synthetic data generation process. Importantly, the lightweight sanity checks and stress tests in Section 4.8 support that the result is not explained by direct target leakage and is not driven by a single favorable split. The additional robustness experiments with noise injection and prevalence shift further contextualize how performance changes under more realistic uncertainty and imbalance settings.

4.8 Robustness and Stress Test Evaluation

This section consolidates diagnostic checks and robustness experiments designed to contextualize near-perfect baseline discrimination on synthetic data. All analyses in this section are diagnostic only and do not affect the deterministic model selection rule.

Sanity checks for near-perfect XGBoost discrimination

Because the tuned XGB model achieves near-perfect discrimination on this synthetic dataset, two lightweight sanity checks, which are diagnostic only and do not affect model selection, were used

to assess whether the observed performance could be explained by trivial artifacts (e.g., direct target leakage or an unusually favorable split).

First, under the original labels, the tuned XGB pipeline achieved a cross-validated ROC-AUC of 0.9992 ± 0.0009 on the training set, confirming the near-perfect tuning-stage discrimination. In a label-permutation test, where training labels were randomly permuted and the model was re-evaluated under the same stratified cross-validation protocol, performance collapsed to chance level (ROC-AUC 0.5011 ± 0.0171), as shown in the figure below. This pattern provides no evidence of direct target leakage, as any spurious predictor–label shortcut would tend to preserve above-chance performance even after permutation.

```
XGB baseline CV AUC (mean±std): 0.9992206018402776 0.0008799416177036603
XGB label-permutation CV AUC (mean±std): 0.5011236080492399 0.017120018599074048
```

Figure 34: XGB label-permutation test

Second, to assess sensitivity to the random partition, repeated stratified hold-out splits were performed within the development set. The resulting AUC values remained consistently near-perfect (0.99986 ± 0.00041 , range 0.99812–1.0000), indicating that the observed discrimination is stable with respect to split variability and not driven by a single favorable split.

```
XGB repeated DEV splits AUC (mean±std): 0.9998576989460541 0.00041389135344267546
XGB repeated DEV splits AUC (min/max): 0.9981179538026514 1.0
```

Figure 35: XGB repeated DEV splits AUC

Feature ablation stress tests

Beyond the above diagnostics, a feature ablation stress test is conducted to quantify how strongly performance depends on two clinically meaningful biomarkers, AFP and LFS.

As mentioned in Section 3.6, the full pipeline is re-run under three ablation scenarios,

- i. Drop AFP.
- ii. Drop LFS.
- iii. Drop AFP and LFS.

In all three experiments, the pipeline remained identical to the baseline setup (same split strategy, CV protocol, tuning method, calibration selection approach and evaluation metrics). The only difference is the predictor variable set, implemented by dropping the specified columns prior to predictor variable typing and preprocessing. This ensures a fair comparison, since no other modeling choices are altered.

Across all stress tests, XGB remains the best-performing model, retaining the highest test ROC-AUC and AP and the lowest Brier score relative to RF and LR. Removing AFP and/or LFS causes a meaningful but controlled degradation of discrimination and probability quality, with the largest degradation observed when both biomarkers are removed simultaneously. For XGB, test ROC-AUC decreases to 0.9743 (drop LFS), 0.9669 (drop AFP) and 0.9362 (drop AFP and LFS), with a corresponding reduction in AP and an increase in Brier score. This pattern is consistent with AFP and LFS carrying substantial predictive signal, while not being the sole determinants of the outcome.

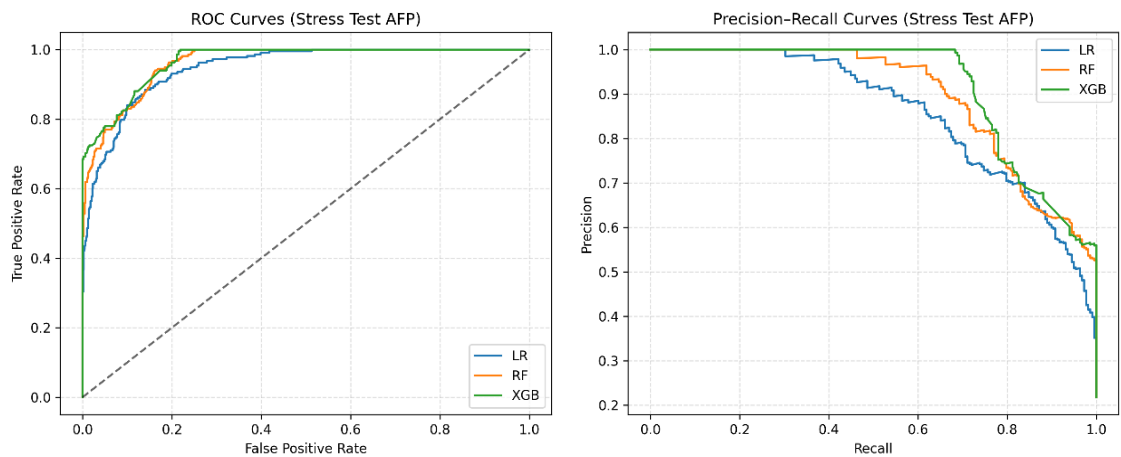
Table 23 summarizes the cross-validated tuning results, calibration-method selection and final test-set performance for each ablation scenario.

Scenario	Model	CV Best ROC-AUC	Validation Brier	Test ROC-AUC	Test AP	Test Brier
Drop AFP	LR	0.9412	0.0752	0.9471	0.8578	0.0735
	RF	0.9523	0.0576	0.9618	0.8946	0.0612
	XGB	0.9620	0.0512	0.9669	0.9110	0.0506
Drop LFS	LR	0.9496	0.0672	0.9514	0.8680	0.0687
	RF	0.9637	0.0463	0.9677	0.9209	0.0510
	XGB	0.9726	0.0307	0.9743	0.9380	0.0374
Drop AFP & LFS	LR	0.9259	0.0799	0.9309	0.8404	0.0780
	RF	0.9282	0.0708	0.9316	0.8454	0.0762
	XGB	0.9350	0.0660	0.9362	0.8580	0.0685

Table 23: Feature ablation stress test summary

Curve-based comparison

Below are the ROC, PR and calibration curves for each scenario,



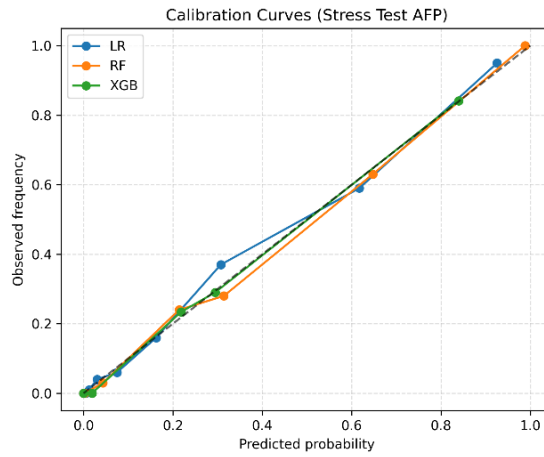


Figure 36: ROC, PR and Calibration curves AFP stress test

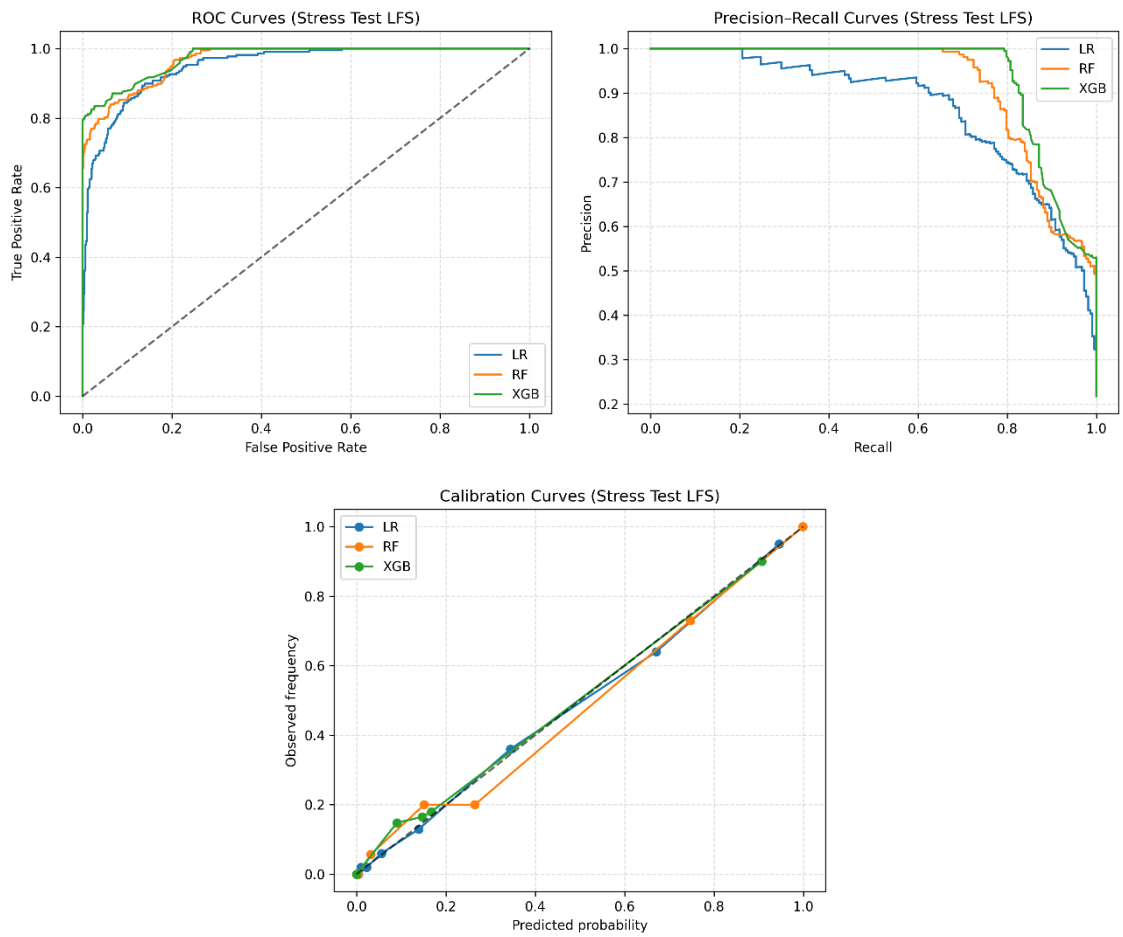


Figure 37: ROC, PR and Calibration curves LFS stress test

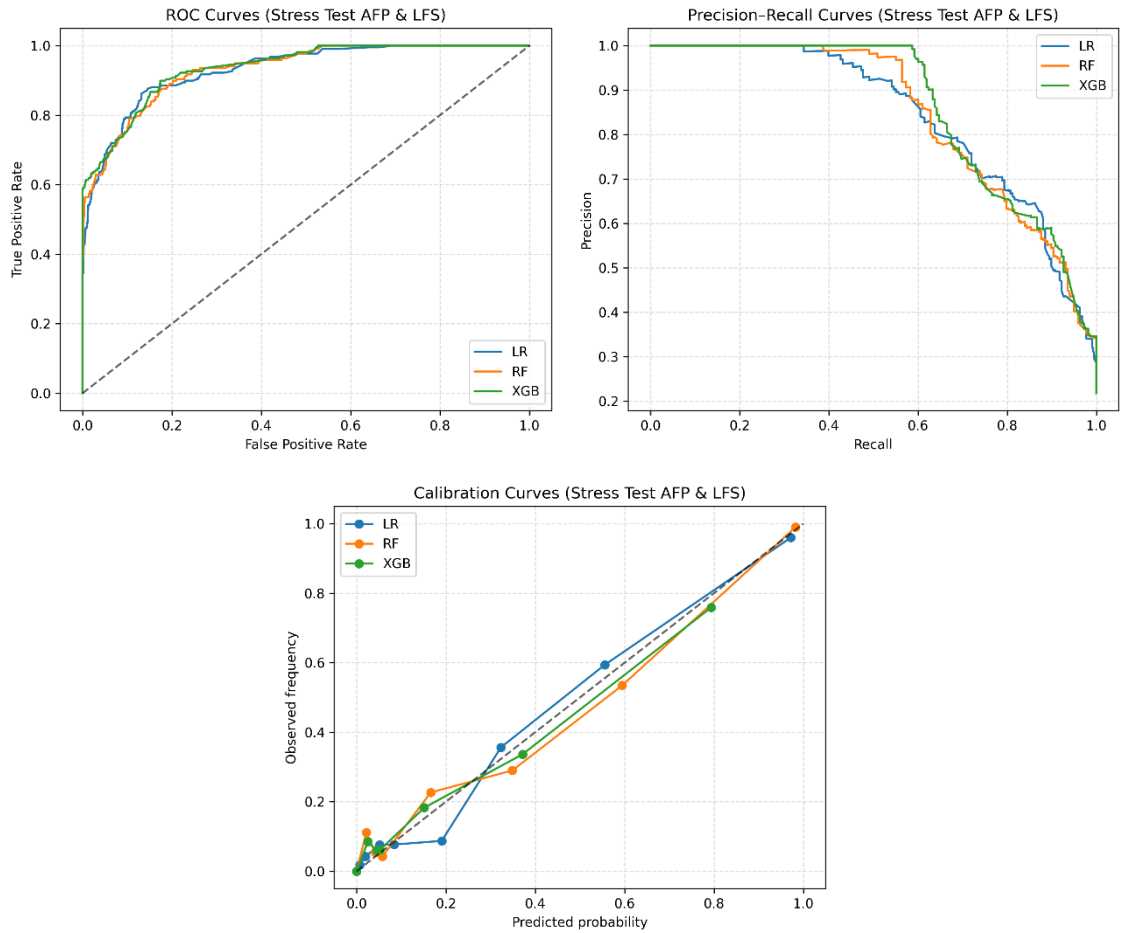


Figure 38: ROC, PR and Calibration curves AFP & LFS stress test

Visually, the stress tests confirm the quantitative results, that XGB remains consistently above RF and LR on ROC/PR and calibration remains generally strong post-calibration, with isotonic regression frequently selected under ablation.

Robustness to measurement uncertainty via noise injection

Under noise injection (Noise Factor = 1.0), cross-validated tuning performance decreases substantially and becomes similar across models.

model	cv_best_roc_auc	best_params
LogisticRegression	0.8545	<code>{'clf__C': 1.0, 'clf__solver': 'lbfgs'}</code>
RandomForest	0.8557	<code>{'clf__max_depth': 8, 'clf__n_estimators': 600}</code>
XGBoost	0.8572	<code>{'clf__learning_rate': 0.03, 'clf__max_depth': ...}</code>

Table 24: Tuning Results (Training with Noise)

This collapse from near-perfect performance to $AUC \approx 0.85\text{--}0.86$ suggests that the baseline near-determinism is sensitive to realistic measurement uncertainty and that a large portion of the

synthetic separability can be removed by moderate to strong noise. Importantly, all models remain substantially above chance-level discrimination, indicating that meaningful predictive signal persists even under degraded measurement quality.

Calibration selection under noise injection remains stable and yields similar validation Brier scores across models.

model	picked	val_brier
LogisticRegression	sigmoid	0.1135
RandomForest	isotonic	0.1144
XGBoost	sigmoid	0.1133

Table 25: Calibration table (noise injection)

These results indicate that under increased uncertainty, probability calibration remains feasible and the calibrated probability quality across models becomes relatively similar.

Robustness under prevalence shift

Final robust shifted-test results (Noise Factor = 1.0, target prevalence 5%) are summarized below,

model	AUC	AP	Brier
LR	0.8626 [0.8234, 0.8997]	0.3732 [0.2245, 0.4850]	0.0640 [0.0577, 0.0713]
RF	0.8733 [0.8367, 0.9043]	0.3845 [0.2411, 0.4966]	0.0617 [0.0575, 0.0676]
XGB	0.8825 [0.8476, 0.9098]	0.4071 [0.2596, 0.4874]	0.0606 [0.0561, 0.0678]

Table 26: Robust test set results

Two main patterns emerge,

- Discrimination remains strong under low prevalence, with $AUC \approx 0.86\text{--}0.88$ and the ranking remains consistent (XGB best, RF second, LR third).
- AP exhibits wider intervals than AUC, reflecting the higher sensitivity of precision–recall metrics to class imbalance and to the reduced number of positives retained after downsampling. This instability is expected and is itself informative, i.e. under low prevalence, the estimated precision–recall performance depends more strongly on the specific positive cases observed.

An example PR curve for the shifted cohort (XGB, prevalence 5%) illustrates a typical screening trade-off, i.e. at low recall, precision can be high, but as recall increases precision declines, implying a rapid increase in false positives when attempting to capture most positives in a low-prevalence population.

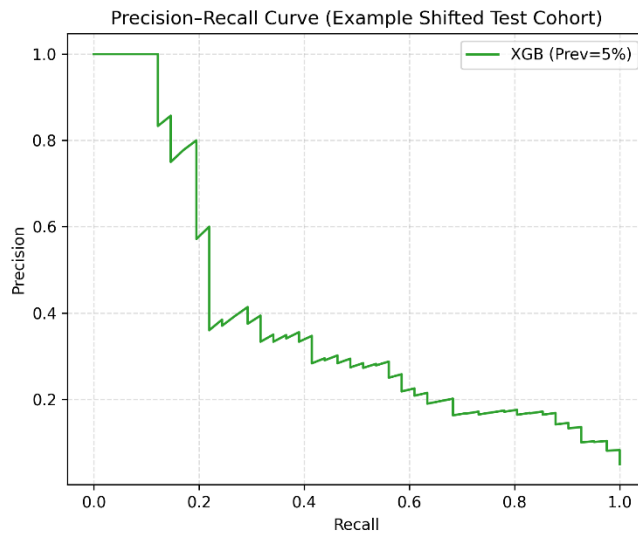


Figure 39: Robust PR curve

Conclusions from robustness evaluation

Overall, the combined robustness evidence supports five conclusions,

- **No evidence of direct target leakage.** The label-permutation test collapses to chance-level discrimination, indicating that near-perfect baseline performance is not explained by leakage or trivial shortcuts.
- **Performance stability across splits.** Repeated stratified DEV splits show consistently near-perfect AUC values, suggesting that the baseline result is not driven by an unusually favorable partition.
- **AFP and LFS are strong contributors, but not sole drivers.** Feature ablation reduces performance in a graded manner, with the largest drop when both biomarkers are removed, yet performance remains above chance, indicating distributed multivariate signal.
- **Near-perfect performance is not robust to measurement uncertainty.** Under strong Gaussian noise (Noise Factor = 1.0), discrimination drops to 0.85–0.86 across models, implying that the synthetic baseline is substantially cleaner than realistic clinical measurements.
- **Under low prevalence, PR performance becomes the critical lens.** While AUC remains relatively stable, AP shows larger sensitivity and wider variability across resampled cohorts, reinforcing that evaluation and operating thresholds must be considered under realistic base rates.

Finally, across the noise-injected and prevalence-shifted evaluations, XGB remains the best overall model (highest AUC/AP and lowest Brier), although margins are reduced compared to the baseline near-perfect setting.

Part II: Decision Optimization

Chapter 5: Introduction: Theoretical Framework and Methodological Foundation

5.1 The Problem of Decision Optimization in Diagnosis

In Part I, a probabilistic risk prediction pipeline was developed, emphasizing not only discrimination performance (ROC-AUC, PR-AUC), but also the reliability of predicted probabilities through calibration and the Brier score. Part II leverages the calibrated risk probabilities $P(Y = 1|x)$ as input to a decision-theoretic framework (decision theory / decision analysis), aiming to transform a continuous probability into a binary, clinically actionable decision.

The central premise of this approach is that high model accuracy (e.g., high ROC-AUC) does not automatically imply optimal clinical practice, because clinical decisions depend not only on how well a model predicts, but also on the relative cost/benefit of outcomes resulting from decision-making (FP, FN), i.e. on diagnostic value in terms of patient outcomes. This distinction is fundamental in the literature on diagnostic test evaluation and decision analysis (Plevritis, 2012). Accordingly, the problem can be formulated as a classic decision optimization problem under uncertainty, as follows,

- **Input:** a risk estimate $P \in [0,1]$, where $P = P(Y = 1 | x)$ and $Y = 1$ denotes Cancer.
- **Output:** a decision $\delta \in \{0,1\}$, where $\delta = 1$ corresponds to Escalate and $\delta = 0$ to No Escalation.

This transition from probability P to decision δ is not performed via arbitrary thresholding, but through the maximum expected utility rule, which determines an optimal decision threshold p^* . In this thesis, the action Escalate ($\delta = 1$) denotes escalation in the diagnostic pathway, such as intensified surveillance, additional imaging, biopsy or specialist referral and it does not imply immediate definitive therapy. Conversely, No Escalation ($\delta = 0$) corresponds to continuing routine surveillance without escalation.

5.2 Utility/Payoff Matrix and Confusion-Matrix Alignment

The methodology defines a utility/payoff matrix, in which,

- reality is described by the health state $Y \in \{0,1\}$,
- the decision is described by the binary action $\delta \in \{0,1\}$.

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer ($Y=1$)	U_{11}	U_{10}
No Cancer ($Y=0$)	U_{01}	U_{00}

Table 27: Payoff Matrix

This structure is fully aligned with the classical confusion matrix in diagnostic evaluation, since each cell corresponds to one of the four fundamental outcomes, TP, FN, FP, TN. This mapping enables a unified perspective, in which statistical concepts (TP/FP/FN/TN) acquire direct clinical meaning through utilities (Varoquaux & Colliot, 2023, Monaghan et al., 2021).

More specifically,

- U_{11} : utility of a TP – correct escalation when cancer is present.
- U_{10} : utility of a FN – failure to escalate when cancer is present.
- U_{01} : utility of a FP – escalation when cancer is absent and associated harm or cost.
- U_{00} : utility of a TN – correct avoidance of escalation when cancer is absent.

In this thesis, the utilities U_{ij} encodes preferences and consequences, such as clinical harm/benefit, quality of life, economic cost or combinations thereof. The critical point is that these utilities weight probabilities, therefore, the quality of probability calibration acquires direct operational importance. Systematic deviation between predicted and true risk (miscalibration) can shift p^* and consequently, lead to suboptimal decisions, such as over-escalation or under-escalation.

In Chapter 8, these optimal thresholds are further operationalized on the held-out test set by reporting the implied Escalation rate and confusion-matrix outcomes (TP/FP/FN/TN) for each scenario.

5.3 The Pauker and Kassirer Threshold Model

The approach adopted in this thesis aligns with the classical work of Pauker and Kassirer (1980) on the threshold approach in clinical medicine. According to this model, the decision to Escalate is rational only when the probability of disease presence P exceeds a specific decision threshold p^* . (Pauker & Kassirer, 1980)

More formally, for each patient with features x , the model estimates the probability $P(Y = 1 | x)$. The clinical decision rule is defined as,

$$\delta(x) = f(x) = \begin{cases} 1, & P(Y = 1 | x) \geq p^* \\ 0, & P(Y = 1 | x) < p^* \end{cases}$$

where p^* is chosen to maximize expected utility.

The optimal decision arises from comparing the expected utilities of the two options $\delta \in \{0,1\}$. Specifically,

- **Expected Utility of Escalate ($\delta = 1$):**

$$\mathbb{E}[U_{\text{escalate}}|P] = \mathbb{E}[U|\delta = 1, P] = P \cdot U_{11} + (1 - P) \cdot U_{01}$$

- **Expected Utility of No Escalation ($\delta = 0$):**

$$\mathbb{E}[U_{\text{no_escalation}}|P] = \mathbb{E}[U|\delta = 0, P] = P \cdot U_{10} + (1 - P) \cdot U_{00}$$

The decision Treat is optimal when,

$$\mathbb{E}[U_{\text{escalate}}|P] \geq \mathbb{E}[U_{\text{no_escalation}}|P]$$

and the threshold p^* corresponds to the indifference point, where the two expected utilities are equal. (Pauker & Kassirer, 1980; Scarffe et al., 2024; Liu et al., 2023)

5.4 Calculation of the Optimal Threshold p^*

Solving the indifference equation with respect to the probability P yields the central formula used throughout Part II to compute p^* .

Starting from,

$$\begin{aligned} \mathbb{E}[U_{\text{escalate}}|P] &= \mathbb{E}[U_{\text{no_escalation}}|P] \Leftrightarrow \\ P \cdot U_{11} + (1 - P) \cdot U_{01} &= P \cdot U_{10} + (1 - P) \cdot U_{00} \end{aligned}$$

We obtain,

$$P \cdot (U_{11} - U_{10}) = (1 - P) \cdot (U_{00} - U_{01})$$

And therefore,

$$\frac{P}{1 - P} = \frac{U_{00} - U_{01}}{U_{11} - U_{10}}$$

Finally, for $P = p^*$,

$$\frac{p^*}{1 - p^*} = \frac{U_{00} - U_{01}}{U_{11} - U_{10}}$$

Which implies,

$$p^* = \frac{U_{00} - U_{01}}{(U_{11} - U_{10}) + (U_{00} - U_{01})}$$

As emphasized in the literature, this form highlights that the threshold does not depend on absolute utility values, but rather on utility differences, which express net benefit and net loss. (Pennello et al., 2016)

Below, we define the two critical quantities,

- **Net Loss of a False Positive**

$$L_{FP} = (U_{00} - U_{01})$$

which represents the utility difference between correctly not escalating (U_{00}) and incorrectly escalating (U_{01}) and can be interpreted as the cost of excessive intervention.

- **Net Benefit of a True Positive**

$$G_{TP} = (U_{11} - U_{10})$$

which represents the utility difference between correctly escalating (U_{11}) and failing to escalate when cancer is present (U_{10}) and can be interpreted as the benefit of timely detection relative to delayed or absent action.

Accordingly, the threshold expression simplifies to,

$$\frac{p^*}{1 - p^*} = \frac{L_{FP}}{G_{TP}} \Leftrightarrow p^* = \frac{L_{FP}}{G_{TP} + L_{FP}}$$

This relationship shows that,

- as the loss from false positives (L_{FP}) increases, p^* increases, meaning greater certainty is required in order to select Escalate,
- as the benefit of true positives (G_{TP}) increases, p^* decreases, meaning Escalate is triggered at a lower disease probability.

The aim of this thesis is to determine, through evidence from the literature, appropriate values for L_{FP} and G_{TP} for each of the six scenarios to be examined, following the rationale that the optimal decision should reflect specific clinical or economic priorities.

Chapter 6: Approach 1: Defining Clinical utilities (QALYs)

6.1 Methodology: QALYs and Disutility Models

The first approach for utility valuation is clinical and uses Quality-Adjusted Life Years (QALYs) as the outcome measure. The QALY is considered a central standard in Cost–Utility Analysis (CUA) in medicine, as it combines life expectancy and quality of life into a single index. Typically, a health state is assigned a utility value $u \in [0,1]$, where $u = 0$ corresponds to death and $u = 1$ corresponds to perfect health, while QALY can be viewed as the time integral of utility over the course of life (McGregor & Caro, 2006).

Although the use of QALYs has been criticized regarding its assumptions and interpretation in specific contexts, it remains one of the most widely adopted tools for the comparative evaluation of interventions at the population and health policy level (Sawhney et al., 2023).

For the application of the threshold model, it is methodologically preferable to define a baseline (reference) state and quantify disutilities rather than relying on absolute utility values. This logic is particularly useful when the goal is to compare decisions (escalate vs no escalation) through utility differences, i.e. through the net impact of an error relative to the usual state. For example, in breast cancer risk analysis, a disutility-based payoff matrix has been used where U_{00} (TN) is set as a reference equal to 0, while all other outcomes receive negative values (as QALY losses) (Abbey et al., 2016).

In the context of HCC surveillance, the reference population is not perfectly healthy ($u = 1$), but consists of high-risk patients with compensated cirrhosis, who undergo regular screening. Therefore, the state U_{00} (TN) does not correspond to perfect health, but to the correct recognition that the patient remains in compensated cirrhosis and continues routine surveillance without escalation (Singal & Marrero, 2008).

A recent cost-effectiveness analysis of HCC surveillance provides the required QALY values (Parikh et al., 2020).

According to that study,

$$\text{Utility (Compensated Cirrhosis)} = 0.78 \text{ QALYs}$$

This value is used as the baseline. Within the disutility model, the baseline value is normalized to zero, i.e. defined as the reference point, so that all other outcomes are expressed as a net loss relative to it.

Therefore, for all clinical scenarios,

- U_{00} (True Negative / Remaining in Compensated Cirrhosis) = 0.0 QALYs (Loss)
- All other utilities (U_{11} , U_{10} , U_{01}) are computed as the negative difference (disutility) from the baseline state, using values from the same bibliographic source (Parikh et al., 2020).

As mentioned before, clinical decision-making is often influenced by the stance of the decision-maker (physician, healthcare system, guidelines) toward different types of errors. That is, some settings prioritize avoiding missed diagnoses (FN), while others emphasize avoiding unnecessary interventions (FP).

To represent this shift in priorities without changing the underlying clinical QALYs, multiplicative weighting coefficients are introduced as a one-way sensitivity analysis. This practice is consistent with decision analysis logic, where benefits/harms are not only biological quantities but also encode preferences over outcomes.

In this thesis, the following factors are used,

- **Risk Aversion Factor (RAF):** amplifies the harm of FN (U_{10})
- **Intervention Aversion Factor (IAF):** amplifies the harm of FP (U_{01})

Typically, for $RAF \geq 1$ and $IAF \geq 1$, the weighted utilities are defined as,

$$U_{10}^{(RAF)} = RAF \cdot U_{10}, \quad U_{01}^{(IAF)} = IAF \cdot U_{01}$$

while all other values remain unchanged, so that the effect of attitude is isolated to a specific type of error.

Overall, the RAF and IAF function as policy levers that map different clinical priorities into different optimal thresholds p^* , enabling comparative evaluation of decision policies under alternative stances toward FN and FP errors.

6.2 Scenario 1: Balanced Clinical Scenario (Balanced QALYs)

This scenario uses mean literature-based QALY values without additional attitude weighting and serves as the neutral reference point, where disutilities are assigned exactly as reported in the source.

Calculations,

i. U_{11} – TP

State: Early detection of HCC (relative to baseline cirrhosis).

Literature value: 0.72 (Parikh et al., 2020).

Disutility,

$$U_{11} = (\text{State QALYs}) - (\text{Baseline QALYs}) \Leftrightarrow$$

$$U_{11} = 0.72 - 0.78 = -0.060 \text{ QALYs}$$

ii. $U_{10} - \text{FN}$

State: Failure to escalate, delayed diagnosis and progression to advanced stage (HCC advanced stage).

Literature value: 0.65 (range 0.52–0.78) (Parikh et al., 2020).

Disutility,

$$U_{10} = 0.65 - 0.78 = -0.130 \text{ QALYs}$$

iii. $U_{01} - \text{FP}$

State: Unnecessary work-up escalation (e.g., biopsy, additional imaging) in a patient without HCC.

Parikh et al. (2020) assume that the evaluation period following a false positive result leads to an absolute 5% reduction in health-state utility during the false positive state, while other studies also support the existence of disutility due to anxiety or burden (Parikh et al., 2020, Lima et al., 2019).

Disutility,

$$U_{01} = -(\text{Baseline QALY} \times \text{Reduction Percentage}) \Leftrightarrow$$

$$U_{01} = -(0.78 \times 0.05) = -0.039 \text{ QALYs}$$

Therefore,

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer ($Y=1$)	$U_{11} = -0.060$	$U_{10} = -0.130$
No Cancer ($Y=0$)	$U_{01} = -0.039$	$U_{00} = 0.000$

Table 28: Payoff Matrix, Scenario 1 – Balanced Clinical (QALYs)

Optimal Threshold p^* Calculation

Using the definitions introduced earlier,

- $L_{FP} = U_{00} - U_{01} = 0.000 - (-0.039) = 0.039$
- $G_{TP} = U_{11} - U_{10} = -0.060 - (-0.130) = 0.070$
- $p^* = \frac{L_{FP}}{G_{TP} + L_{FP}} = \frac{0.039}{0.070 + 0.039} = \frac{0.039}{0.109} = 0.357$

Thus, the optimal threshold p^* (Balanced Clinical Scenario) is 35.7%. In other words, under a balanced clinical scenario, escalation is justified when the estimated probability of cancer exceeds 35.7%.

6.3 Scenario 2: Conservative Clinical Scenario (Risk-Averse QALYs)

This scenario simulates the stance of a clinician or a healthcare system that assigns particularly high weight to avoiding FN. In diagnosis, a false negative is often considered a much more severe error than a false positive, because it may delay escalation and timely detection for a serious disease. The literature highlights that false negative results may entail substantial clinical risk, while factors such as fear of medical liability and ethical concerns may also be involved (Jackson, 2008, Korenstein et al., 2022).

To reflect this preference, as described at the beginning of the chapter, a RAF is applied as a multiplicative coefficient to the FN disutility.

$$U_{10}^{(\text{RAF})} = \text{RAF} \cdot U_{10}, \quad \text{RAF} \geq 1$$

This is a standard sensitivity analysis technique that does not alter the underlying objective QALYs, but functions as a loss amplification model for errors with high clinical severity.

A value of $\text{RAF} = 5$ is selected as a strong but not extreme choice, for three reasons,

- i. $\text{RAF} = 5$ implies that the loss from FN is treated as five times larger than the neutral literature-based estimate, strengthening the emphasis on not missing cases.
- ii. With $\text{RAF} = 5$, the optimal threshold p^* shifts to lower values, making it empirically clear how decision-making changes when the decision-maker primarily fears FN.
- iii. It avoids extreme and unrealistic values, such as 20 or 50, which would tend to recommend escalation for almost any probability.

Section 6.5 presents a one-way sensitivity analysis on the various values that RAF can take.

Calculations,

- i. $U_{00} - \text{TN}$

$$U_{00} = 0.000 \text{ (unchanged)}$$

- ii. $U_{11} - \text{TP}$

$$U_{11} = -0.060 \text{ (unchanged)}$$

iii. $U_{01} - \text{FP}$

$$U_{01} = -0.039 \text{ (unchanged)}$$

iv. $U_{10} \text{ (Conservative)} - \text{FN}$

$$U_{10}^{(\text{RAF}=5)} = (\text{Initial Loss } U_{10}) \times \text{RAF} \Leftrightarrow$$

$$U_{10}^{(\text{RAF}=5)} = -0.130 \times 5 = -0.650 \text{ QALYs}$$

Therefore,

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer ($Y=1$)	$U_{11} = -0.060$	$U_{10} = -0.650$
No Cancer ($Y=0$)	$U_{01} = -0.039$	$U_{00} = 0.000$

Table 29: Payoff Matrix, Scenario 2 – Conservative Clinical (QALYs)

Optimal Threshold p^* Calculation

- $L_{\text{FP}} = U_{00} - U_{01} = 0.000 - (-0.039) = 0.039$
- $G_{\text{TP}} = U_{11} - U_{10} = -0.060 - (-0.650) = 0.590$
- $p^* = \frac{L_{\text{FP}}}{G_{\text{TP}} + L_{\text{FP}}} = \frac{0.039}{0.590 + 0.039} = \frac{0.039}{0.629} = 0.062$

Thus, the optimal threshold p^* (Conservative Clinical Scenario) is 6.2%. In other words, risk aversion toward missing disease dramatically lowers the threshold, with escalation being selected even at relatively low probabilities, in order to minimize the risk of false negatives.

6.4 Scenario 3: Aggressive Clinical Scenario (Risk-Tolerant QALYs)

This scenario simulates a stance in which decision-makers place greater weight on avoiding FP, i.e. on reducing unnecessary downstream work-up escalation. In contemporary clinical practice, perspectives that emphasize avoiding non-essential decisions, such as unnecessary tests or interventions, are associated with psychological and financial burden. The literature indicates that attitudes toward aggressive diagnostic testing may vary substantially (Korenstein et al., 2022, Baxi et al., 2017).

To represent this stance, an IAF is applied as a multiplicative coefficient to the disutility of U_{01} (FP).

$$U_{01}^{(\text{IAF})} = \text{IAF} \cdot U_{01}, \quad \text{IAF} \geq 1$$

The IAF simulates an intervention-averse stance, in which FP is assumed to cause greater real-world harm than what is captured by the mean literature-based reduction in QALYs, due to additional consequences such as unnecessary procedures or psychological burden, as discussed earlier.

A value of $IAF = 5$ is selected as a symmetric and comparable stress test relative to RAF, in order to clearly examine how amplifying the cost of FP shifts the optimal threshold p^* in the opposite direction.

Section 6.6 presents a one-way sensitivity analysis on the various values that IAF can take.

Calculations,

i. $U_{00} - TN$

$$U_{00} = 0.000 \text{ (unchanged)}$$

ii. $U_{11} - TP$

$$U_{11} = -0.060 \text{ (unchanged)}$$

iii. $U_{10} - FN$

$$U_{10} = -0.130 \text{ (return to the neutral value)}$$

iv. $U_{01} - FP$

$$U_{01}^{(IAF=5)} = (\text{Initial Loss } U_{01}) \times IAF \Leftrightarrow$$

$$U_{01}^{(IAF=5)} = -0.039 \times 5 = -0.195 \text{ QALYs}$$

Therefore,

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer ($Y=1$)	$U_{11} = -0.060$	$U_{10} = -0.130$
No Cancer ($Y=0$)	$U_{01} = -0.195$	$U_{00} = 0.000$

Table 30: Payoff Matrix, Scenario 3 – Aggressive Clinical (QALYs)

Optimal Threshold p^* Calculation

- $L_{FP} = U_{00} - U_{01} = 0.000 - (-0.195) = 0.195$
- $G_{TP} = U_{11} - U_{10} = -0.060 - (-0.130) = 0.070$
- $p^* = \frac{L_{FP}}{G_{TP} + L_{FP}} = \frac{0.195}{0.070 + 0.195} = \frac{0.195}{0.265} = 0.735$

Thus, the optimal threshold p^* (Aggressive Clinical Scenario) is 73.5%. In other words, strong aversion to false positives substantially increases the threshold, with the decision-maker requiring a high disease probability before selecting intervention, accepting a higher risk of false negatives as the trade-off for reducing unnecessary downstream work-up.

6.5 One-way Sensitivity Analysis: p^* vs RAF (Clinical/QALYs)

To systematically quantify the effect of FN aversion beyond a single illustrative policy point (RAF=5 in Scenario 2), p^* is computed for a range of RAF values, holding all other payoffs constant (Scenario 1 baseline) and scaling only U_{10} .

Under this sensitivity analysis, L_{FP} remains constant while G_{TP} increases with RAF, implying a monotonic decrease in p^* .

Multiplier	Value	U_{10}	p^*	p^* (%)
RAF	1	-0.130	0.358	35.78%
RAF	2	-0.260	0.163	16.32%
RAF	3	-0.390	0.106	10.57%
RAF	5	-0.650	0.062	6.20%
RAF	8	-1.040	0.038	3.83%
RAF	20	-2.600	0.015	1.51%

Table 31: Sensitivity table RAF

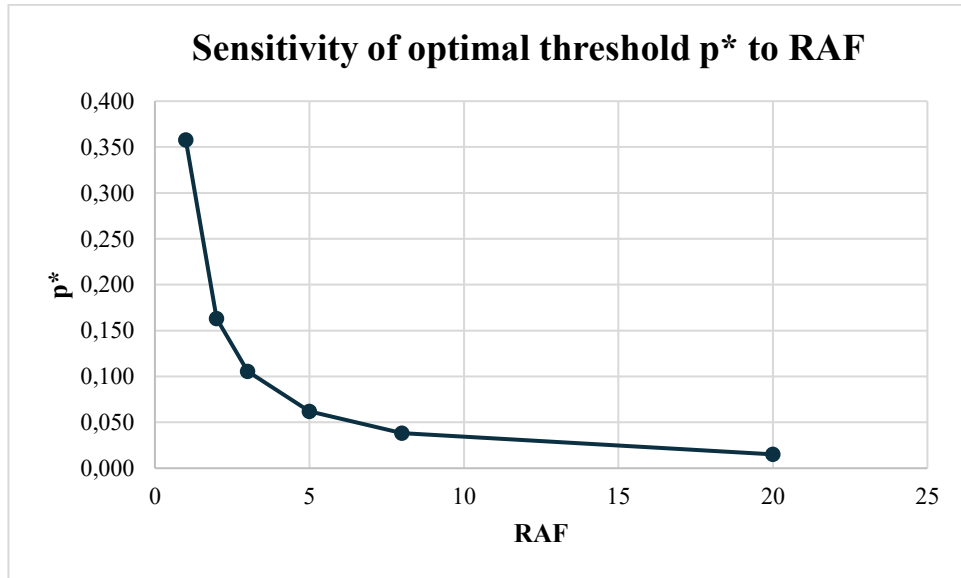


Figure 40: p^* vs RAF (Clinical/QALYs)

As RAF increases, the disutility of a false negative is amplified while remaining utilities remain unchanged. This makes U_{10} more negative and increases G_{TP} , which decreases the optimal threshold p^* . Consequently, a more FN-averse decision-maker is willing to escalate at

progressively lower estimated cancer probabilities, with p^* decreasing from 35.78% (RAF=1) to 1.51% (RAF=20).

The relationship is non-linear, as the largest absolute drop occurs at low-to-moderate RAF values, whereas beyond RAF=8 the threshold approaches a low-probability regime and additional increases yield diminishing absolute reductions. The extreme value RAF=20 is included as a stress test to illustrate the limiting behavior of the rule, where very strong FN aversion drives p^* toward 0, meaning that escalation would be recommended for nearly any non-zero predicted risk.

6.6 One-way Sensitivity Analysis: p^* vs IAF (Clinical/QALYs)

Analogously, to quantify the effect of FP aversion beyond a single illustrative policy point (IAF=5 in Scenario 3), p^* is computed for a range of IAF values, holding all other payoffs constant (Scenario 1 baseline) and scaling only U_{01} .

Under this sensitivity analysis G_{TP} remains constant while L_{FP} increases with IAF, implying a monotonic increase in p^* .

Multiplier	Value	U_{01}	p^*	p^* (%)
IAF	1	-0.039	0.358	35.78%
IAF	2	-0.078	0.527	52.70%
IAF	3	-0.117	0.626	62.57%
IAF	5	-0.195	0.736	73.58%
IAF	8	-0.312	0.817	81.68%
IAF	20	-0.780	0.918	91.76%

Table 32: Sensitivity table IAF

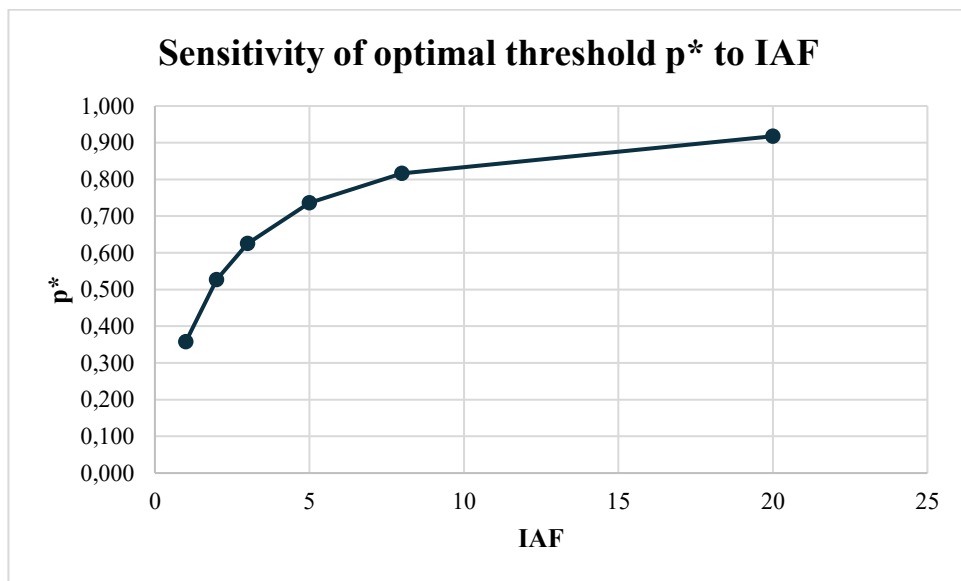


Figure 41: p^* vs IAF (Clinical/QALYs)

Increasing IAF amplifies the perceived harm of false positives, making U_{01} more negative and increasing L_{FP} , while leaving G_{TP} unchanged. Therefore, the optimal threshold p^* increases monotonically, p^* rises from 35.78% (IAF=1) to 91.76% (IAF=20). This quantifies how intervention aversion can push the policy toward conservative escalation decisions, requiring progressively higher certainty before escalating in order to minimize unnecessary downstream work-up.

As with RAF, the extreme value IAF=20 is included as a stress test illustrating the limiting behavior of the decision rule, as $IAF \rightarrow \infty$, $p^* \rightarrow 1$, implying that escalation would be recommended only at near-certainty of cancer.

Chapter 7: Approach 2: Defining Economic Utilities (Cost-Benefit Analysis, CBA)

7.1 Methodology: Cost-Benefit Analysis (CBA) as a Payoff Matrix

The second approach for defining utilities is economic and is based on CBA, where benefits are expressed in monetary units. To maintain full consistency with the expected-utility maximization framework $\max(\mathbb{E}[U])$, utility is defined as the negative of cost,

$$U \equiv -\text{Cost}$$

Under this definition, maximizing expected utility $\max(\mathbb{E}[U])$ is mathematically equivalent to minimizing expected cost, $\min(\mathbb{E}[\text{Cost}])$ (POLARIS, 2024, Office for Health Improvement and Disparities, 2020).

This choice allows the exact same threshold model from Chapter 5 to be used, with the only difference being the interpretation of the payoff matrix U_{ij} .

The cost analysis utilizes data from the U.S. National Cancer Institute (NCI) expressed in USD, chosen for the availability of high-quality, stage-specific cost estimates which are often fragmented in other jurisdictions. While absolute monetary values differ significantly between the U.S. and European systems (e.g., Greece), the decision-theoretic framework remains robust. The optimal threshold p^* is mathematically determined not by absolute magnitudes, but by the ratio of the cost of intervention to the benefit of early treatment. Consequently, assuming that the relative cost structure specifically, the cost multiplier of treating advanced versus early-stage disease is comparable across developed healthcare systems, the derived thresholds can retain cross-border validity. Furthermore, the proposed model serves as a universal template, as local health authorities need only substitute domestic cost parameters to derive valid, nation-specific optimal policies.

Economic Interpretation of U_{ij}

Within the economic framework, utilities are defined as negative costs, in full correspondence with the confusion matrix,

- i. **U_{00} – TN:** baseline reference state. The patient does not have cancer and is correctly monitored under No Escalation. The incremental cost of this outcome is defined as zero,

$$U_{00} = 0$$

- ii. **$U_{01} - \text{FP}$** : direct medical cost of unnecessary work-up escalation due to a false positive indication (e.g., additional imaging, biopsy, or other invasive evaluation),

$$U_{01} = -\text{Cost}(\text{FP})$$

- iii. **$U_{11} - \text{TP}$** : cost when cancer is detected early following appropriate escalation and is managed at an early stage,

$$U_{11} = -\text{Cost}(\text{Early stage management})$$

- iv. **$U_{10} - \text{FN}$** : cost at an advanced stage as a consequence of a missed escalation,

$$U_{10} = -\text{Cost}(\text{Advanced stage management})$$

Based on the definitions introduced in Chapter 5, the quantities L_{FP} and G_{TP} obtain a direct economic meaning,

- $L_{\text{FP}} = U_{00} - U_{01} = 0 - (-\text{Cost}(\text{FP})) = \text{Cost}(\text{FP})$

That is, the loss equals the direct cost of a FP.

- $G_{\text{TP}} = U_{11} - U_{10} = -\text{Cost}(\text{Early}) - (-\text{Cost}(\text{Advanced})) \Leftrightarrow$
 $G_{\text{TP}} = \text{Cost}(\text{Advanced}) - \text{Cost}(\text{Early})$

That is, the benefit of a TP equals the cost savings achieved when diagnosis occurs at an early stage rather than an advanced stage.

Therefore, the decision threshold in the economic setting is defined as,

$$p^* = \frac{\text{Cost}(\text{FP})}{\text{Cost}(\text{Advanced}) - \text{Cost}(\text{Early}) + \text{Cost}(\text{FP})}$$

This relationship represents a fundamental form in the economic evaluation of diagnostic tests and threshold-based decision policies and highlights that the threshold is driven by the trade-off between the cost of unnecessary work-up escalation (FP) and the cost savings achieved by

preventing progression from early to advanced-stage disease through timely diagnosis (Plevritis, 2012, Pennello et al., 2016).

7.2 Scenario 4: Balanced Economic Scenario (Balanced CBA)

This scenario uses direct medical expenditures as reported in the literature, without additional weighting for social costs or behavioral factors. The goal is to capture the basic economic relationship between the cost of FP versus the savings from avoiding progression from early to advanced stage.

Calculations,

i. $U_{01} - \text{FP}$

The literature provides various estimates for the cost of an FP. For example, one study on breast cancer screening (mammography) estimates a cost of \$500–\$564, while another study on general cancer screening (lung, prostate, etc.) reports an adjusted mean difference of \$1,171 (Chubak et al., 2010, Lafata et al., 2004).

The value \$1,171 is adopted as a more general estimate of follow-up costs.

$$U_{01} = -1171 \$$$

ii. $U_{11} - \text{Early Stage Management Cost}$

Data from the U.S. NCI regarding “initial care” cost (the first year after diagnosis) for liver cancer report approximately \$62,775.7. Other sources confirm that early-stage disease is clearly less expensive than advanced-stage disease (National Cancer Institute, n.d., Karim et al., 2022).

The value \$62,775 is selected.

$$U_{11} = -62775 \$$$

iii. $U_{10} - \text{Advanced Stage Management Cost}$

A missed diagnosis is often associated with diagnosis at an advanced stage, while costs increase substantially in the end-of-life phase. NCI data report approximately \$92,133.5 (National Cancer Institute, n.d.; Zou et al., 2022).

The value \$92,133 is selected.

$$U_{10} = -92133 \$$$

Therefore,

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer (Y=1)	$U_{11} = -62775$	$U_{10} = -92133$
No Cancer (Y=0)	$U_{01} = -1171$	$U_{00} = 0$

Table 33: Payoff Matrix, Scenario 4 – Balanced Economic (USD)

Optimal Threshold p^* Calculation

- $L_{FP} = U_{00} - U_{01} = 0.000 - (-1171) = 1171$
- $G_{TP} = U_{11} - U_{10} = -62775 - (-92133) = 29358$
- $p^* = \frac{L_{FP}}{G_{TP} + L_{FP}} = \frac{1171}{29358 + 1171} = \frac{1171}{30529} = 0.038$

Thus, the optimal threshold p^* (Balanced Economic Scenario) is 3.8%. This threshold is much lower than the balanced clinical threshold (35.7%), because in the economic setting the FP cost is relatively small (~\$1.1k) compared to the savings achieved by avoiding progression from early to advanced stage (~\$29k). Therefore, even a relatively small probability of cancer may justify escalation from a purely economic perspective.

7.3 Scenario 5: Conservative Economic Scenario (Societal Cost/Risk-Averse)

This scenario adopts a societal perspective, as is commonly used in Health Technology Assessment (HTA). Under this perspective, evaluation includes not only direct medical expenditures, but also indirect costs, such as productivity losses due to morbidity or premature mortality, as well as the cost of informal care (Gondek et al., 2008, Ufere et al., 2022, The costs of liver cancer, 2021).

It is characterized as conservative / risk-averse, because it expands the true economic burden of a FN, where a missed escalation may lead to substantial societal harm beyond hospital expenditures. In this scenario, the values of U_{11} and U_{10} are modified to incorporate indirect costs, while U_{01} (FP) remains unchanged, since the literature suggests that the non-medical costs of an FP (e.g., time loss) are relatively small compared to those associated with true disease (Lafata et al., 2004). A European-style summary of the economic burden of liver cancer reports indicative figures (The costs of liver cancer, 2021),

- Direct cost: 1.2 billion
- Indirect cost (mortality): 1.9 billion
- Indirect cost (morbidity): 0.3 billion
- Total indirect cost: 2.2 billion

The ratio of indirect to direct cost is,

$$\frac{\text{Indirect Cost}}{\text{Direct Cost}} = \frac{2.2}{1.2} = 1.83$$

Thus, an approximate overall societal cost multiplier is defined as,

$$(1 + 1.83) = 2.83$$

That is,

$$\text{Cost}_{\text{societal}} \approx 2.83 \times \text{Cost}_{\text{direct}}$$

The multiplier 2.83 is selected as the most explicitly documented estimate (The costs of liver cancer, 2021).

Calculations,

i. $U_{00} - \text{TN}$

$$U_{00} = 0 \text{ \$ (Unchanged)}$$

ii. $U_{01} - \text{FP}$

$$U_{01} = -1171 \text{ \$ (Unchanged)}$$

iii. $U_{11} - \text{Societal Cost (Early Stage)}$

$$U_{11}^{(\text{Soc})} = (\text{Direct Cost } U_{11}) \times 2.83 = -62775 \times 2.83 = -177658 \Leftrightarrow$$

$$U_{11}^{(\text{Soc})} = -177658 \text{ \$}$$

iv. $U_{10} - \text{Societal Cost (Advanced Stage)}$

$$U_{10}^{(\text{Soc})} = (\text{Direct Cost } U_{10}) \times 2.83 = -92133 \times 2.83 = -260737 \Leftrightarrow$$

$$U_{10}^{(\text{Soc})} = -260737 \text{ \$}$$

Therefore,

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer (Y=1)	$U_{11} = -177658$	$U_{10} = -260737$
No Cancer (Y=0)	$U_{01} = -1171$	$U_{00} = 0$

Table 34: Payoff Matrix, Scenario 5 – Conservative Economic (USD)

Optimal Threshold p^* Calculation

- $L_{FP} = U_{00} - U_{01} = 0.000 - (-1171) = 1171$
- $G_{TP} = U_{11} - U_{10} = -177658 - (-260737) = 83079$
- $p^* = \frac{L_{FP}}{G_{TP} + L_{FP}} = \frac{1171}{83079 + 1171} = \frac{1171}{84250} = 0.014$

Thus, the optimal threshold p^* (Conservative Economic Scenario) is 1.4%. In other words, when the full societal cost is considered, the cost savings from early versus advanced diagnosis become much larger (~\$83k), resulting in an extremely strong economic incentive for early intervention, justifying escalation even at very low estimated probabilities.

7.4 Scenario 6: Aggressive Economic Scenario (Cost-Containment/Risk-Tolerant)

This scenario simulates the behavior of a stakeholder (e.g., a private insurer or a hospital operating under a strict budget) that prioritizes cost containment and the avoidance of overuse of healthcare services (Baxi et al., 2017).

The stance is aggressive with respect to the future risk of FN, which affects fewer individuals and occurs later, but it is highly averse to the immediate, system-level cost of FP. Although FN is costly on a per-patient basis, it concerns fewer patients. In contrast, FP tends to have smaller cost per case but potentially much higher frequency, generating substantial total expenditure at the population level.

The literature has estimated that national expenditures due to false positives and overdiagnosis can reach extremely high levels (e.g., billions annually in certain screening programs), which explains why a system manager may focus disproportionately on reducing FP (Ong & Mandl, 2015).

To represent this administrative/systemic pressure, an Overuse Multiplier (OM) is introduced in U_{01} , increasing the perceived cost of FP.

$$U_{01}^{(Agg)} = OM \cdot U_{01}, \quad OM \geq 1$$

OM does not represent biological harm, but functions as a shadow price of overuse. That is, it models the fact that organizations often penalize excessive follow-up much more than the purely medical per-case cost, because it is volume-driven and creates systemic congestion and waste of resources.

A value of $OM = 20$ is selected, intentionally larger than RAF and IAF, as an upper-bound stress test, since the objective here is to explore when a policy transitions from “escalate at low P” to “escalate only at high P” due to excessive penalization of FP.

Section 7.5 presents a one-way sensitivity analysis on the various values that OM can take.

The values U_{11} and U_{10} remain at the Scenario 4 levels (direct cost), reflecting a stakeholder focused on direct expenditures rather than societal costs (Ong & Mandl, 2015).

Calculations,

i. $U_{00} - TN$

$$U_{00} = 0 \$ \text{ (Unchanged)}$$

ii. $U_{11} - \text{Early Stage Management Cost}$

$$U_{11} = -62775 \$ \text{ (Unchanged)}$$

iii. $U_{10} - \text{Advanced Stage Management Cost}$

$$U_{10} = -92133 \$ \text{ (Unchanged)}$$

iv. $U_{01} - FP \text{ Cost}$

$$U_{01}^{(OM=20)} = (\text{Initial Cost } U_{01}) \times OM = -1171 \times 20 \Leftrightarrow$$

$$U_{01}^{(OM=20)} = -23420 \$$$

Therefore,

Reality / Decision	Escalate ($\delta=1$)	No Escalation ($\delta=0$)
Cancer ($Y=1$)	$U_{11} = -62775$	$U_{10} = -92133$
No Cancer ($Y=0$)	$U_{01} = -23420$	$U_{00} = 0$

Table 35: Payoff Matrix, Scenario 6 – Aggressive Economic (USD)

Optimal Threshold p^* Calculation

- $L_{FP} = U_{00} - U_{01} = 0.000 - (-23420) = 23420$
- $G_{TP} = U_{11} - U_{10} = -62775 - (-92133) = 29358$
- $p^* = \frac{L_{FP}}{G_{TP} + L_{FP}} = \frac{23420}{29358 + 23420} = \frac{23420}{52778} = 0.444$

Thus, the optimal threshold p^* (Aggressive Economic Scenario) is 44.4%. In other words, a stance that penalizes FP disproportionately for cost-containment reasons substantially increases the decision threshold. The resulting threshold (44.4%) approaches the balanced clinical scenario

(35.7%), but for an entirely different reason, in the clinical scenario the dominant factor is the clinical harm/disutility of FP, whereas here the dominant factor is the direct economic burden of FP at the population level.

7.5 One-way Sensitivity Analysis: p^* vs OM (Economic/CBA)

To quantify the effect of cost-containment pressure beyond a single illustrative value (OM=20 in Scenario 6), a one-way sensitivity analysis on OM is computed, holding all other payoffs constant (Scenario 4 baseline) and scaling only the FP cost.

Under this analysis, G_{TP} remains constant while L_{FP} increases with OM, implying a monotonic increase in p^* .

Multiplier	Value	U_{01}	p^*	p^* (%)
OM	1	-1171	0.038	3.84%
OM	5	-5855	0.166	16.63%
OM	10	-11710	0.285	28.51%
OM	20	-23420	0.444	44.37%
OM	40	-46840	0.615	61.47%

Table 36: Sensitivity table OM

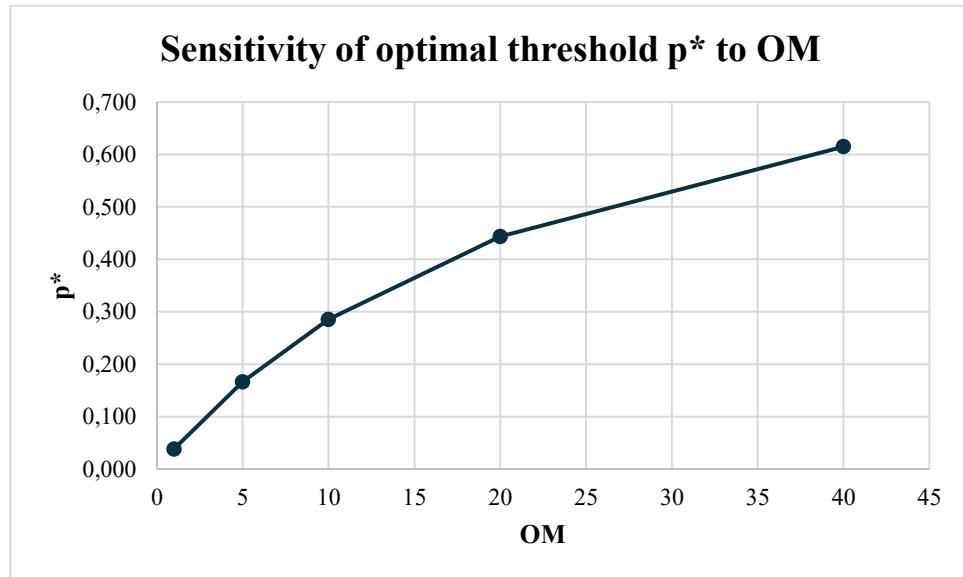


Figure 42: p^* vs OM (Economic/CBA)

Increasing OM amplifies the effective cost of false positives, increasing L_{FP} while leaving G_{TP} unchanged. Therefore, p^* increases monotonically, from 3.8% at OM=1 (direct-cost baseline) to 61.5% at OM=40. This quantifies how strongly system-level cost-containment policies can

reshape escalation rules even if the risk model and early vs advanced cost differences remain the same.

Chapter 8: Consolidation and Operationalization of Decision Thresholds

8.1 Consolidated Payoffs and Optimal Thresholds

Table 37 consolidates the six representative policies developed in Chapters 6 and 7. For the clinical (QALYs) approach, payoffs are defined as disutilities (QALY losses) relative to the baseline state $U_{00} = 0$. For the economic (CBA) approach, payoffs are defined as negative costs (USD), again with $U_{00} = 0$ as the reference point.

Scenario	Approach	Risk Attitude	U_{11} (TP)	U_{10} (FN)	U_{01} (FP)	U_{00} (TN)	p^*
1	Clinical (QALYs)	Balanced	-0.060	-0.130	-0.039	0.000	35.7%
2	Clinical (QALYs)	Conservative	-0.060	-0.650	-0.039	0.000	6.2%
3	Clinical (QALYs)	Aggressive	-0.060	-0.130	-0.195	0.000	73.5%
4	Economic (USD)	Balanced	-62775	-92133	-1171	0	3.8%
5	Economic (USD)	Conservative	-177658	-260737	-1171	0	1.4%
6	Economic (USD)	Aggressive	-62775	-92133	-23420	0	44.4%

Table 37: Consolidated Payoffs & Thresholds

The consolidated results illustrate how changing the relative penalty of FP versus FN reshapes p^* ,

- In the clinical framework, strong FN aversion (Scenario 2) increases $G_{TP} = U_{11} - U_{10}$ and substantially reduces p^* , while strong FP aversion (Scenario 3) increases $L_{FP} = U_{00} - U_{01}$ and substantially increases p^* .
- In the economic framework, p^* is generally low when the FP work-up cost is small relative to the early vs advanced cost gap (Scenarios 4–5) but can increase markedly when FP is penalized via an overuse multiplier (Scenario 6), reflecting system-level cost containment pressures.

8.2 Operationalization on the Held-Out Test Set

The consolidated thresholds in Table 37 become actionable only when applied to patient-level predicted risks.

This section operationalizes each policy by applying the threshold rule $P(Y = 1 | x) \geq p^*$ on the test set probabilities produced by the final calibrated model from Part I.

- The operational evaluation uses the held-out test set ($n = 1000$) with prevalence $P(Y = 1) = 0.218$ (218 cancer cases).
- Calibrated risk estimates $P(Y = 1 | x)$ are produced by the selected best-performing model (XGB) from Part I.
- For each scenario, applying p^* yields,
 - Escalate_n and Escalate_%, i.e. how many patients are escalated and the resulting escalation rate,
 - a confusion matrix (TP, FP, FN, TN),
 - standard operational metrics (Sensitivity, Specificity, PPV, NPV),
 - the realized expected utility on the test set under that scenario's payoffs.

The code implementing this operationalization is provided in Appendix E and it uses as inputs the calibrated test-set probabilities produced by the pipeline in Appendix B.

Tables 38 and 39 reports the operational results on the test set for each scenario.

Scenario	p^*	Escalate_n	Escalate_%	TP	FP	FN	TN
S5_Econ_Conservative	0.0139	238	0.238	218	20	0	762
S4_Econ_Balanced	0.0384	226	0.226	218	8	0	774
S2_Clinical_Conservative	0.0620	224	0.224	218	6	0	776
S1_Clinical_Balanced	0.3578	220	0.220	218	2	0	780
S6_Econ_Aggressive	0.4437	220	0.220	218	2	0	780
S3_Clinical_Aggressive	0.7358	218	0.218	217	1	1	781

Table 38: Escalation rate & Payoff Matrix

Scenario	Sensitivity	Specificity	PPV	NPV	EU_per_patient	EU_total
S5_Econ_Conservative	1.0000	0.9744	0.92	1	-38752.8640	-3.8753E+07
S4_Econ_Balanced	1.0000	0.9898	0.96	1	-13694.3180	-1.3694E+07
S2_Clinical_Conservative	1.0000	0.9923	0.97	1	-0.0133	-1.3314E+01
S1_Clinical_Balanced	1.0000	0.9974	0.99	1	-0.0132	-1.3158E+01
S6_Econ_Aggressive	1.0000	0.9974	0.99	1	-13731.7900	-1.3732E+07
S3_Clinical_Aggressive	0.9954	0.9987	0.995	0.9987	-0.0133	-1.3345E+01

Table 39: Operational Metrics

Key Observations

- 1) Policy thresholds translate into escalation rates close to the disease prevalence. Across scenarios, the escalation rate ranges narrowly from 21.8% to 23.8%, which is close to the test set prevalence (21.8%). Operationally, this means that the decision rules produce escalation volumes that are broadly consistent with the underlying burden of disease in this test set.
- 2) The confusion matrices are largely stable across a wide range of thresholds. Despite thresholds spanning from $p^* = 0.0139$ (very permissive escalation) to $p^* = 0.7358$ (very restrictive escalation), the induced confusion matrices are very similar. In five scenarios, FN = 0 and sensitivity is 1.0000, while FP remains small (2–20). Only the most restrictive clinical policy (Scenario 3) produces FN = 1, reflecting the intended trade-off of strong FP aversion, i.e. fewer escalations and fewer FP, at the cost of potentially missing a case.
- 3) Economic policies differ mainly in FP volume when p^* is extremely low. Scenario 5 (societal perspective) yields the lowest threshold (1.39%) and the highest escalation rate (23.8%), increasing FP to 20. Scenario 4 (balanced economic) still achieves zero FN but with fewer FP (8). Scenario 6 (cost containment) pushes p^* upward (44.4%), yielding the same operational outcomes as Scenario 1 in this test set (TP=218, FP=2, FN=0), indicating that the model's predicted probabilities separate cases so strongly that these different thresholds fall in a region where few observations switch sides.
- 4) Realized expected utility should be interpreted within each framework (units differ). The clinical utilities are expressed in QALY disutility units, whereas economic utilities are expressed in USD costs (negative utility). Therefore, EU values are not directly comparable across clinical vs economic scenarios, but they provide a consistent policy evaluation measure within each scenario's payoff definition.

Chapter 9: Summary and Conclusions

9.1 Thesis Summary and Main Contributions

This thesis developed an integrated framework that bridges probabilistic risk prediction with clinically actionable decision policies. In Part I, the emphasis was placed not only on model discrimination performance, but also on the reliability of predicted probabilities through calibration, so that the estimates $P(Y = 1 | x)$ can be used operationally. In Part II, the calibrated probabilities were incorporated into an explicit decision-theoretic model of the Pauker–Kassirer type, where the transition from prediction to action is achieved through maximum expected utility and results in interpretable optimal decision thresholds p^* .

The analysis demonstrated that the decision threshold is not a fixed technical characteristic of the model but depends critically on the utility/cost structure and on the decision maker’s risk attitude toward false positive and false negative errors. Beyond the six representative scenarios, one-way sensitivity analyses (RAF/IAF/OM) demonstrated that p^* varies monotonically and often non-linearly as error aversion or cost-containment pressure increases, confirming that threshold selection is fundamentally a policy choice.

Accordingly, the resulting decision policies span a wide threshold range across plausible clinical and economic perspectives, from very low thresholds in strongly FN-averse or societal-cost settings to high thresholds under strong aversion to FP or overuse penalties. In this way, the thesis supports the conclusion that the meaningful clinical value of an ML model emerges only when performance is evaluated in relation to the consequences of the decisions it induces, rather than exclusively through metrics such as ROC-AUC or PR-AUC.

A further contribution of this thesis is the explicit contextualization of near-perfect predictive performance on synthetic data through diagnostic robustness evaluation. In addition to sanity checks and biomarker ablation stress tests, two robustness experiments were introduced to emulate real-world deployment threats, measurement uncertainty via Gaussian noise injection in numerical variables and low-prevalence evaluation via prevalence shift (5% target prevalence) on the test set with repeated resampling. These experiments demonstrate that discrimination and probability quality can change substantially under degraded measurement quality or altered base rates, reinforcing that internal near-perfect performance should not be interpreted as clinical readiness.

The outcome is a generalizable and transparent methodological framework that can be adapted to different clinical or economic priorities and can support the development of consistent, evidence-based decision policies for surveillance and early diagnosis of HCC.

9.2 Key Findings of Part I

Part I of this thesis developed a complete and reproducible framework for data-driven liver cancer risk prediction, using a public synthetic dataset and modern probabilistic ML methods. The study began with structured dataset exploration, combining descriptive statistics with EDA, aiming to characterize variable distributions, class balance and potential associations between predictor variables and the target variable. EDA provided a justified basis for preprocessing and modeling choices, while highlighting clinically plausible patterns in both numerical and categorical variables.

Building on this foundation, Part I introduced a unified model comparison pipeline for three widely used classifiers, LR as a transparent linear baseline, RF as a robust ensemble approach and XGB as a gradient-boosted tree method capable of capturing nonlinearities and higher-order interactions. A central methodological choice was to treat the models as probabilistic predictors, i.e. mechanisms that produce calibratable risk probabilities, rather than as systems that assign labels through an arbitrary decision threshold.

Accordingly, evaluation was not limited to discrimination performance, but was extended to include probability quality. Specifically, performance was measured using ROC-AUC and PR-AUC (implemented as AP) for discrimination, as well as calibration diagnostics and the Brier score for probabilistic accuracy. Hyperparameter optimization was performed using stratified CV, while post hoc calibration was applied using either sigmoid (Platt scaling) or isotonic regression, selecting the method that minimized the validation Brier score for each model. Final evaluation was conducted on a held-out test set, with uncertainty quantified via bootstrap CI for ROC-AUC and AP.

Empirical results showed strong performance across all three models, with XGB emerging as the best-performing model under the adopted selection rule on development data. Robustness was further assessed through feature-ablation stress tests, where the pipeline was re-run after removing AFP, LFS and both jointly. These analyses showed graded performance degradation under biomarker removal, indicating that predictive signal is not solely driven by a single dominant predictor variable, while also confirming that near-perfect baseline performance is not explained by trivial target leakage.

The results from the two robustness experiments to simulate realistic development threats showed that, when Gaussian noise was injected into numerical features (Noise Factor = 1.0) within the pipeline (leakage-free), the cross-validated tuning ROC-AUC dropped from near-perfect levels to approximately 0.85–0.86 across models. This result indicates that the synthetic dataset’s separability is sensitive to measurement uncertainty and that model performance becomes substantially more realistic under degraded signal-to-noise ratio. Calibration method selection remained stable with similar validation Brier scores suggesting that reliable probability

calibration remains feasible even under noise. Under a simulated low-prevalence test cohort (5% prevalence) repeated across 30 downsampling seeds, discrimination remained strong and model ranking was preserved, while AP exhibited larger variability across seeds. These findings highlight that under screening-like base rates, precision–recall performance becomes more fragile and more dependent on cohort composition, reinforcing the necessity of evaluating models under realistic prevalence conditions rather than relying solely on ROC-AUC under enriched datasets. Model selection was performed using training cross-validated ROC-AUC as the primary criterion, with validation Brier score as a tie-breaker after calibration-method selection, while the test set was reserved exclusively for final unbiased reporting. Therefore, within the limitations of the synthetic dataset, Part I concludes that the combined evaluation of discrimination and probability quality (calibration), augmented with diagnostic robustness under measurement uncertainty and prevalence shift, provides a more clinically appropriate basis for risk prediction than relying exclusively on discrimination metrics and classification accuracy.

9.3 Key Findings of Part II

In Part II, the calibrated probabilities $P(Y = 1 | x)$ were incorporated into a maximum expected utility rule. A utility/payoff matrix was defined in direct alignment with the confusion matrix outcomes (TP, FP, FN, TN) and the optimal threshold p^* was calculated as the indifference point between the decisions Escalate and No Escalation.

The analysis was implemented under two utility environments, a clinical framework based on QALYs (Cost–Utility), where outcomes were defined as disutilities relative to a baseline state and an economic framework based on CBA, where utilities were defined as $U = -\text{Cost}$. Within each framework, six scenarios (balanced, conservative and aggressive) were examined. In addition, one-way sensitivity analyses (RAF/IAF/OM) systematically quantified how p^* shifts as stakeholder aversion to FN, aversion to FP or cost-containment pressure increases.

The critical finding is that p^* is not an inherent property of the predictive model, but rather a derivative of the utility/cost structure and the preferences of the decision-making entity. The same predicted probability can lead to different decisions depending on whether the objective prioritizes maximizing QALYs, avoiding missed diagnoses, minimizing direct expenditures or limiting resource overuse. Sensitivity results further confirmed that these preference shifts translate into monotonic and in some regimes non-linear changes in the operational threshold, reinforcing the interpretation of p^* as a policy parameter rather than a technical constant.

Finally, the consolidated thresholds were operationalized on the held-out test set by applying the rule $P(Y = 1 | x) \geq p^*$ and reporting escalation rates and confusion-matrix outcomes for each scenario, linking the theoretical threshold model directly to test-set operational consequences.

9.4 Integrated Interpretation and Implications

An integrated reading of the two parts leads to three practical conclusions.

First, statistical capability does not guarantee clinical utility. A model with near-perfect discrimination ($AUC \approx 1.0$) remains operationally inert if the decision threshold is set arbitrarily (e.g. at 0.5) rather than being derived from the specific risk-benefit profile of the intervention. The analysis demonstrates that the best model is useless if the decision threshold represents a risk attitude incompatible with the patient's prognosis or the system's constraints.

Second, the action threshold should be treated as a policy parameter and justified based on the FP/FN consequences, as shown by the sensitivity analysis, rather than adopted as a default value of 0.5. The payoff/utility framework enables transparent comparison of alternative decision policies applied to the same predicted probabilities, explicitly revealing the structural tension between clinically prudent and economically efficient criteria.

Third, the robustness results from Part I reinforce that both predictive performance and the downstream operating threshold can be sensitive to realistic deployment threats. Injecting measurement noise reduces discrimination substantially compared to the baseline synthetic setting, while prevalence shift (5% prevalence) increases the instability of precision–recall performance and emphasizes the importance of evaluating models under realistic base rates. Since Part II relies on calibrated probabilities as inputs, shifts that alter calibration or base rate can affect the operational meaning of any fixed threshold and motivate recalibration or prevalence-aware adaptation before clinical use.

Consequently, the meaningful clinical value of an ML model emerges only when performance is assessed in relation to the consequences of the decisions it induces, rather than exclusively through metrics such as ROC-AUC or PR-AUC. The decision layer acts as the necessary final step that transforms prediction into implementable, interpretable and justified clinical action.

9.5 Limitations

Dataset Realism and Generalizability

The most important limitation is that the dataset used in this thesis is synthetic. Synthetic datasets are highly useful for methodological development, yet they may incorporate simplified relationships, reduced measurement noise or cleaner class separation compared to real clinical data. As a result, strong performance on such data does not guarantee comparable performance under real-world conditions, where labels may be noisy, clinical measurements may contain error and patient samples may exhibit distributional heterogeneity. Therefore, the findings of Part I should not be interpreted as evidence of clinical readiness, but rather as an indication of successful modeling within the scope of the chosen experimental setup.

Near-Perfect Performance

Near-perfect discrimination on a synthetic dataset should be interpreted primarily as evidence of strong separability induced by the data-generation process, rather than as evidence that the clinical prediction problem is intrinsically easy. Synthetic generation can reduce measurement noise, encode overly deterministic predictor–outcome mappings and impose structural regularities that sharpen class boundaries. Under such conditions, models such as XGB may achieve $AUC \approx 1$ even when real-world surveillance data would contain label uncertainty, heterogeneous patient trajectories and imperfect biomarker measurements. The robustness experiments introduced in this thesis help contextualize this limitation but remain simplified proxies. Therefore, the results of Part I should be read as methodological feasibility within the chosen experimental setup, not as clinical readiness.

The most consequential threats to generalization are forms of dataset shift that alter either the distribution of predictors or the predictor–outcome relationship, such as case-mix shift (different cirrhosis etiologies, staging, demographics, surveillance adherence), prevalence (prior probability) shift across settings and time, measurement shift (assay changes, lab calibration, missingness practices) and label shift (different diagnostic criteria or verification intensity). Importantly, these shifts affect not only discrimination but also probability calibration and thus can move the operational decision threshold p^* derived in Part II. For this reason, external validation is necessary before translating near-perfect internal performance into clinical action.

Model Selection

The pipeline selects the best model using development data, with training cross-validated ROC-AUC as the primary criterion and validation Brier score as a tie-breaker after calibration-method selection, while the test set remains reserved exclusively for final, unbiased reporting. This design reduces the risk of selection optimism that would arise if the test set were used for model selection. Nevertheless, some degree of uncertainty remains because any single train/validation/test split can yield optimistic or pessimistic estimates depending on the particular partition. Future work can further strengthen robustness through repeated splits, nested cross-validation for hyperparameter tuning and model comparison or an additional external hold-out dataset and multi-center external validation.

Calibration Evaluation Constraints

Calibration method selection is based on the validation Brier score on the held-out validation set, which is an appropriate and coherent criterion. However, calibration itself is sensitive to the size and representativeness of the calibration set. Isotonic regression, in particular, may overfit when calibration data are limited, whereas sigmoid calibration may underfit when miscalibration is structurally complex. In addition, binned calibration curves may obscure local miscalibration in

sparse probability regions. These constraints motivate reporting multiple calibration summary measures in future extensions and performing stability checks under alternative resampling schemes.

Interpretability and Clinical Transparency

While LR offers high interpretability through model coefficients, tree-based models and particularly XGB, are more complex and may require additional interpretability tools (e.g. feature importance or SHAP-style explanations) to support clinical trust and accountability. In this thesis, interpretability was partially addressed through feature-ablation stress testing, which quantified the dependence of model performance on key biomarkers, however, a full explanation layer remains future work. Part I prioritizes predictive performance and probability quality, however, a clinically oriented extension should incorporate interpretability analysis and fairness checks, especially if the model is intended for use in contexts that directly affect patients or resource allocation.

Economic Data and External Validity

Cost values are primarily drawn from the U.S. healthcare system and expressed in USD, implying that absolute costs may differ substantially across other systems. Nevertheless, the methodology remains valid because p^* is driven mainly by relative cost differences (e.g., $\text{Cost}(\text{Advanced}) - \text{Cost}(\text{Early})$) rather than the absolute monetary scale.

QALY Values and Estimation Variability

Utility values for HCC health states depend on the target population, the valuation method (e.g. Time Trade-Off, Standard Gamble), access to treatment options and the modeled time horizon. Consequently, numerical QALY estimates may differ across studies, introducing uncertainty into disutilities and therefore into the final p^* . This limitation is consistent with broader critiques of QALY-based modeling, where structural assumptions may influence interpretation.

Risk-Attitude Multipliers (RAF/IAF/OM) as Sensitivity Scenarios

The multipliers $\text{RAF} = 5$, $\text{IAF} = 5$ and $\text{OM} = 20$ are not derived from a single empirical source, but represent methodological choices intended for sensitivity analysis and simulation of decision-maker preferences and incentives. They should be interpreted as plausible but deliberately strong assumptions designed to highlight how decision attitudes can shift p^* . As shown by the one-way sensitivity analyses, varying these multipliers produces monotonic and in some regimes non-linear shifts in p^* , reinforcing the interpretation of the threshold as a policy-dependent quantity. Nevertheless, this analysis remains deterministic and future work could extend it via preference

elicitation or probabilistic sensitivity analysis, treating utilities and costs as distributions rather than fixed values.

Binary Decision Structure and Simplification of Clinical Processes

The model is formulated as a binary decision $\delta \in \{0,1\}$ (Escalate vs No Escalation). In reality, many clinical decisions unfold in multi-step pathways (e.g. intensified surveillance, additional imaging, biopsy, referral). The binary formulation is useful as a first principled foundation of the threshold model, but it does not fully represent the sequential nature of real diagnostic processes.

9.6 Future Work

Building on the limitations discussed above, the natural next step of this thesis is to move from a proof-of-concept framework toward a clinically and organizationally deployable decision policy, where probabilistic risk prediction is systematically translated into actionable diagnostic strategies.

Application to real-world clinical data and external validation

A critical next step is to evaluate the proposed framework on real HCC surveillance data, in order to test generalization across different case-mix profiles and to quantify the risk of dataset shift.

In practice, this requires,

- Strict study design with train/validation/test separation, using temporal and/or geographic splits to better emulate prospective deployment conditions.
- Assessment of bias sources that can distort risk estimation (e.g. verification bias, spectrum bias) and thereby affect downstream decision thresholds.
- Systematic evaluation not only of discrimination, but also calibration in external cohorts, with the option of recalibration before any clinical use.
- Subgroup analyses (e.g. etiology, cirrhosis stage, age/sex), to determine whether local subgroup-specific thresholds are necessary or whether distinct models are required for clinically heterogeneous strata.
- Incorporation of Explainable AI mechanisms to avoid “black-box” decision support and improve clinician trust and accountability, particularly when complex models such as boosted trees are used.

Updating and localizing cost and QALYs inputs to the target population

For the decision thresholds to reflect real clinical pathways, the clinical and economic objective functions must be tied to the local context of care delivery.

This includes,

- Updating cost inputs (direct expenditures, DRG-based reimbursements, imaging/biopsy follow-up costs, medications, hospitalizations) in a local framework (e.g. Greece/EU), rather than relying exclusively on U.S.-based estimates.
- Explicit definition of the economic perspective (payer vs societal), the time horizon and discounting assumptions, which are central components in cost-benefit and health-economic evaluation.
- Incorporating uncertainty through ranges/distributions instead of point estimates, to support sensitivity analysis and more robust policy conclusions.
- Harmonizing QALY estimates with available treatment options and with the true baseline state of the surveillance population (e.g. compensated cirrhosis), ensuring that utility decrements reflect realistic disease trajectories.

As shown by the one-way sensitivity analyses, p^* can shift substantially as stakeholder preferences and system incentives change. Future work should therefore focus on grounding these inputs in locally valid evidence and uncertainty ranges so that thresholds emerge from realistic incremental trade-offs rather than transferred estimates.

Extending sensitivity analysis to probabilistic uncertainty

While this thesis used deterministic one-way sensitivity analyses, as mentioned before, a natural extension is probabilistic sensitivity analysis (PSA). In PSA, key utilities and costs are treated as distributions, allowing uncertainty to propagate to a distribution over p^* rather than a single point estimate. This would enable reporting of uncertainty intervals for thresholds and provide a more formal basis for robustness claims.

Modeling multi-action policies and sequential decision-making

The binary decision Escalate vs No Escalation provides a clear theoretical foundation, but real clinical workflows often involve escalation pathways, such as intensified monitoring, additional imaging, biopsy or specialist referral.

Future work can therefore expand the decision layer into,

- A multi-action framework (e.g. 3–4 possible actions, each with its own cost/utility structure), where instead of a single p^* , multiple thresholds emerge (e.g. p_1 for additional imaging, p_2 for invasive work-up).

- A dynamic, time-dependent decision model (Markov models, microsimulation or MDP-like formulations), where decisions at each visit depend on disease progression, prior test outcomes and accumulated costs/benefits, something that better fits the longitudinal nature of HCC surveillance.

Linking to clinical utility tools (Decision Curve Analysis and related measures)

To connect the decision-theoretic definition of p^* with practical clinical assessment, it is useful to present model value as clinical net benefit across a range of thresholds, enabling direct comparison with baseline strategies such as escalate-all and escalate-none.

This supports,

- Identification of threshold regions where the model provides meaningful clinical utility (i.e. improved benefit-risk trade-offs).
- Alignment of the six decision scenarios with interpretable threshold intervals corresponding to different stakeholders (risk-averse clinician, minimal-intervention policy, cost-containment payer).
- Evaluation beyond discrimination, ensuring that predictive models are judged by their impact on decisions and outcomes rather than AUC alone.

In this way, the proposed framework can be extended into a fully decision-aware validation strategy, where the model's contribution is quantified directly in terms of patient and system relevant outcomes.

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Appendix A: Exploratory Data Analysis (EDA) Code

This appendix contains the code that has been used to compute the descriptive statistics and generate the figures reported in Chapter 2.

```
Import os
import re
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

# Global display / theme
pd.set_option("display.max_columns", 120)
pd.set_option("display.width", 160)
pd.set_option("display.float_format", lambda x: f'{x:0.4f}')

# seaborn theme + unified palette
sns.set_theme(style="whitegrid", font_scale=1.05)
PALETTE_CONT = "rocket" # continuous colormap (matches heatmap hues)
PALETTE_DISC = sns.color_palette("rocket", 6) # discrete palette for bars/lines
CLASS_PALETTE = sns.color_palette("rocket", 2) # for target=0/1

plt.rcParams["figure.figsize"] = (7.0, 5.0)
plt.rcParams["axes.titlesize"] = 12
plt.rcParams["axes.labelsize"] = 11
```

Code.A1: Imports and global configuration

```
# Load data
df = pd.read_csv("data/synthetic_liver_cancer_dataset.csv")
TARGET = "liver_cancer"

# File for Tables & Figures
out_dir = "eda_figs"
os.makedirs(out_dir, exist_ok=True)

y = df[TARGET].astype(int)
X = df.drop(columns=[TARGET])

num_cols = X.select_dtypes(include=[np.number]).columns.tolist()
cat_cols = X.select_dtypes(include=["object", "category", "bool"]).columns.tolist()
```

Code.A2: Load data and define feature lists

```
print("First Rows:")
df.head()

print("Shape (rows, cols):", df.shape)
print("\nDtypes:\n", df.dtypes)
```

Code.A3: Dataset preview, dimensions and data types

```
# Missingness & class balance
missing_tbl = (df.isna().sum()
               .to_frame("missing_count")
               .assign(missing_rate=lambda d: d["missing_count"]/len(df))
               .sort_values("missing_rate", ascending=False))
display(missing_tbl)

balance = (y.value_counts().rename_axis(TARGET)
           .to_frame("count")
           .assign(ratio=lambda s: s["count"]/len(y))
           .sort_index())
display(balance)
```

Code.A4: Missingness audit and target distribution

```
# Descriptive statistics for numeric variables
desc_num = df[num_cols].describe().T
desc_num["skew"] = df[num_cols].skew(numeric_only=True)
desc_num["kurtosis"] = df[num_cols].kurtosis(numeric_only=True)
display(desc_num)
```

Code.A5: Descriptive statistics for numeric features

```
# Descriptive statistics for categorical variables
rows = []
for c in cat_cols:
    vc = df[c].value_counts(dropna=False)
    rows.append({
        "variable": c,
        "n_unique": df[c].nunique(dropna=True),
        "n_missing": int(df[c].isna().sum()),
        "top_category": (str(vc.index[0]) if len(vc) else np.nan),
        "top_freq": (int(vc.iloc[0]) if len(vc) else np.nan)
    })
desc_cat = pd.DataFrame(rows).sort_values("n_unique", ascending=False)
display(desc_cat)
```

Code.A6: Descriptive statistics for categorical features

```
# Zero-scan (numeric)
zero_tbl = pd.DataFrame({
    "variable": num_cols,
    "n_zero": [int((df[c] == 0).sum()) for c in num_cols]
})
zero_tbl["zero_rate"] = zero_tbl["n_zero"] / len(df)
display(zero_tbl.sort_values("zero_rate", ascending=False))
```

Code.A7: Zero-value scan across numeric features

```
# Univariate EDA — numeric (histogram + KDE) with thesis hues

for c in num_cols:
    plt.figure()
    sns.histplot(df, x=c, bins=30, kde=True, color=PALETTE_DISC[1], edgecolor="white")
    plt.title(f"Distribution — {c}")
    plt.xlabel(c)
    plt.ylabel("Count")
    plt.tight_layout()
    plt.savefig(f"{out_dir}/hist_{c}.png", dpi=150)
    plt.show()
```

Code.A8: Univariate numeric distributions (histograms & KDE)

```
# Numeric vs Target — box/violin (legend outside)

for c in num_cols:
    plt.figure()
    sns.violinplot(data=df, x=TARGET, y=c, palette=CLASS_PALETTE, cut=0,
inner="quartile")
    plt.title(f"{c} by Target")
    plt.xlabel("Target (0=no cancer, 1=cancer)")
    plt.tight_layout()
    plt.savefig(f"{out_dir}/violin_{c}_by_target.png", dpi=150)
    plt.show()
```

Code.A9: Numeric features by target (violin plots)

```
# Categorical — countplots (overall) and by target (stacked via hue)

for c in cat_cols:
    # overall
    plt.figure()
    sns.countplot(data=df, x=c, palette=PALETTE_DISC)
    plt.title(f"Counts — {c}")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.savefig(f"{out_dir}/count_{c}.png", dpi=150)
    plt.show()

    # by target
    plt.figure()
    sns.countplot(data=df, x=c, hue=TARGET, palette=CLASS_PALETTE)
    plt.title(f"Counts by Target — {c}")
    plt.xticks(rotation=45)
    plt.tight_layout()
    plt.savefig(f"{out_dir}/count_{c}_by_target.png", dpi=150)
    plt.show()
```

Code.A10: Categorical frequencies (overall and by target)

```
# Target-wise numeric summary table

grp_num = df.groupby(TARGET)[num_cols].agg(["count", "mean", "std", "median", "min",
"max"])
display(grp_num)
```

Code.A11: Descriptive contrasts by outcome

```
# Correlations — Spearman (numeric) + heatmap with 'rocket' hues

if len(num_cols) >= 2:
    corr = df[num_cols].corr(method="spearman")
    mask = np.triu(np.ones_like(corr, dtype=bool))
    plt.figure(figsize=(7.5, 5.5))
    sns.heatmap(corr, mask=mask, cmap=PALETTE_CONT, vmin=-1, vmax=1, center=0,
                annot=True, fmt=".2f", linewidths=0.6, linecolor="white",
                cbar_kws={"label": "Spearman  $\rho$ "})
    plt.title("Spearman Correlation — Numeric Variables")
    plt.tight_layout()
    plt.savefig(f"{out_dir}/spearman_corr.png", dpi=150)
    plt.show()
```

Code.A12: Spearman correlation heatmap (numeric features)

```
# Point-biserial (numeric vs target) — Pearson with {0,1} target

pb_vals = []
yt = y.astype(float).values
for c in num_cols:
    xv = df[c].values
    mask = ~pd.isna(xv)
    xv = xv[mask]; yv = yt[mask]
    if xv.size > 2 and np.std(xv) > 0:
        r = np.corrcoef(xv, yv)[0,1]
    else:
        r = np.nan
    pb_vals.append((c, r, abs(r)))

pb_df = (pd.DataFrame(pb_vals, columns=["variable", "r", "abs_r"])
        .sort_values("abs_r", ascending=False))
display(pb_df)

plt.figure(figsize=(8,5))
sns.barplot(data=pb_df, x="abs_r", y="variable", palette=PALETTE_DISC)
plt.title("Point-biserial correlation |numeric vs target|")
plt.xlabel("|r|")
plt.ylabel("")
plt.tight_layout()
plt.savefig(f"{out_dir}/point_biserial_barplot.png", dpi=150)
plt.show()
```

Code.A13: Point-biserial correlations (numeric vs target)

```

# Chi-square & Cramér's V (categorical vs target) — with seaborn bars

chi_rows = []
for c in cat_cols:
    ct = pd.crosstab(df[c], y)
    obs = ct.values.astype(float)
    if obs.size == 0:
        continue
    row_sum = obs.sum(axis=1, keepdims=True)
    col_sum = obs.sum(axis=0, keepdims=True)
    total = obs.sum()
    exp = (row_sum @ col_sum) / total
    with np.errstate(divide='ignore', invalid='ignore'):
        chi2 = np.nansum((obs - exp)**2 / exp)
    r, k = obs.shape
    phi2 = chi2 / total if total > 0 else np.nan
    # bias-corrected Cramér's V
    if total > 1:
        phi2corr = max(0, phi2 - ((k-1)*(r-1))/(total-1))
        rcorr = r - ((r-1)**2)/(total-1)
        kcorr = k - ((k-1)**2)/(total-1)
        denom = max(1e-12, min((kcorr-1), (rcorr-1)))
        V = np.sqrt(phi2corr / denom)
    else:
        V = np.nan
    chi_rows.append({"variable": c, "chi2": chi2, "dof": (r-1)*(k-1), "cramers_v": V})

cat_assoc = pd.DataFrame(chi_rows).sort_values("cramers_v", ascending=False)
display(cat_assoc)

plt.figure(figsize=(8,5))
sns.barplot(data=cat_assoc, x="cramers_v", y="variable", palette=PALETTE_DISC)
plt.title("Cramér's V — Categorical vs Target")
plt.xlabel("Cramér's V")
plt.ylabel("")
plt.tight_layout()
plt.savefig(f"{out_dir}/cramersV_barplot.png", dpi=150)
plt.show()

```

Code.A14: Chi-square and Cramér's V (categorical vs target)

```

# Stacked proportions by target (visual association) — legend on right

for c in cat_cols:
    ct = pd.crosstab(df[c], y, normalize="index")
    plt.figure()
    bottoms = np.zeros(len(ct))
    for k_idx, k in enumerate(ct.columns):
        plt.bar(ct.index.astype(str), ct[k].values, bottom=bottoms,
                label=f"{TARGET}={int(k)}", color=CLASS_PALETTE[k_idx])
        bottoms += ct[k].values
    plt.title(f"Within-category proportions by target — {c}")
    plt.xlabel(c)
    plt.ylabel("Proportion")
    plt.xticks(rotation=45)
    plt.legend(title="Target", loc="upper left", bbox_to_anchor=(1.0, 0.5))
    plt.tight_layout()
    plt.savefig(f"{out_dir}/stacked_prop_{c}.png", dpi=150)
    plt.show()

```

Code.A15: Stacked within-category proportions by target

Appendix B: Pipeline Code (Model Comparison and Calibration)

This appendix provides the complete Python implementation of the model comparison pipeline used in Chapters 3 and 4. The code executes the following steps: dataset loading, stratified splitting into TRAIN/VALIDATION/TEST (60/20/20), leakage-safe preprocessing, cross-validated hyperparameter tuning (ROC-AUC) on TRAIN, probability calibration (sigmoid vs isotonic selected by Brier on VALIDATION), refitting on the full development set (TRAIN+VALIDATION), TEST set evaluation (ROC-AUC, AP, Brier), bootstrap confidence intervals and generation of ROC / PR / calibration plots.

```
# Import libraries
import os
import warnings
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split, StratifiedKFold, GridSearchCV
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.impute import SimpleImputer
from sklearn.utils import resample
from sklearn.base import clone

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import (
    roc_auc_score, average_precision_score, brier_score_loss,
    roc_curve, precision_recall_curve
)
from sklearn.calibration import CalibratedClassifierCV, calibration_curve

from xgboost import XGBClassifier

# Basic playability/display settings
warnings.filterwarnings("ignore")
RANDOM_STATE = 42
np.random.seed(RANDOM_STATE)
pd.set_option("display.precision", 4)
```

Code.B1: Imports and global settings

```

# File for Tables & Figures

out_dir = "pipeline_figs"
os.makedirs(out_dir, exist_ok=True)

# Load the dataset.
# TARGET: the name of the target column (binary collation: 0/1).
# Check column types and class proportions, so we know
# if the problem is unbalanced and what preprocessing is needed.
TARGET = "liver_cancer"
df = pd.read_csv("data/synthetic_liver_cancer_dataset.csv")

# Separation into X (features) and y (target)
y = df[TARGET].astype(int).values
X = df.drop(columns=[TARGET])

# Identifying feature types
cat_cols = X.select_dtypes(include=["object", "category"]).columns.tolist()
num_cols = X.select_dtypes(include=[np.number]).columns.tolist()

# Quick summary
display(pd.DataFrame({
    "numeric_cols_count": [len(num_cols)],
    "categorical_cols_count": [len(cat_cols)],
    "class_0_ratio": [(y == 0).mean()],
    "class_1_ratio": [(y == 1).mean()],
}))

```

Code.B2: Data loading and feature typing

```

# Train / Validation / Test split (60/20/20)
# - TEST is a final hold-out set (never used for model selection).
# - VALIDATION is used for calibration method selection (Brier).
# - TRAIN is used for CV tuning and model selection (ROC-AUC).
# We create a ColumnTransformer:
# - Numerical: missing values (median) + normalization (StandardScaler).
# - Categorical: filling (most_frequent) + One-Hot Encoding.
# We put them in a Pipeline so that the "fit" is done ONLY on the train
# folds (avoid leakage).

# Step 1: create TEST (20%)
X_dev, X_test, y_dev, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=RANDOM_STATE
)

# Step 2: split DEV into TRAIN (60% total) and VAL (20% total)
# 0.25 of 0.80 = 0.20 of total
X_train, X_val, y_train, y_val = train_test_split(
    X_dev, y_dev, test_size=0.25, stratify=y_dev, random_state=RANDOM_STATE
)

print("TRAIN size:", X_train.shape, "VAL size:", X_val.shape, "TEST size:", X_test.shape)

# Preprocessing
numeric_tf = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])

categorical_tf = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore"))
])

preprocessor = ColumnTransformer([
    ("num", numeric_tf, num_cols),
    ("cat", categorical_tf, cat_cols)
])

```

Code.B3: Train/validation/test split and preprocessing pipeline

```

# We define three models:
# - Logistic Regression: with class_weight='balanced' for unbalanced data.
# - Random Forest: typical robust baseline.
# - XGBoost: robust gradient boosting model. We define scale_pos_weight = (#negatives /
#positives).
# We also create the hyper-parameter grids for GridSearchCV.

log_reg = LogisticRegression(
    max_iter=5000,
    class_weight="balanced",
    solver="lbfgs"
)

rf = RandomForestClassifier(
    random_state=RANDOM_STATE,
    n_jobs=-1
)

pos_weight = (y_train == 0).sum() / max((y_train == 1).sum(), 1)
xgb = XGBClassifier(
    objective="binary:logistic",
    eval_metric="auc",
    random_state=RANDOM_STATE,
    n_estimators=500,
    learning_rate=0.05,
    max_depth=4,
    subsample=0.9,
    colsample_bytree=0.9,
    reg_lambda=1.0,
    n_jobs=-1,
    scale_pos_weight=pos_weight
)

# Pipelines (preprocessing + model)
pipe_lr = Pipeline([
    ("prep", preprocessor),
    ("clf", log_reg)
])

pipe_rf = Pipeline([
    ("prep", preprocessor),
    ("clf", rf)
])

pipe_xgb = Pipeline([
    ("prep", preprocessor),
    ("clf", xgb)
])

# Simple but essential grids for tuning
param_grid_lr = {
    "clf__C": [0.1, 1.0, 3.0],
    "clf__solver": ["lbfgs", "liblinear"]
}

```

```

param_grid_rf = {
    "clf__n_estimators": [300, 600],
    "clf__max_depth": [None, 8],
}

param_grid_xgb = {
    "clf__n_estimators": [300, 600],
    "clf__max_depth": [3, 5],
    "clf__learning_rate": [0.03, 0.07],
}

```

Code.B4: Model definitions and hyperparameter grids

```

# 5-fold cross-validation for fair assessment of training performance
cv = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=RANDOM_STATE
)

# We run GridSearchCV with ROC-AUC as the main metric for each model.
# GridSearch does internal cross-validation on the train data.
# We keep the best estimator (best_estimator_) for each model.

def tune(pipe, grid, name):
    gs = GridSearchCV(
        estimator=pipe,
        param_grid=grid,
        scoring="roc_auc",
        cv=cv,
        n_jobs=-1,
        refit=True,
        verbose=0
    )
    gs.fit(X_train, y_train)
    return {"model": name, "cv_best_roc_auc": gs.best_score_, "best_params":
gs.best_params_, gs.best_estimator_}

res_lr, best_lr = tune(pipe_lr, param_grid_lr, "LogisticRegression")

res_rf, best_rf = tune(pipe_rf, param_grid_rf, "RandomForest")

res_xgb, best_xgb = tune(pipe_xgb, param_grid_xgb, "XGBoost")

# Table with tuning results (CV)
tuning_tbl = pd.DataFrame([res_lr, res_rf, res_xgb])
display(tuning_tbl)

```

Code.B5: Hyperparameter tuning via cross-validation

```

# Sanity checks for near-perfect performance (XGBoost)
# simple checks, NOT used for model selection
# 1) Label permutation test (TRAIN only, CV ROC-AUC should collapse to ~0.5)
# We keep the tuned hyperparameters fixed (best_xgb) and evaluate with the same CV
object.
# Baseline CV AUC (tuned XGB, true labels)

xgb_cv_aucs = []
for tr_idx, te_idx in cv.split(X_train, y_train):
    m = clone(best_xgb)
    m.fit(X_train.iloc[tr_idx], y_train[tr_idx])
    proba = m.predict_proba(X_train.iloc[te_idx])[:, 1]
    xgb_cv_aucs.append(roc_auc_score(y_train[te_idx], proba))

print("XGB baseline CV AUC (mean±std):", float(np.mean(xgb_cv_aucs)),
      float(np.std(xgb_cv_aucs)))

# Permuted-label CV AUC distribution (mean AUC over folds per permutation)

rng = np.random.RandomState(RANDOM_STATE)
perm_cv_auc_means = []
N_PERM = 30

for _ in range(N_PERM):
    y_perm = rng.permutation(y_train)
    fold_aucs = []
    for tr_idx, te_idx in cv.split(X_train, y_perm):
        m = clone(best_xgb)
        m.fit(X_train.iloc[tr_idx], y_perm[tr_idx])
        proba = m.predict_proba(X_train.iloc[te_idx])[:, 1]
        fold_aucs.append(roc_auc_score(y_perm[te_idx], proba))
    perm_cv_auc_means.append(float(np.mean(fold_aucs)))

print("XGB label-permutation CV AUC (mean±std):", float(np.mean(perm_cv_auc_means)),
      float(np.std(perm_cv_auc_means)))
display(pd.DataFrame({"perm_cv_auc_mean": perm_cv_auc_means}))

# 2) Repeated stratified splits inside DEV (exclude TEST), fixed tuned XGB
# We repeatedly split DEV into train/hold-out and check whether AUC stays consistently
high.

rep_split_aucs = []
N_SPLITS = 20

for seed in range(N_SPLITS):
    X_tr_s, X_te_s, y_tr_s, y_te_s = train_test_split(
        X_dev, y_dev, test_size=0.20, stratify=y_dev, random_state=seed
    )
    m = clone(best_xgb)
    m.fit(X_tr_s, y_tr_s)
    proba = m.predict_proba(X_te_s)[:, 1]
    rep_split_aucs.append(float(roc_auc_score(y_te_s, proba)))

rep_split_aucs = np.array(rep_split_aucs)

```

```
print("XGB repeated DEV splits AUC (mean±std):", float(rep_split_aucs.mean()),  
      float(rep_split_aucs.std()))  
print("XGB repeated DEV splits AUC (min/max):", float(rep_split_aucs.min()),  
      float(rep_split_aucs.max()))  
display(pd.DataFrame({"repeat_split_auc": rep_split_aucs}))
```

Code.B6: Sanity checks (label permutation + repeated splits)

```

# We use an explicit StratifiedKFold object also for the calibration step
# (instead of passing cv=5). This keeps the procedure stratified and reproducible.
cv_cal = StratifiedKFold(
    n_splits=5,
    shuffle=True,
    random_state=RANDOM_STATE
)
# To make the probabilities "fair", we calibrate the models with CalibratedClassifierCV.
# We test two methods: 'isotonic' and 'sigmoid' (Platt scaling).
# Selection is made based on the Brier score on a validation set.
# - We now pass an explicit StratifiedKFold (cv_cal) to the calibrator.
# This makes calibration:
# (a) stratified
# (b) reproducible (shuffle + random_state)
def pick_calibration(model, X_tr, y_tr, X_val, y_val, cv_cal):
    methods = {}
    scores = {}
    for m in ["isotonic", "sigmoid"]:
        cal = CalibratedClassifierCV(model, method=m, cv=cv_cal)
        cal.fit(X_tr, y_tr)
        proba_val = cal.predict_proba(X_val)[: , 1]
        scores[m] = brier_score_loss(y_val, proba_val)
        methods[m] = cal
    best_m = min(scores, key=scores.get) # smaller Brier = better calibration
    return methods[best_m], best_m, scores

# Calibration method selection is based directly on the VAL set.
cal_lr, m_lr, s_lr = pick_calibration(best_lr, X_train, y_train, X_val, y_val, cv_cal)

cal_rf, m_rf, s_rf = pick_calibration(best_rf, X_train, y_train, X_val, y_val, cv_cal)

cal_xgb, m_xgb, s_xgb = pick_calibration(best_xgb, X_train, y_train, X_val, y_val, cv_cal)

# Table: which method was chosen and what Brier gave in validation
calibration_tbl = pd.DataFrame([
    {"model": "LogisticRegression", "picked": m_lr, "val_brier": s_lr[m_lr]},
    {"model": "RandomForest", "picked": m_rf, "val_brier": s_rf[m_rf]},
    {"model": "XGBoost", "picked": m_xgb, "val_brier": s_xgb[m_xgb]},
])

# Display the calibration-method selection and its quality on the VALIDATION set
# (lower Brier = better probability calibration)
display(calibration_tbl)

# Final fit of the "winners" over ALL development data (TRAIN + VAL)
# After selecting the calibration option on VALIDATION,
# we refit each calibrated model on the full development set (TRAIN + VAL)
# to maximize training signal, while keeping TEST untouched.
X_dev_final = pd.concat([X_train, X_val], axis=0)
y_dev_final = np.concatenate([y_train, y_val], axis=0)

cal_lr.fit(X_dev_final, y_dev_final)
cal_rf.fit(X_dev_final, y_dev_final)
cal_xgb.fit(X_dev_final, y_dev_final)

```

Code.B7: Probability calibration selection and development refit


```

# Here we define:
# - bootstrap_ci: calculates confidence intervals (percentile) for a metric (e.g. AUC or AP)
# - evaluate: returns AUC, AP (PR-AUC), Brier in the test set and the vector with
probabilities

def bootstrap_ci(y_true, proba, metric_fn, n_boot=2000, alpha=0.05,
rng=RANDOM_STATE):
    rng = np.random.RandomState(rng)
    n = len(y_true)
    stats = []
    for _ in range(n_boot):
        idx = rng.randint(0, n, n) # bootstrap sample with resampling
        stats.append(metric_fn(y_true[idx], proba[idx]))
    lo, hi = np.percentile(stats, [100*alpha/2, 100*(1-alpha/2)])
    return float(np.mean(stats)), (float(lo), float(hi))

def evaluate(model, name):
    proba = model.predict_proba(X_test)[: , 1]
    auc = roc_auc_score(y_test, proba)
    ap = average_precision_score(y_test, proba)
    brier = brier_score_loss(y_test, proba)
    # 95% CI with bootstrap for AUC and AP
    _, auc_ci = bootstrap_ci(y_test, proba, roc_auc_score)
    _, ap_ci = bootstrap_ci(y_test, proba, average_precision_score)
    return {"model": name, "auc": auc, "auc_ci": auc_ci, "ap": ap, "ap_ci": ap_ci, "brier":
brier}, proba

# TEST evaluation (strictly for reporting and not for selection)
m_lr, p_lr = evaluate(cal_lr, "LR")

m_rf, p_rf = evaluate(cal_rf, "RF")

m_xgb, p_xgb = evaluate(cal_xgb, "XGB")

# Summary table for the test set
summary = pd.DataFrame([m_lr, m_rf, m_xgb])[["model", "auc", "auc_ci", "ap", "ap_ci",
"brier"]]
summary = summary.sort_values("auc", ascending=False)
display(summary)

```

Code.B8: Test-set evaluation with bootstrap confidence intervals

```

# Here we produce 3 single charts (one for each metric).
# Each model has a different color.
colors = {"LR": "C0", "RF": "C1", "XGB": "C2"}

# 1) ROC (a chart with all the curves)
plt.figure(figsize=(6,5))
for name, proba in [("LR", p_lr), ("RF", p_rf), ("XGB", p_xgb)]:
    fpr, tpr, _ = roc_curve(y_test, proba)
    plt.plot(fpr, tpr, label=name, color=colors[name])
plt.plot([0,1], [0,1], "--", color="black", alpha=0.6) # random classifier
plt.title("ROC Curves (Test)")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.legend()
plt.grid(True, linestyle="--", alpha=0.4)
plt.tight_layout()

save_path = os.path.join(out_dir, "roc_curve.png")
plt.savefig(save_path, dpi=300, bbox_inches='tight')
print(f'Saved: {save_path}')

plt.show()

# 2) Precision-Recall (a diagram with all the curves)
plt.figure(figsize=(6,5))
for name, proba in [("LR", p_lr), ("RF", p_rf), ("XGB", p_xgb)]:
    prec, rec, _ = precision_recall_curve(y_test, proba)
    plt.plot(rec, prec, label=name, color=colors[name])
plt.title("Precision-Recall Curves (Test)")
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.legend()
plt.grid(True, linestyle="--", alpha=0.4)
plt.tight_layout()

save_path = os.path.join(out_dir, "precision_recall_curve.png")
plt.savefig(save_path, dpi=300, bbox_inches='tight')
print(f'Saved: {save_path}')

plt.show()

# 3) Calibration (a diagram with all the curves)
plt.figure(figsize=(6,5))
for name, proba in [("LR", p_lr), ("RF", p_rf), ("XGB", p_xgb)]:
    prob_true, prob_pred = calibration_curve(y_test, proba, n_bins=10, strategy="quantile")
    plt.plot(prob_pred, prob_true, marker="o", label=name, color=colors[name])
plt.plot([0,1], [0,1], "--", color="black", alpha=0.6) # perfect calibration
plt.xlabel("Predicted probability")
plt.ylabel("Observed frequency")
plt.title("Calibration Curves (Test)")
plt.legend()
plt.grid(True, linestyle="--", alpha=0.4)
plt.tight_layout()

save_path = os.path.join(out_dir, "calibration_curve.png")

```

```
plt.savefig(save_path, dpi=300, bbox_inches='tight')
print(f'Saved: {save_path}')

plt.show()
```

Code.B9: Performance plots (ROC/PR/Calibration)

```
# Model Selection
# We use two criteria, each evaluated on the appropriate split:
# 1) Discrimination -> CV ROC-AUC on TRAIN (from GridSearchCV)
# ROC-AUC tells us how well the model ranks positives above negatives.
# 2) Probability quality -> Brier score on VALIDATION (after calibration)
# Brier tells us how good the predicted probabilities are (calibration + sharpness).
# - The TEST set is NOT used at all for selecting the model.
# - TEST is used ONLY once, for the final unbiased report.

# Join the tuning (CV) results with the validation calibration quality (Brier)
model_selection_tbl = tuning_tbl.merge(
    calibration_tbl[["model", "val_brier"]],
    on="model",
    how="left"
)
display(model_selection_tbl)

# Selection rule:
# - Primary objective: maximize CV ROC-AUC (discrimination).
# - Secondary objective: minimize VAL Brier (probability quality).
# This is a simple deterministic multi-objective selection:
# sort by AUC (descending) and then by Brier (ascending).

winner_row = model_selection_tbl.sort_values(
    ["cv_best_roc_auc", "val_brier"],
    ascending=[False, True]
).iloc[0]

winner_cv = winner_row["model"]
print("Best model (TRAIN CV ROC-AUC + VAL Brier tie-break):", winner_cv)
```

Code.B10: Model selection (TRAIN CV ROC-AUC + VAL Brier tie-break)

Appendix C: Features Stress Tests

This appendix provides the minimal code modifications required to reproduce the feature ablation (stress test) experiments described in Part I. The goal is to evaluate the robustness of the model comparison pipeline when removing key numeric biomarkers from the feature set, `alpha_fetoprotein_level` (AFP) and `liver_function_score`.

All remaining steps of the pipeline (data split, preprocessing, cross-validated tuning, calibration method selection and final test evaluation) remain identical to Appendix B.

```
# Stress-test setting
SCENARIO = "drop_AFP"
DROP_COLS = ["alpha_fetoprotein_level"]

# Separation into X (features) and y (target)
y = df[TARGET].astype(int).values
X = df.drop(columns=[TARGET] + DROP_COLS)
```

Code C.1: Stress test — drop AFP (`alpha_fetoprotein_level`)

```
# Stress-test setting
SCENARIO = "drop_LFS"
DROP_COLS = ["liver_function_score"]

# Separation into X (features) and y (target)
y = df[TARGET].astype(int).values
X = df.drop(columns=[TARGET] + DROP_COLS)
```

Code C.2: Stress test — drop LFS (`liver_function_score`)

```
# Stress-test setting
SCENARIO = "drop_AFP_&_LFS"
DROP_COLS = ["alpha_fetoprotein_level", "liver_function_score"]

# Separation into X (features) and y (target)
y = df[TARGET].astype(int).values
X = df.drop(columns=[TARGET] + DROP_COLS)
```

Code C.3: Stress test — drop AFP & LFS

Appendix D: Robustness / Stress-test Additions (Noise Injection & Prevalence Shift)

The full code of the basic pipeline is listed in Appendix B. This Appendix includes only the additions/modifications that implement the robustness/stress tests i.e. Gaussian noise injection as a transformer within the numeric preprocessing pipeline and prevalence shift in the test set (target prevalence 5%) with iteration on multiple seeds for robust estimation of the metrics.

```
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.utils.validation import check_is_fitted
from sklearn.utils import check_random_state

class GaussianNoiseInjection(BaseEstimator, TransformerMixin):
    """
    Injects Gaussian noise into numerical features to simulate measurement error / aleatoric
    uncertainty.

    Key points:
    - fit(): learns per-feature standard deviation ONLY from the training data it receives
      (i.e. within each CV fold).
    - transform(): adds noise ~ N(0, (std * noise_factor)).
    """
    def __init__(self, noise_factor=1.0, random_state=42):
        self.noise_factor = noise_factor
        self.random_state = random_state

    def fit(self, X, y=None):
        # X is expected to be numeric numpy array (after imputer)
        X = np.asarray(X)
        # ddof=0 for population std; consistent and stable
        self.stds_ = X.std(axis=0, ddof=0)
        return self

    def transform(self, X):
        # Ensure transformer has been fitted
        check_is_fitted(self, "stds_")
        X = np.asarray(X)

        # Create sklearn-style RNG (deterministic for reproducibility)
        rng = check_random_state(self.random_state)

        # Noise scale per feature
        scale = self.stds_ * self.noise_factor

        # Draw noise with same shape as X
        noise = rng.normal(loc=0.0, scale=scale, size=X.shape)
        return X + noise
```

Code D.1: Gaussian Noise Injection Transformer

```

numeric_tf = Pipeline(steps=[
    ("imputer", SimpleImputer(strategy="median")),
    ("noise", GaussianNoiseInjection(noise_factor=NOISE_FACTOR,
    random_state=RANDOM_STATE)),
    ("scaler", StandardScaler())
])

```

Code D.2: Numeric preprocessing branch modification

```

def adjust_test_prevalence_seeded(X, y, target_prevalence, seed):
    """
    Downsamples positive class to reach target prevalence (fixed-N constraint).
    - Keeps all negatives (0)
    - Randomly subsamples positives (1) to match target prevalence:
        n1_needed / (n0 + n1_needed) = target_prevalence
    Uses local RNG for reproducibility without affecting global state.
    """
    rng = np.random.default_rng(seed)

    y_arr = y if isinstance(y, np.ndarray) else np.asarray(y)
    idx_0 = np.where(y_arr == 0)[0]
    idx_1 = np.where(y_arr == 1)[0]

    n_0 = len(idx_0)
    n_1_needed = int(n_0 * target_prevalence / (1 - target_prevalence))
    n_1_needed = min(n_1_needed, len(idx_1))

    # Ensure both classes exist (roc_auc_score requires both)
    if n_0 < 1 or n_1_needed < 1:
        raise ValueError("Not enough samples to form shifted test set with both classes.")

    idx_1_selected = rng.choice(idx_1, size=n_1_needed, replace=False)
    new_indices = np.concatenate([idx_0, idx_1_selected])

    # Shuffling makes the cohort look like a realistic mixed test set.
    # Note: with deterministic noise in the pipeline, shuffling can change which sample
    receives
    # which noise draw (because noise is generated per row order). This is OK if you interpret
    # results as "noise + prevalence variability combined".
    rng.shuffle(new_indices)

    return X.iloc[new_indices], y_arr[new_indices]

```

Code D.3: Prevalence shift function

```

def robust_evaluate(model, X_test_base, y_test_base, n_seeds=30, target_prev=0.05):
    """
    Runs evaluation multiple times with different downsampling seeds to simulate low
    prevalence.
    Collects ROC-AUC, AP, and Brier.

    Output summary:
    - mean
    - empirical 95% interval across seeds (2.5th to 97.5th percentiles)
    This is better here than "mean  $\pm$  1.96*SE" because each seed produces a different shifted
    cohort.
    """
    records = []

    for seed in range(n_seeds):
        # 1) Shift prevalence
        X_shifted, y_shifted = adjust_test_prevalence_seeded(
            X_test_base, y_test_base, target_prevalence=target_prev, seed=seed
        )

        # 2) Predict probabilities (noise is applied automatically via pipeline inside the model)
        proba = model.predict_proba(X_shifted)[ :, 1]

        # 3) Metrics
        auc = roc_auc_score(y_shifted, proba)
        ap = average_precision_score(y_shifted, proba)
        brier = brier_score_loss(y_shifted, proba)

        records.append({"auc": auc, "ap": ap, "brier": brier})

    df_metrics = pd.DataFrame(records)

    # Summarize with mean + empirical 95% interval
    stats = {}
    for col in df_metrics.columns:
        stats[f"{col}_mean"] = df_metrics[col].mean()
        stats[f"{col}_lo"], stats[f"{col}_hi"] = np.percentile(df_metrics[col], [2.5, 97.5])

    return stats, df_metrics

results_robust = []
all_metrics = {}

for model_name, model in [("LR", cal_lr), ("RF", cal_rf), ("XGB", cal_xgb)]:
    stats, df_metrics = robust_evaluate(
        model, X_test, y_test, n_seeds=N_SEEDS, target_prev=TARGET_PREV
    )
    all_metrics[model_name] = df_metrics

    row = {"model": model_name}
    row.update(stats)
    results_robust.append(row)

summary_robust = pd.DataFrame(results_robust)

```

```

# Nicely formatted display table
summary_display = summary_robust[["model"]].copy()

summary_display["AUC"] = summary_robust.apply(
    lambda x: f"{x['auc_mean']:.4f} [{x['auc_lo']:.4f}, {x['auc_hi']:.4f}]",
    axis=1
)

summary_display["AP"] = summary_robust.apply(
    lambda x: f"{x['ap_mean']:.4f} [{x['ap_lo']:.4f}, {x['ap_hi']:.4f}]",
    axis=1
)

summary_display["Brier"] = summary_robust.apply(
    lambda x: f"{x['brier_mean']:.4f} [{x['brier_lo']:.4f}, {x['brier_hi']:.4f}]",
    axis=1
)

print("\n--- FINAL ROBUST TEST SET RESULTS ---")
display(summary_display)

```

Code D.4: Robust evaluation loop & summary

```

# We generate a single shifted test set for plotting a representative PR curve.
X_test_plot, y_test_plot = adjust_test_prevalence_seeded(X_test, y_test, TARGET_PREV,
seed=RANDOM_STATE)
proba_xgb_plot = cal_xgb.predict_proba(X_test_plot)[: , 1]

plt.figure(figsize=(6, 5))
prec, rec, _ = precision_recall_curve(y_test_plot, proba_xgb_plot)
plt.plot(rec, prec, label=f"XGB (Prev={TARGET_PREV:.0%})", color="C2")
plt.title("Precision–Recall Curve (Example Shifted Test Cohort)")
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.legend()
plt.grid(True, linestyle="--", alpha=0.4)
plt.tight_layout()

save_path = os.path.join(out_dir, "robust_pr_curve.png")
plt.savefig(save_path, dpi=300, bbox_inches="tight")
plt.show()

```

Code D.5: Example PR curve plot

Appendix E: Operationalization of the Optimal Decision Threshold (p^*) on the Test Set

This appendix provides the additional code used to operationalize the optimal decision threshold p^* derived in Part II on the held-out test set. Specifically, it reports the resulting Treat rate, the corresponding confusion matrix (TP/FP/FN/TN) and standard operational metrics and the realized expected utility on the test set under each payoff scenario. All preceding pipeline steps required to obtain the calibrated test-set probabilities (data split, preprocessing, model training/selection, calibration, and test evaluation) are implemented in Appendix B and are not repeated here.

```
from sklearn.metrics import confusion_matrix

# Winner model probabilities on TEST (already computed in the pipeline)
proba_map = {
    "LogisticRegression": p_lr,
    "RandomForest": p_rf,
    "XGBoost": p_xgb
}
proba_test = proba_map[winner_cv]

# Payoff scenarios (use the exact values from your Part II tables)
scenarios = {
    # Clinical (QALYs)
    "S1_Clinical_Balanced": {"U11": -0.060, "U10": -0.130, "U01": -0.039, "U00":
0.000},
    "S2_Clinical_Conservative": {"U11": -0.060, "U10": -0.650, "U01": -0.039, "U00":
0.000},
    "S3_Clinical_Aggressive": {"U11": -0.060, "U10": -0.130, "U01": -0.195, "U00":
0.000},

    # Economic (USD)
    "S4_Econ_Balanced": {"U11": -62775.0, "U10": -92133.0, "U01": -1171.0, "U00": 0.0},
    "S5_Econ_Conservative": {"U11": -177658.0, "U10": -260737.0, "U01": -1171.0, "U00":
0.0},
    "S6_Econ_Aggressive": {"U11": -62775.0, "U10": -92133.0, "U01": -23420.0, "U00":
0.0},
}
```

Code E.1: Inputs & payoff scenarios

```

def p_star_from_payoffs(U11, U10, U01, U00):
    # Pauker-Kassirer:  $p^* = \text{LFP} / (\text{GTP} + \text{LFP})$ , where  $\text{LFP} = U_{00} - U_{01}$  and  $\text{GTP} = U_{11} - U_{10}$ 
    LFP = U00 - U01
    GTP = U11 - U10
    denom = GTP + LFP
    if denom <= 0:
        raise ValueError(f"Invalid payoffs: GTP+LFP must be > 0, got {denom:.6f}")
    return float(LFP / denom)

def policy_report_row(y_true, proba, thr, U):
    """
    Apply policy: Escalate if proba >= thr.
    Return one row with Escalate rate, confusion matrix, operational metrics,
    and realized expected utility on TEST given the same Uij payoffs.
    """
    y_pred = (proba >= thr).astype(int)
    tn, fp, fn, tp = confusion_matrix(y_true, y_pred, labels=[0, 1]).ravel()

    n = len(y_true)
    escalate_n = int(y_pred.sum())
    escalate_rate = escalate_n / n

    sens = tp / (tp + fn) if (tp + fn) else np.nan
    spec = tn / (tn + fp) if (tn + fp) else np.nan
    ppv = tp / (tp + fp) if (tp + fp) else np.nan
    npv = tn / (tn + fn) if (tn + fn) else np.nan

    # Realized total utility on TEST under this policy (counts-weighted)
    EU_total = (tp * U["U11"]) + (fn * U["U10"]) + (fp * U["U01"]) + (tn * U["U00"])
    EU_per_patient = EU_total / n

    return {
        "p*": float(thr),
        "Escalate_n": escalate_n,
        "Escalate_%": float(escalate_rate),
        "TP": int(tp), "FP": int(fp), "FN": int(fn), "TN": int(tn),
        "Sensitivity": float(sens),
        "Specificity": float(spec),
        "PPV": float(ppv),
        "NPV": float(npv),
        "EU_per_patient": float(EU_per_patient),
        "EU_total": float(EU_total),
    }

```

Code E.2: Threshold & policy evaluation functions

```

# Single-line summary (keep or remove)
print(f'Operational policy table on TEST | winner={winner_cv} | n={len(y_test)} |
prevalence(y=1)={(y_test==1).mean():.4f}')

rows = []
for scen_name, U in scenarios.items():
    thr = p_star_from_payoffs(U["U11"], U["U10"], U["U01"], U["U00"])
    row = policy_report_row(y_test, proba_test, thr, U)
    row["Scenario"] = scen_name
    rows.append(row)

op_tbl = pd.DataFrame(rows)[[
    "Scenario", "p*", "Escalate_n", "Escalate_%",
    "TP", "FP", "FN", "TN",
    "Sensitivity", "Specificity", "PPV", "NPV",
    "EU_per_patient", "EU_total"
]].sort_values("p*")

display(op_tbl)

```

Code E.3: Operational policy table on test set